

2019 K-Innovation ODA Program with Indonesia

Policy Consultation on STI Policy and R&D Governance

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Title	Policy Consultation on STI Policy and R&D Governance
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Policy Consultation on STI Policy and R&D Governance

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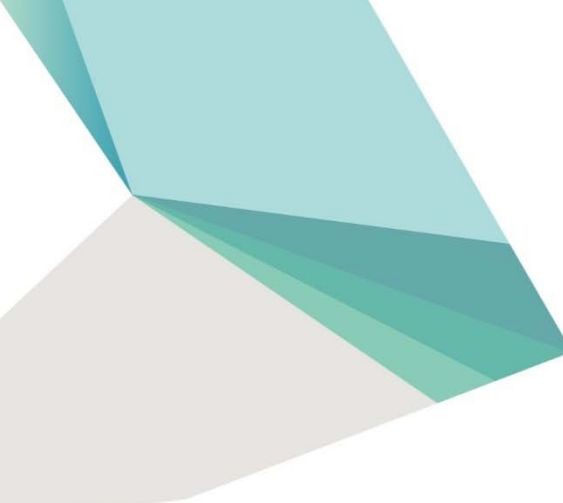
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Executive Summary

The Science and Technology Policy Institute (STEPI) has established itself as a leading global think tank in science, technology, and innovation (STI) policy. To serve as the hub of Korea's ODA in science, technology, and innovation, STEPI opened the International Innovation Cooperation Center (IICC) in 2014. Since its establishment, IICC has successfully performed three key roles -- policy consulting, training, and network building in the field of STI policy - - while working closely with partner countries and organizations around the globe. Among its activities, IICC's representative program is the Science, Technology, and Innovation Policy (STIP) Program, designed to meet the demands of developing countries with customized consultancy and training for the specific country concerned (Wangdong Kim et al., 2018).

As the element of the STIP program for Indonesia, the project is being implemented with the title "STEPI-LIPI K-Innovation Project 2018: Policy Consultation on 1) R&D Strategy Plan and 2) Science and Technology Park" in the first year of the project. In response to the demand from Indonesia, the project theme for 2019 was changed. The title of this year's project is "STEPI-LIPI K-Innovation Project 2019: Policy Consultation on STI Policy and R&D Governance" (Wangdong Kim et al., 2018).

Before the STEPI-LIPI Project 2018, STEPI signed an MOU with LIPI in 2015 to strengthen bilateral cooperation. On the basis of the MOU, STEPI utilized the STI Policy Program (STIP) and designed the STEPI-LIPI project in partnership with LIPI. Indonesia has been one of Korea's major partner countries in STI policy cooperation, and the project is expected to strengthen cooperation between the two institutes(Wangdong Kim et al., 2018).

The project mainly aims to strengthen the capacity of LIPI, but it also seeks to diagnose Indonesia's STI Policy and R&D governance status. The research team was formed to diagnose -- in partnership with LIPI -- the Indonesian STI Policy and R&D governance system,

which will provide the basis for the development of Science, Technology, and Innovation policy in Indonesia. For the sharing of the relevant Korean experience during the Capacity-building Workshop and provision of constructive comments for the progress evaluation of the program, the research team identified and selected additional experts deemed most appropriate based on their education, experience, and expertise in the jointly selected topic.

To achieve the abovementioned goals, the STEPI research team initiated the kick-off meeting and feasibility study before the start of the project. A capacity-building local workshop was completed in May 2018, December 2018, and July 2019. STEPI held the Dissemination Seminar in Jakarta on October 22, 2019 with several participants such as Indonesian government officials and researchers from the Ministry of Science, Technology, and Higher Education, Ministry of National Development Planning, LIPI, BPPT, etc.

The workshops and seminar were very effective in deriving future action plans for LIPI to develop the Science and Technology Innovation policy and improve the R&D governance system in Indonesia. The STEPI team shared the Korean experiences and policy directions related to governance, especially in the national science and technology innovation system being promoted by the new Indonesian Government. Notably, Indonesia's Basic Act on Science and Technology was passed in August 2019 based on the policy advisory and is currently in the process of revision. Moreover, there were subsequent requests from a department of the Indonesian government, i.e., Audit Board of the Republic of Indonesia, that institutes want to visit STEPI to benchmark the Korean experience in the National Innovation System and STI policy instruments.

Most of all, the STEPI research team is striving to share the experience of Korea in STI Policy and R&D Governance with an Indonesian partner organization. We expect the project to pave the way for Indonesia and LIPI to promote the country's technology innovation capabilities and strengthen collaboration with STEPI.

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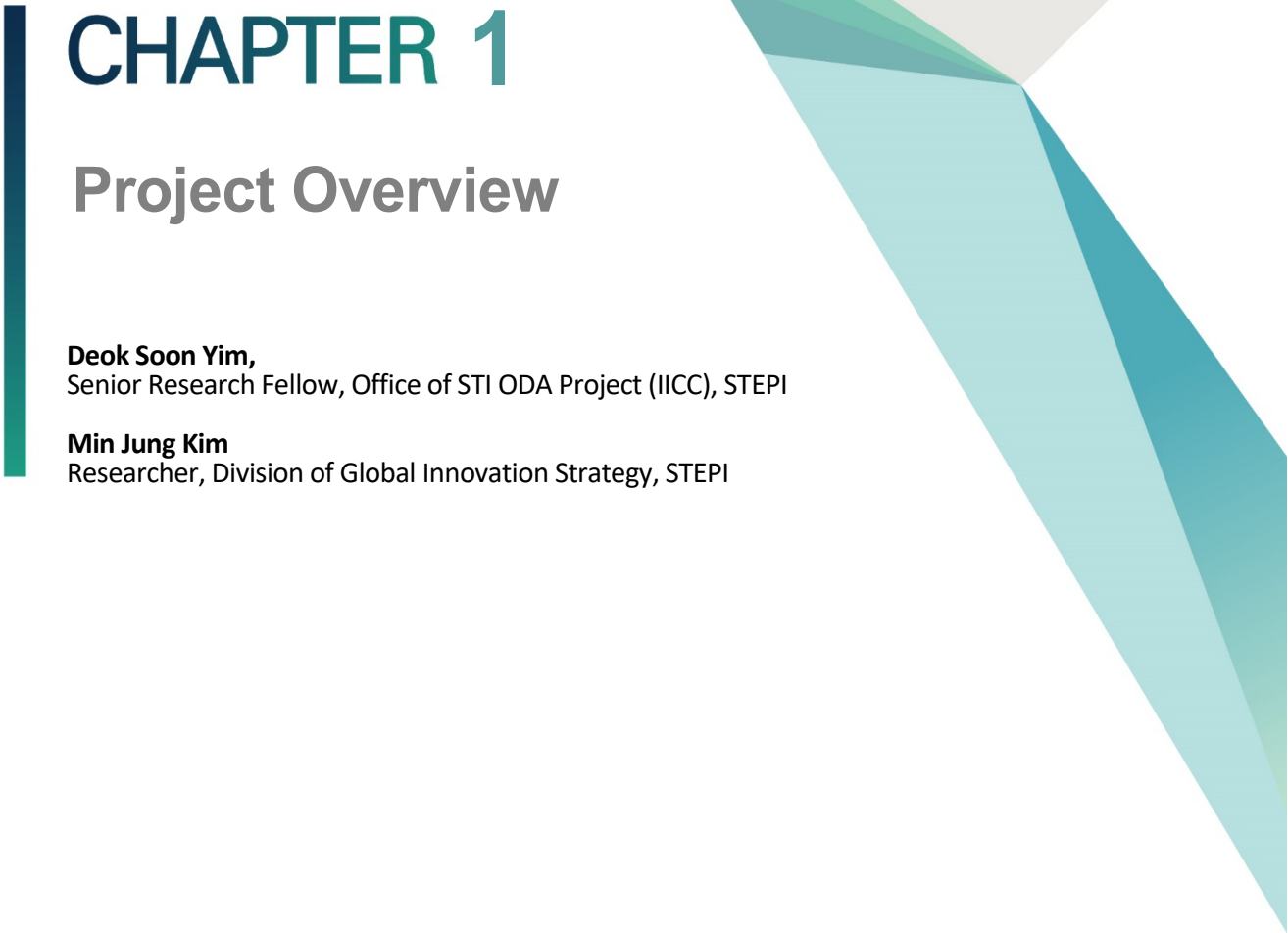
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CHAPTER 1

Project Overview

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Chapter 1. Project Overview

1. Introduction

During the global financial crisis and the slowdown of the world economy, Indonesia maintained relatively high GDP growth, averaging 5.9% between 2009 and 2013 (OECD, 2014). Despite the weak global context and lower export prices, growth has remained relatively robust. Indonesia's Gross Domestic Product (GDP) was 1015.54 billion US dollars in 2017 (OECD, 2015), representing 1.64 percent of the world economy (OECD, 2015; OECD, 2016). Indonesia has made great strides in improving its economic and social outcomes. Note, however, that the industry in Indonesia is mainly composed of primary industries such as agriculture, forestry, and fisheries (Wangdong Kim et al., 2018). Industrial advancement through foreign capital, etc. in this country was also slow compared to other ASEAN countries such as Thailand, Cambodia, and Singapore (CRDS, 2015). As shown in Figure 1-2, the main science and innovation indices of Indonesia are lower than the median values of OECD, except the network, cluster, and transfer indices. The poverty issue is also being addressed by expanding various social programs, although food policies protect the most vulnerable well. The Indonesian government is strongly pushing for technological innovation policies for continuous growth via advanced industry and poverty eradication (Wangdong Kim et al., 2018).

During the 1980s to the 1990s, Indonesia implemented ambitious S&T policies, developing specific industries from aircraft to shipbuilding through its state-owned corporations. Having enough financial resource especially from oil money, the Indonesian government had no difficulty mobilizing resources to fund the development of the so-called "strategic industries." Technically, these efforts have led to technological progress, especially in the

aircraft industry. Starting from manufacturing licensed aircraft, state-owned aircraft company IPTN (currently known as PTDI) proceeded to develop aircraft jointly with Spanish aircraft company CASA. Finally, Indonesia pushed its ambition even further by developing its own designed aircraft, N250, a twin-engine aircraft that can accommodate 50 passengers. Nonetheless, the success in engineering has not been accompanied by the success in business, which shows that technological development is much more than just creating technological artifact.

Despite the current position in S&T development, Indonesia has a relatively long history of research. The richness of its biodiversity has attracted overseas researchers for more than two hundred years. The Dutch colonial government has established research institutions to explore and exploit its rich natural resources (Goss, 2011). After the country gained its independence, the research institutions were steadily taken over by local researchers. While buildings could be easily being taken over and occupied, however, the same could not be said for knowledge and scientific culture.

The missing link of knowledge and scientific culture during the transition from Dutch researchers to their Indonesian counterparts cannot be solely blamed for the current S&T condition of the country. Countries such as South Korea, even without a rich S&T institution heritage from their colonial government, could also achieve technological progress.

As a matter of fact, Indonesia has maintained its low R&D intensity in the past. R&D intensity was falling steadily from 0.54% in 1982 to around 0.14% of GDP in 2013. The government sector contributes to around 80% of national R&D activity, and the rest is the share of the private sector (Pappiptek-LIPI, 2013, 2011, 2007, 2006). The private industrial sector mostly generated innovation through learning without formal R&D. It suggests that Indonesia had been less successful in developing STI capacity in the past, including intellectual capital formation, knowledge mastery accumulation, technical capabilities acquisition, and STI policy governance. Low STI capacity resulted in low absorptive and learning capacity by actors in the system of innovation (Schot, J. and Steinmueller, W.E., 2018).

The system of innovation may not work effectively due to problems in the elements of innovation system and its functioning. A key element of the innovation system is institution, where institutional failure creates a systemic problem in the system of innovation. Creating an adequate institutional framework facilitates collective learning among actors of those elements of the innovation system (Intarakumnerd, P. and Chaminade, C., 2007). The collective learning among actors is more than just R&D activities but focuses more on STI capacity building and networking. In short, the role of the institution is important in promoting STI and R&D governance.

STI policy and R&D governance are essentially about STI capacity building, which deals with the problems of building of scientist and engineer competency, physical technological infrastructure, entrepreneurial and managerial capacity, and scientific knowledge accumulation through R&D. STI capacity building is expected to solve the pressing social and economic problems of the country, transform its societies, and positively impact the standards of living and quality of life (Watkins, A. and Ehst, M., 2008).

Discussing STI policy issues and R&D governance problems in Indonesia has been documented in literature -- Sigurdson, J. and Polanka, K. (2002), OECD (2012), OECD (2012), Naudé, W. (2013), and ADB (2019), among others. STI policy issues and R&D governance problems need a “fitting way” (with the Indonesian STI realities) of thinking to provide policy alternatives and governance solutions.

The objective of this study is to find the way forward in promoting STI and R&D governance in Indonesia and provide policy consultation based on the lessons of the Korean experience. This report will discuss the mid-/long-term STI policy of Indonesia and the governance and institutional aspects of its S&T development while comparing with the Korean experience of STIP and R&D governance system.

The discussion of S&T governance in Indonesia should proceed in a different direction from that in developed countries. In Europe, for example, people are no longer concerned with advancing S&T development and industrial utilization since S&T has been developed

intensively. People are more concerned with the societal and environmental impacts of S&T, so they have "to take into account the expanding range of stakeholders and public assessments (including consumer choices and more diffused expressions of anxiety)" (Hagendijk, Healey, Horst, & Irwin, 2005, p. 7). In Indonesia, S&T governance is needed to steer and push S&T development more effectively given the scarce resources that S&T communities can utilize. It does not imply that technology has no societal and environmental impact. There have even been heated debates and confrontation between the government and the opponents of nuclear plant when the plant was planned to be built in the Middle Java area. It has engaged local people assisted by NGO activists, including those knowledgeable in nuclear energy (Amir, 2010). Ultimately, the plan of building the nuclear plant was canceled. Currently, there is also strong dispute in the development of a cement factory, which is being opposed by many local people. Nonetheless, those cases will not be discussed in this report, because they are caused by technologies that were not developed in Indonesia. There are various strategies being pursued by the Indonesian government to develop its R&D capacity and innovation system (OECD, 2014: 344-356).

This paper will focus on highlighting 1) the comparison and analysis of long-/mid-term STIP of Indonesia and Korea and 2) review of the status and challenges of R&D Governance in Indonesia and provide some lessons from the Korean case.

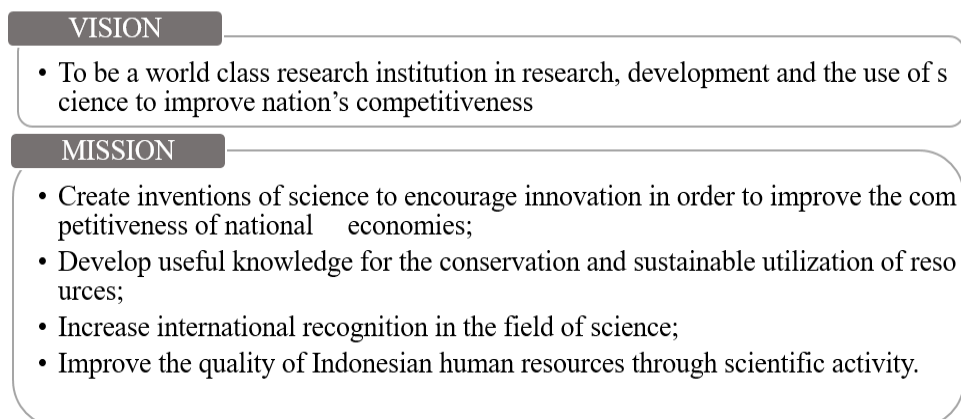
2. Objectives

The project is titled "STEPI-LIPI Project 2019 Policy Consultation on STI Policy and R&D Governance." Before the STEPI-LIPI Project 2019, STEPI signed an MOU with LIPI in 2015 to strengthen cooperation. This project was developed based on the MOU signed by STEPI and LIPI in 2015 to strengthen cooperation between the two organizations. STEPI will operate the STI Policy Program (STIP) in partnership with LIPI. Indonesia is also one of Korea's major partner countries in development cooperation (Wangdong Kim et al., 2018).

The Science and Technology Policy Institute (STEPI) is one of the government-funded research institutes in Korea and a major leading global think tank conducting research on policy issues in science, technology, and innovation. STEPI works closely with various ministries, regional governments, and relevant institutes to support policy design. It also has a long experience in providing STI policy consultation for developing countries, and it recently launched the International Innovation Cooperation Center (IICC) with the aim of conducting STI development cooperation activities with partner countries. The program duration is from January 2, 2019 to December 31, 2019(Wangdong Kim et al., 2018).

The Indonesian Institute of Sciences (LIPI) is a non-ministerial government research institution reporting to the President under the auspices of the Ministry of Research and Technology. The vision of LIPI is to be a world-class research institution in research, development, and use of science to improve the nation's competitiveness. In order to achieve said mission of the institution and to strengthen the capacity of the agency, LIPI has requested the Science and Technology Policy Institute (STEPI) to evaluate the STI policy and its implementation in Indonesia and also review the STP policy and management (Wangdong Kim et al., 2018).

[Figure 1-1] Vision and Mission of LIPI



The objectives of the STEPI-LIPI Project 2018 are to: 1) review and evaluate the STI policy and management in Indonesia (based on the Indonesian Mid-term Development Plan 2015-2019); 2) assist and evaluate the STI policy and management in Indonesia (based on the Indonesian Long-term Development Plan 2020-2025); and 3) review and provide consulting on the Science and Technology Park (STP) Policy and management in Indonesia on a full scale (Wangdong Kim et al., 2018).

Especially, this project sought to support the development of R&D strategy and STP competitiveness building in Indonesia in the following aspects (Wangdong Kim et al., 2018):

- 1) Review and evaluation of STI policy and STP management in Indonesia
- 2) Advice on research methodology for the main components including the following (but not limited to):
 - i. Policy recommendations for better STI policy and its implementation
 - ii. Data survey of STI policy implementation
 - iii. Review and evaluation of STI policy and STP management in Indonesia
 - iv. Visiting research program
 - v. Capacity building for researchers, policymakers, and STP managers
- 3) Progress assessment
- 4) Sharing the Korean experience relevant to Indonesia

This project provided comments and advice on the conceptual and logical frameworks of the R&D strategic plan and Science Technology Park Policy in Indonesia. In other words, the project included the review and evaluation of the Science, Technology, and Innovation (STI) policy and management in Indonesia, based on the Indonesian Mid-term Development Plan 2015-2019. The project assists in and evaluates the STI policy and management of Indonesia based on Indonesia's Long-term Development Plan 2020-2025. Based on these analyses, policy implications were derived. Capacity-building workshops were conducted for the purpose of sharing the relevant Korean experience and methodologies for the program development of LIPI (Wangdong Kim et al., 2018).

3. Project Framework

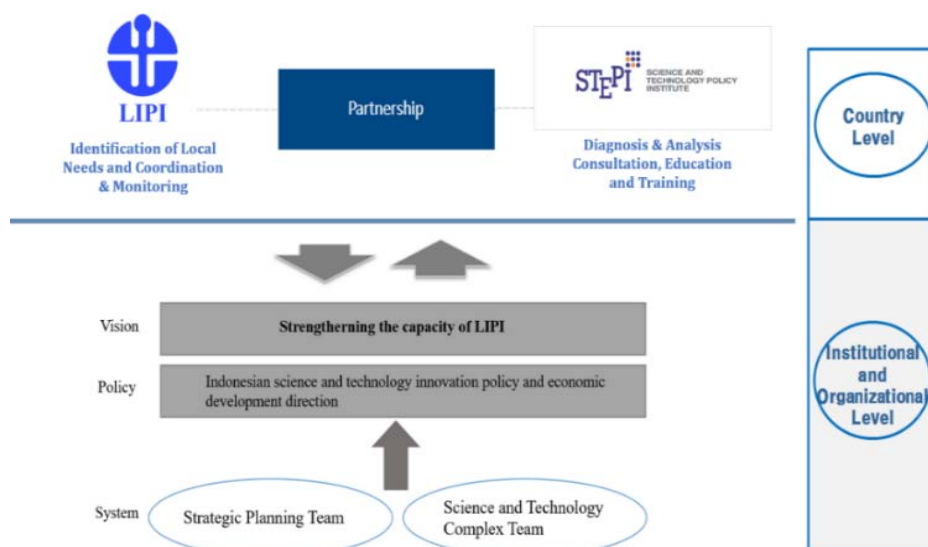
3.1 First year, K-Innovation Project 2018

For the first year, two themes were covered: STP (Science and Technology Park) and R&D Strategic Planning. The project proceeded by dividing the members into two teams and covered research topics in parallel (Wangdong Kim et al., 2018).

The Strategic Planning Team will seek strategies for LIPI based on forecasting methodology and lay the groundwork for the manpower management and evaluation of LIPI researchers. The Science and Technology Complex Team will identify the status of science and technology complexes under the jurisdiction of LIPI and seek directions for development (Wangdong Kim et al., 2018).

With these diagnoses and evaluations, the findings of the research suggested the strategic plan and provided remedy for the development of STP (Science and Technology Park) and R&D strategic plan (Wangdong Kim, et al, 2018).

[Figure 1-2] Conceptual Framework of the 2018 Project



3.2 Second year, K-Innovation Project 2019

After the internal meeting between STEPI and LIPI in Jakarta in December 2018, the demand of LIPI was found to have been changed. Thus, the main themes of the 2019 project are adjusted to Science and Technology Innovation policy review and giving of some lessons on the national R&D governance system.

As mentioned before, the process of the project mainly focuses on strengthening the capacity of LIPI but also aims to diagnose the Indonesian STI Policy and R&D governance status.

[Figure 1-3] Conceptual Framework of the 2019 Project



4. Project Team

4.1 Research Team

The research team is composed of two full-time experts: Dr. Deok Soon Yim serves as team leader, and the program coordinator is Ms. Min Jung Kim a researcher of STEPI (Wangdong Kim et al., 2018).

In year 2018, the following additional experts were identified for the delivery of the Capacity-building workshop and expert advice based on the topics to be covered: Seong Won Park, Kibeom Park, Jaehoon Rhee, Young Sub Kwon, and Yoo Hyung Won (Wangdong Kim et al., 2018).

In year 2019, the STEPI team invited three experts who specialize in STI policy and R&D governance. Prof. Seong-soo Kim is a professor in public administration of HUFS, and he has researched on STI policy. Dr. Youngrak Choi is the former president of STEPI, and he knows a lot about the history and trends of the Korean STI policy. Dr. Youngjoo Ko has special job experience as a member of the Evaluation Committee of the National Science & Technology Council 2013~2016, so he is the suitable person for transferring Korea's experience and know-how in R&D governance.

[Table 1-1] Research Team of the K-Innovation Project 2018

Name	Organization & Position	Roles
Deok Soon Yim	Senior Research Fellow of STEPI	Project Manager
Seong Won Park	Research Fellow of the National Assembly Futures Institute	Consulting on R&D Strategic Plan Evaluation
Jaehoon Rhee	CEO of Gyeongbuk Technopark	Consulting on the Operation of Science and Technology Park
Yoo Hyung Won	Head of the Policy Department in the Korea Institute of Science and Technology	Consulting on R&D Strategic Plan Evaluation
Young Sub Kwon	Senior Research Fellow of the Korea Research Institute for Human Settlements	Consulting on the Selection and Evaluation of the Location of the Science and Technology Complex
Min Jung Kim	Researcher of STEPI	Project Coordinator

[Table 1-2] Research Team of the K-Innovation Project 2019

Name	Organization & Position	Roles
Deok Soon Yim	Senior Research Fellow of STEPI	Project Manager
Dr. Youngrak Choi	Senior Research Fellow Emeritus, STEPI	Consulting on STI Policy
Dr. Seong-soo Kim	Professor of Hankuk University of Foreign Studies (HUFS)	Consulting on STI Policy
Dr. Youngjoo Ko	Principal Researcher of the Korea Research Institute of Chemical Technology (KRICT)	Consulting on R&D Governance
Min Jung Kim	Researcher of STEPI	Project Coordinator

The research team is responsible for the diagnosis of the Indonesian STI Policy and R&D governance system in partnership with LIPI, which provide the basis for the development of the Science, Technology, and Innovation policy in Indonesia. For the sharing of the relevant Korean experience during the Capacity-building workshop and provision of constructive comments for the progress evaluation of the program, the research team will identify and select additional experts deemed most appropriate based on their education, experience, and expertise in the jointly selected topic (Deok Soon Yim, et al, 2016).

5.

Project's Main Activities

5.1 Activity 1: Preparation of the project

From August to October 2016, the STEPI team conducted a literature review related to 1) R&D Strategy Plans and 2) Science and Technology Park management of Indonesia. Any relevant documents on the innovation system of Indonesia to enhance understanding should be provided prior to “Activity 2.” A project proposal and the program for demand survey and field study were developed based on the project concept (Deok Soon Yim, et al, 2016; Wangdong Kim et al., 2018).

5.2 Activity 2: Pre-feasibility study

The STEPI team conducted a demand analysis and a pre-feasibility study for the project from December 23, 2017 to February 8, 2018. The study sought to analyze the feasibility of the STEPI-LIPI project 2018 through detailed demand analysis and literature review (Wangdong Kim et al., 2018).

It concluded that the STEPI-LIPI project 2018 is very relevant considering the Indonesian R&D and innovation environment (Wangdong Kim et al., 2018).

- Chi-Woong Song, research fellow of STEPI, attended the 6th National Conference on Science and Technology in Indonesia (Indonesia) to present the subject of experience and implications of the Korean science and technology policy; policy advisory projects were discussed with the LIPI researchers during the conference.
- From November 20 to December 3, 2016, two researchers from LIPI (two persons) stayed for two weeks at STEPI-IICC, visiting researcher programs and consulting on policy advisory projects.
- Consultation was held on the STP (Science and Technology Park) policy for regional innovation.
- Conducted interviews and surveys with Indonesian government officials and researchers.

5.3 Activity 3: Local capacity-building workshop

A research team consisting of 5 people carried out the 1st dispatch and led a local capacity-building workshop on May 1–6, 2018 (including the departure / arrival date) (Wangdong Kim et al., 2018).

[Table 1-3] Photos of the Workshop in Indonesia

Attendees of the Strategic Planning Team Workshop	Attendees of the Science and Technology Complex Team Workshop
	

The workshop had three main purposes: 1) the workshop focused on selected topics that were organized during business trips in December 2017; 2) the workshop aimed to derive the future strategic plan and human resource management methods of the 2020-2024 LIPI; 3) the workshop sought to consult on the establishment and operation of LIPI's Science and Technology Park (STP) (Wangdong Kim et al., 2018).

STP (Science and Technology Park) diagnosis was an important topic of the workshop at the same time. Additionally, it included simulation and strategy forecast methods to identify potential problems that may arise during the advisory stage and uncover solutions for such potential problems (Wangdong Kim et al., 2018).

5.4 Activity 4: Capacity-building Workshop in Korea with LIPI and KU

The Korea Institute of Science and Technology Policy (STEPI) held a capacity-building training session for seven days from December 2 to 8. The training program -- which was conducted with 7 researchers from the Indonesian Institute of Sciences (LIPI) and 2 professors from Kathmandu University (KU) -- is part of the K-Innovation 2018 aimed at strengthening the capacity of both LIPI and KU (Wangdong Kim et al., 2018).

As part of the “STEPI-LIPI Project 2018 Policy Consultation on 1) R&D Strategy Plan and 2) Science and Technology Park” and the “Project for the Integrated Rural Development of Nepal by Strengthening the Research and Development Capacity of Kathmandu University,” the team from LIPI and NTIC was invited to Korea to participate in the capacity-building workshop. Through the workshop, the participants were able to deepen their understanding, particularly of the role of the technopark (Korean version of Science Technology Park) in Korea and R&D strategy in general, including how their functions and programs can be applied to LIPI (Indonesian Institute of Sciences) and KU (Kathmandu University) as well as the necessary policy measures (Wangdong Kim et al., 2018).

The lectures served as a theoretical and practical intensive course designed for Nepalese government officials and qualified practitioners looking to broaden their knowledge in the areas of Science Technology Park by participating in practical training. Field visits were also provided in two practical areas: for example, Gyeongbuk Technopark, Daegu Technoparks, and Start-up promotion in Hanbat University (Wangdong Kim et al., 2018).

Overall, the workshop was composed of two methods: 1) lectures and 2) field visits. The participants were expected to participate actively in the course and derive implications and lessons for developing the Science Technology Park Policy, which could be applied to LIPI and KU (Wangdong Kim et al., 2018).

**Opening Ceremony and Lecture by Dr.
Sunyang Chung, professor of Konkuk
University**



**Lecture by Dr. Jaehoon Lee,
CEO of Gyeongbuk Technopark**



6. Project Schedule

6.1 Project Schedule

The entire schedule of the project is as follows (Wangdong Kim et al., 2018):

[Table 1-4] Project Schedule

Schedule	Contents
Pre-feasibility Study (November 20—December 3, 2017)	- Analyze the feasibility of the R&D Strategic Plan - Evaluate the consulting project through detailed demand survey and detailed survey
Field Study and Local Capacity-building Workshop (May 1—6, 2018)	- Derive the future strategic plan and human resource management methods of the 2020-2024 LIPI - STP (Science and Technology Park) diagnosis
Request review the draft of strategic plan (October 2018)	- Review the drafts of the LIPI 2020-2025 Strategic Plan and STP policy; provide comments and policy implications
Invitation Training (December 3—8, 2018)	- Through the capacity building invitation training, the participants were able to deepen their understanding particularly of the role of the Science Technology Park in Korea, including how their functions and programs can be applied to LIPI as well as the necessary policy measures. - Seven outstanding LIPI researchers were invited to participate in the highly condensed, seven-day capacity-building workshop. The workshop was conducted in the forms of 1) lectures and 2) field visits.
Final Report and Demand Survey Meeting (December 9—14)	- Final Report of the “Policy Consultation on the R&D Strategy Plan and Science and Technology Park” - Workshop for the relevant stakeholders for the successful implementation of programs such as civil servants, scholars, and businessmen in related industries

7. Project Outputs

STEPI provided the following project deliverables:

Deliverable 1: Presentation and training materials for the Invitation capacity-building workshop with LIPI and KU in Korea (December 2018)

Deliverable 2: Presentations and training materials for the Local capacity-building workshop (May 2018)

Deliverable 3: Final Report on the Policy Consultation on R&D Strategy Plan and S&T Policy (December 2018)

Deliverable 4: Presentation and training materials for the Local capacity-building workshop in Jakarta (March 2019)

Deliverable 5: Presentation and training materials for the Invitation capacity-building workshop with Indonesia and Uganda in Korea (July 2019)

Deliverable 6: Presentation and training materials for the Local capacity-building workshop in Jakarta (October 2019)

Deliverable 7: Final Report on the Policy Consultation on STI Policy and R&D Governance (December 2019)



CHAPTER 2

Current Status of STI Policy and R&D Governance in Indonesia

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Chapter 2. Current Status of STI Policy and R&D Governance in Indonesia

1. Overview of the STI Policy in Indonesia

1.1 Background

Indonesia has maintained its low R&D intensity in the past. R&D intensity was falling steadily from 0.54% in 1982 to around 0.14% of GDP in 2013. The government sector contributes to around 80% of the national R&D activity, and the rest is the share of the private sector (Pappiptek-LIPI, 2013, 2011, 2007, 2006). The private industrial sector mostly generated innovation through learning without formal R&D. It suggests that Indonesia had been less successful in developing STI capacity in the past, including intellectual capital formation, knowledge mastery accumulation, technical capabilities acquisition, and STI policy governance. Low STI capacity resulted in low absorptive and learning capacity by actors in the system of innovation. (Schot, J. and Steinmueller, W.E., 2018)

The system of innovation may not work effectively due to problems in the elements of innovation system and its functioning. A key element of the innovation system is institution, where institutional failure creates a systemic problem in the system of innovation. Creating an adequate institutional framework facilitates collective learning among actors of those elements of the innovation system. (Intarakumnerd, P. and Chaminade, C., 2007) The collective learning among actors is more than just R&D activities but focuses more on STI capacity building and networking. In short, the role of the institution is important in promoting STI and R&D governance.

STI policy and R&D governance are essentially about STI capacity building, which deals with the problems of building of scientist and engineer competency, physical technological infrastructure, entrepreneurial and managerial capacity, and scientific knowledge accumulation through R&D. STI capacity building is expected to solve the pressing social and economic problems of the country, transform its societies, and positively impact the standards of living and quality of life (Watkins, A. and Ehst, M., 2008).

Discussing STI policy issues and R&D governance problems in Indonesia has been documented in literature -- Sigurdson, J. and Polanka, K. (2002), OECD (2012), OECD (2012), Naudé, W. (2013), and ADB (2019), among others. STI policy issues and R&D governance problems need a “fitting way” (with the Indonesian STI realities) of thinking to provide policy alternatives and governance solutions. The objective of study is to find the way forward in promoting STI and R&D governance in Indonesia. From the viewpoint of institution and public policy, this paper constructed an operational model of STI policy and R&D governance system, which had three characteristics: long-term strategic perspective; top STI policy direction, and synergistic R&D governance. The model enriched the three frames of innovation of long term-growth, global competitiveness, and sustainable living system as claimed by Schot, J. and Steinmueller, W.E. (2018).

1.2 Issue and problem

1.2.1 STI policy issue

The lack of priority setting and policy discontinuity were two events that generally occurred during the course of STI policy implementation in Indonesia. The events were related to the distribution and division of STI resources across sectoral boundaries. Ministries and institutions compete to secure as many resources as possible from the limited STI budget allocated by the central government. Consequently, the choice for picking the winner in STI as a far-reaching sector so far cannot be executed because all ministries and institutions are considered the priority, or all are picked to be winners due to the competition among sectors and short-term orientations. As a result, policy

discontinuity is a chronic failure. Given the limited STI budget, the key question is how to focus priorities and implement an effective STI policy for economic development especially industrial technology development. It is the first STI policy issue.

The second STI policy issue is how to make use of the public R&D outputs in the area of STI to contribute to fulfilling societal needs, mainly economic actors, especially in the industrial sector. Economic development is mostly driven by industrial growth, especially manufacturing. S&T-based industrial innovation is the key for moving up toward hi-tech industries. Given the pattern of Indonesia's industrial output, around 70%, comes from low- and medium-tech industries, namely SMEs and micro-scale businesses. Moreover, Indonesian industries, mostly in the private sector, are still dominated by innovation with low and/or without R&D activities. Indonesian industries need to rise from low-tech producers to medium-tech industries, and the medium-tech class should go up to high-tech industries as well. It is imperative to make use of public R&D outputs in the area of STI to contribute to raising the technology capability of low- and medium-tech industries.

1.2.2 R&D governance problem

R&D activities in the public sector face the stalled process of downstreaming from R&D results into products or commercial innovation. R&D activities in the public sector are separated by specific disciplines and split according to the sectoral orientation of institutions. R&D activities tend to ignore the importance of integral sequencing from upstream to downstream, consisting of research, technology, and innovation for the global market. Downstreaming the outputs of R&D in the fields of STI into commercial products is a multi-disciplinary, inter-disciplinary, and cross-disciplinary activity. R&D activities should discover new knowledge, invent new technology, develop new design, and create innovation, which manifest into commercial marketing activities and expand operations in the global market. The first problem of R&D governance is how to integrate and downstream R&D in the fields of STI into commercial products.

Since Indonesian industries — mostly those in the private sector — are still dominated by innovation with low and/or without R&D activities, solving the problem of innovation

without R&D is constrained by the lack of capacity to absorb technological capability. Consequently, industries face difficulties in moving up the technological capability ladder, namely S&T-based industrial innovation. This weakness is aggravated by the global environment, where the innovation race is very rapid and the competition among innovative products is very tight. The lack of industrial R&D efforts to catch up with S&T-based industrial innovation has made Indonesia lag way behind, because the global S&T world is running faster. The second problem of R&D governance is how to move up the technological capability ladder toward creating industrial innovation based on S&T research.

1.3 Methodology and structure of the paper

This is a desk study by collecting, using, and analyzing the available documents, literature, data, and evidence (mostly from the author's previous works) related to the problem and issue. The methodology used in this study is qualitative modeling. The issue and problem are assessed and brought into solutions and recommendations by using qualitative modeling analysis. The stages of modeling that have been carried out are as follows: i. formulation of issue and problem; ii. problem conceptualization as seen from a particular approach or perspective; iii. model construction as a tool or an analytical framework to explain the issues and problems conceptually; iv. provision of data and evidence on the issue and problem; v. analysis of data and evidence by using a workable model or an applied analytical framework; vi. conclusion and recommendation.

The rest of this paper is organized as follows. Following this introductory section, the next section will describe data and facts concerning policy actions and instruments that have been applied in the past. The outcomes of these government policy efforts are presented in the form of statistics/quantitative science indicators supplemented with the results of case studies on R&D activities in the manufacturing industry. Furthermore, based on the statistical data and case study facts, more detailed elements and factors influencing STI capacity and R&D governance are described. Explaining the Indonesian STI capacity R&D governance was confirmed by making a comparison of countries. Furthermore, based on

thorough description of Indonesian STI policy issues and R&D governance, the policy analysis of STI challenges, operation, and expected outcomes was explained by using the operational model of STI Policy and R&D governance system. The conclusion will be supplemented with policy recommendations to achieve the desired outcomes.

2. STI Policy and R&D Governance Framework

Based on policy issues and governance problems (section 1.2), we construct the way of thinking needed to provide policy alternative and governance solution. Viewed from the perspective of “institution and public policy,” we constructed the STI policy and R&D governance framework, which have three characteristics as described below

2.1 Long-term strategic perspective

As for the nature and scope of STI for economic development, they are cross-disciplinary and cross-sectoral, with broad spectrum and far-reaching perspective. In the future, the application of STI for economic competitiveness needs to be demonstrated by creating innovative products and services resulting from S&T-based industrial innovation. The innovative products and services should not only target the domestic market but also develop into global production networks and global innovation networks. Given the very rapid S&T development race for global innovation, the competition among global innovative products is very tight. Indonesia's low R&D efforts in catching up in terms of S&T development have made the country lag behind in the fast-paced global S&T race. Such S&T lag has been exacerbated by Indonesia's past experiences in policy discontinuity on S&T. The availability of long-term strategic choice to maintain policy continuity determines the effective allocation of STI resources to spur and achieve the nation's S&T-based industrial innovation in the future.

2.2 Top STI policy direction

In handling the cross-disciplinary, cross-sectoral, global, and far-reaching nature of STI, all need to be driven and controlled by the highest STI policy direction for coordinating the country's STI policy. The STI policy needs to be decided, coordinated, and supervised by the top executive institution at the national level under the direct control of the President, by which its authority must be followed by all stakeholders in STI affairs. For the comparison on the importance of the highest controlling institution in the field of STI, in

Japan, the science and technology policy is set by the Prime Minister's Council for Science, Technology, and Innovation (CSTI), and its members are leading ministers, experts, and industrialists. In Malaysia, the science and technology policy was established by the Malaysian Industry-Government Group for High Technology (MIGHT) led by the Prime Minister. In Korea, the Korean Science and Technology Council (KSTC) is chaired by the president.

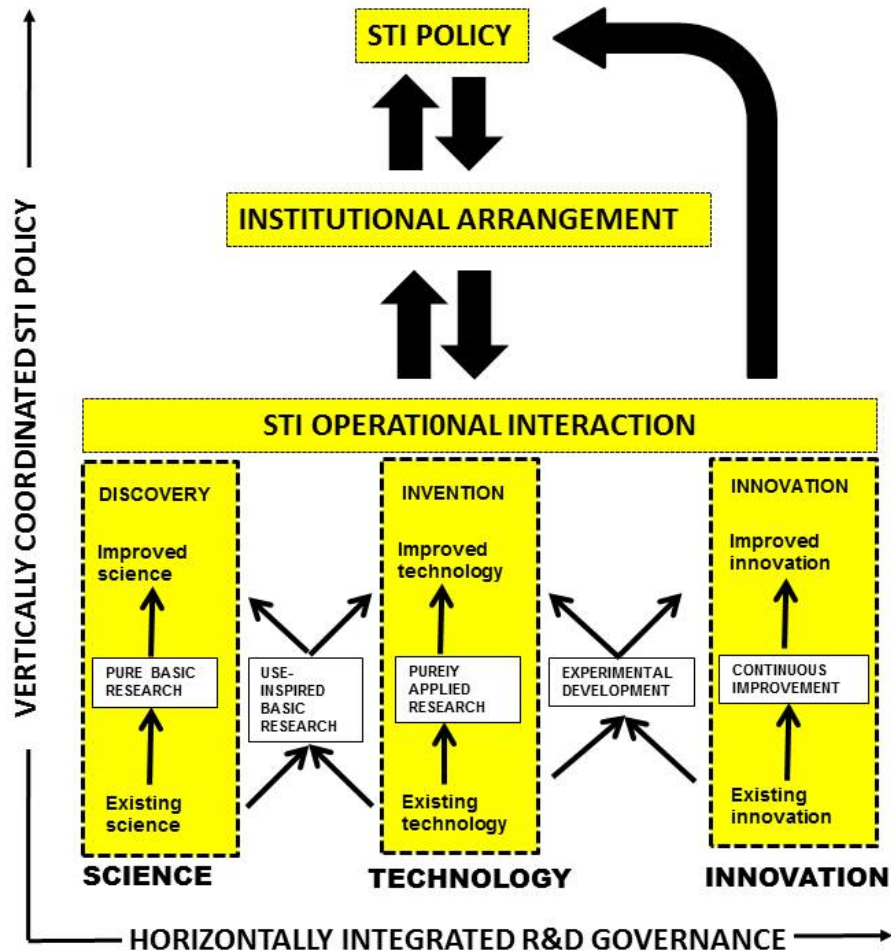
2.3 Synergistic R&D governance

R&D activities in the fields of STI are under the sectoral jurisdiction and authority of each ministry and institution. In the future, a central R&D controlling board is needed to implement a coordination mechanism to integrate sectors and institutions. The board task is assigned by the STI policy council to execute policy implementation, R&D program, budget planning, and evaluation in a clear, straightforward manner. The cross-ministerial and institutional R&D programs should be governed to meet societal needs and economic development. The R&D program needs to solve the real problems faced by Indonesia with S&T solutions and innovations in the field of basic needs (adequacy of food, health, sanitation, nutrition, medicine, energy, etc.) to meet the Sustainable Development Goal (SDGs). Synergistic R&D governance by the central R&D controlling board is more than the formulation of a national R&D master plan. Effective R&D governance is needed, including: (i) R&D policy design to produce innovative product and services in each stage of development, and (ii) R&D program and evaluation in conducting cross-disciplinary, inter-disciplinary, and multi-disciplinary R&D for creating innovative products and services.

Conceptual building that contains the three characteristics above is basically the combination of vertically coordinated STI policy and horizontally integrated R&D governance. STI policy coordination consists of three levels: STI policy, institutional arrangement, and operational interaction. The STI policy determines how institutions set the rules of works for STI operational interaction, and institutional arrangement provides feedback for STI policy enhancement. The policy should provide institutional arrangement rules on how the elements of the STI operation system interact institutionally. Furthermore, the STI operation system performance provides feedback for the

improvement of institutional arrangement (Figure 2-1).

[Figure 2-1] STI Policy and R&D Governance Framework (constructed by the author)



On the other hand, R&D governance integration contains all of the STI operation system's elements that interact with each other to realize the final goal of creating innovative products and services. The interaction between the elements of science, technology, and innovation are the continuum activities from pure basic research, use-inspired basic research, pure applied research, experimental development, and continuous improvement. The relationship between science and technology is connected by use-inspired basic research, and the relationship between technology and innovation is connected by experimental development. This framework is constructed by combining the

concept of D.W. Bromley (1989) on the role of institution in public policy with the enriched concept of D.E. Stokes (1996) on Pasteur's quadrant on basic science and technological innovation, where Pasteur's quadrant on science and technology was enriched by and connected with the author's quadrant on technology and innovation.

The interaction between science, technology, and innovation is described as follows. (i) Basic research contributes to science discovery by adding to the existing science or changing old paradigms. A classic example is the discovery of electron and radioactive by Niels Bohr, followed by many contemporary discoveries in field life sciences and material sciences. (ii) Use-inspired basic research contributes to both science discovery and technology invention, which can be discovered or invented from the convergence of existing technology with existing science. A famous example is Louis Pasteur's invention of microbiological techniques to stop bacterial contamination. The discovery of Pasteurization comes from the uses of microbiology in existing food technology. The invention contributed to enriching life sciences, and the application of invention in the food industry resulted in the latter's improvement. (iii) Pure applied research contributes to technology invention by inventing improved technology from an existing one. A well-known example is the invention of the long-lasting incandescent light bulb by Thomas Alva Edison to replace the impractical existing electric lamp. (iv) Developmental experiment contributes to both technology invention and product innovation. It can be invented from combining the existing technology with existing innovation. As an example, Bukaka made an aerobridge using the existing principles of science inherent in the existing product technology, and then improved the design of technology product according to market preferences in a more efficient, quality manner so that it competes with similar products in the global market. (v) Continuous improvement contributes to product or process innovation by creating improved product or process innovation from the existing or early innovation. A classic example is the application of kaizen in improving the production process.

This framework was used to analyze issues and problems and to guide the solving of issues of the Indonesian STI policy and the problems of R&D governance. The solutions, having three characteristics of long-term strategic perspective, top policy direction, and

synergistic R&D governance, are expected to give the following outcomes: (i) Indonesia would achieve the status of STI-based economy in the long run, i.e., economic prosperity supported by technological excellence, industrial development, and growth of manufacturing sector, which would all be achieved through science advancement, technology development, and continuous innovation; (ii) the STI-based economy would be achieved through STI policy continuity maintained by the function of STI policy council, which is capable of coordinating science, technology, and innovation resources in an integrated manner; (iii) the integration of science, technology, and innovation resource are governed by the R&D governance body, which is effectively implementing STI policies in the programs of multi-disciplinary, interdisciplinary, and cross-disciplinary R&D activities to create innovations for the advancement of the national economy, social development, and sustainability of living systems.

Before doing the analysis on STI policy and R&D governance, we first describe the background and status with some facts and figures as well as a case study reflecting the STI policy evolution and past R&D intensity in Indonesia.

3. Indonesia's STI Evolution and Past R&D Activity

3.1 Indonesia's STI Evolution

Indonesia started building its STI capacity by accessing, utilizing, and improving knowledge and technology in the 1970s, but it has not yet fully succeeded in reaping the benefit of S&T in fulfilling societal needs and increasing citizen welfare. Indonesia has shown gradual improvement in almost all STI building indicators, except scientific knowledge accumulation through R&D. The Indonesian industrial sector largely consists of the low- and medium-technology manufacturing and natural resources-based industries, which were able to engage in innovation without formal R&D. This is one explanation as to why Indonesia has maintained its low R&D intensity in the last 30 years.

Policy shifts behind the STI development process in Indonesia had occurred before and after the 1997 economic crisis. First, innovation policy for the development of technological capability prior to the 1997 economic crisis emphasized industrial policy, especially the development of the high-technology industry. The policy orientation placed the government in a dominant role by applying a strategy of state-led industrial transformation based on technological innovation. The deployed policy instrument was preferential treatment given to ten high-technology strategic companies.¹ The implementation of innovation policy on the developing high-technology industry was completed under strong government support. It had successfully brought about industrial transformation² but resulted in policy discontinuity, succumbing to the effects of the 1997

¹ These ten strategic public-owned companies (POWs) were: IPTN, now known as DI (aircraft); PAL (shipbuilding); INKA (railroad wagons); LEN (electronics); INTI (telecommunications); PINDAD (light weapons and ammunition); DAHANA (explosives); BBI (engines and machinery); BARATA (heavy equipment and construction material); and KRAKATAU (steel) (see Aminullah, E. and Fizzanty, T., 2016).

² An example of successful transformation in stages is the production of aircraft at IPTN, starting from the assembly of various small planes and helicopters in 1976, followed by the integration of previous knowledge gained in the production of CN 235 airplane with double engines and capacity of 35 passengers, designed in cooperation with CASA (air-tech) in 1984, followed in turn by the improvement of aircraft technology through R&D activities to make the N250 aircraft with double engines and capacity of 50 passengers fully designed by IPTN in 1989. The final stage was the full

economic crisis due to weak public support.

Second, innovation policy after the 1997 economic crisis emphasized R&D policy as a national priority aimed at solving nationally urgent problems. The policy was strongly oriented toward market-led industrial development through the creation of a conducive climate for technological innovation. The Indonesian government has launched several STI policy instruments to intensify R&D between 2001 and 2011. In 2001, the Indonesian Ministry of Finance launched Regulation No. 231/2001, aimed at providing compensation for imported goods used for research and science purposes, as well as exemptions from value-added tax and luxury goods tax. The regulation was restated in 2007 when the Indonesian Ministry of Finance launched Regulation No. 51/2007, aimed at providing compensation for imported goods used for R&D purposes. The next incentive was launched in 2010, where Regulation No. 93/2010 stated clearly that R&D expenditure (for private companies) was to be included in deductible expenses, and that any income-sourced donation from private companies to selected R&D institutions was to be declared fully deductible. In 2011, Regulation No. 52/2011 was launched in order to promote a one-year investment allowance for private companies who spend their income on R&D activities in national R&D institutions (Aminullah, E. and T. Fizzanty, 2016).

Although these STI policy instruments exist, the Indonesian innovation system is not well-established due to some STI policy issues and R&D governance problems as explained in Section 1.1. The negligence in STI policy toward industrial technology development (section 1.1.i) has been the underlying factor of Indonesia's STI backwardness since the 1990s. Despite the implementation of S&T policy using various policy instruments for supporting and stimulating R&D in the private sector, the subsequent statistical evidence and case studies (Section 2.2) confirmed that such policy instruments appeared not to have affected the increase of R&D activities significantly in private industrial sectors. In spite of the lack of attention and confidence of the private sector to innovate based on R&D, as a matter of fact, some industries experienced growth without R&D; still, this is not

production of a jet aircraft with capacity of 100 passengers. Nonetheless, the termination of production was unavoidable because IPTN was hit by the 1997 economic crisis. (see Aminullah, E. and Fizzanty, T., 2016)

a justification to keep Indonesia positioned as a low R&D country in the future.

The shortcomings of past STI policies had impacted the emergence of problems related to intellectual capital, knowledge accumulation, technical capability, and STI governance. First, low intellectual capital formation had delayed the development of science and technology culture with the support of educated people in society. Second, low knowledge mastery accumulation had left Indonesia behind the global competition toward a science-based economy. Third, low technical capability acquisition had resulted in low business efficiency and productivity, which in turn weakened economic resilience in facing global competition. Fourth, weak R&D governance had led to low government commitment to build the nation's future competitiveness based on science, technology, and innovation capacities. In short, past STI policies failed to deal with low STI capacity and R&D intensity, which will be explained further.

3.2 Past R&D activity

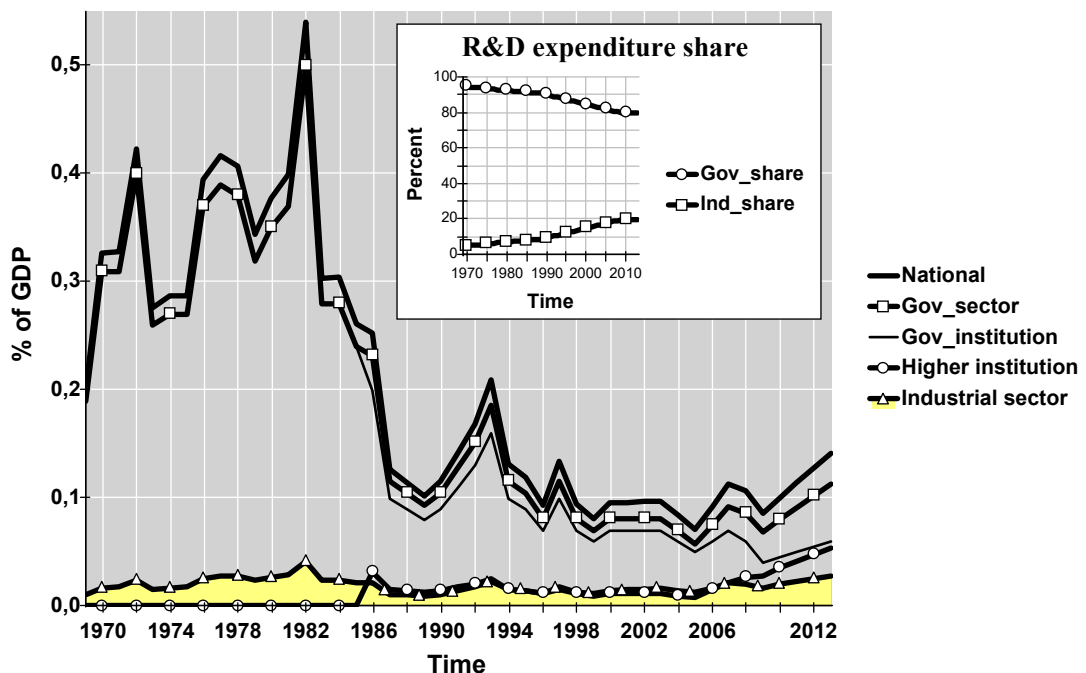
Innovation with low R&D activity in the private industrial sector had occurred for a long time, since the mid-1980s. It demonstrated that the Indonesian STI policy framework and its implementation had not gained attention and the confidence of the industrial sector to conduct innovation based on R&D activities. Certainly, the industrial sector as a driver of industrialization had contributed to Indonesia's economic growth. Nonetheless, the industrialization should be visible to accumulate the knowledge and technology acquired through industrial R&D activities. So far, it has yet to be materialized in the course of industrialization in Indonesia.

3.2.1 Statistical evidence

The statistical evidence showed that Indonesia has practiced and maintained its status as a low R&D country for a long time (Figure 2). The declining expenditure on R&D in the Indonesian economy was seen from the steadily falling national R&D intensity -- from 0.54% in 1982 to around 0.14% of GDP in 2013. Expenditure on industrial R&D only made up around 0.028% of GDP, or around 20% of the overall national R&D in 2013. While total government R&D expenditure (excluding R&D expenditure in higher education) was

around 0.067% of GDP in 2013 – accounting for up to 80% of national R&D activity -- it has little meaning due to the fact that it rarely produced commercial innovation. The Indonesian economy continued to grow even though its national R&D intensity continued to decline. For the period 1990–2010, GDP rose by thirty times at nominal value and grew consistently at 7% per year before the 1997 Asian financial crisis, and at 5% per year thereafter. On the other hand, national R&D expenditure rose by 16 times, but R&D intensity declined from 0.13% in 1990 to 0.09% of GDP in 2010 (Aminullah, E., 2012).

[Figure 2-2] Trends in Indonesian R&D activities (1969-2013)

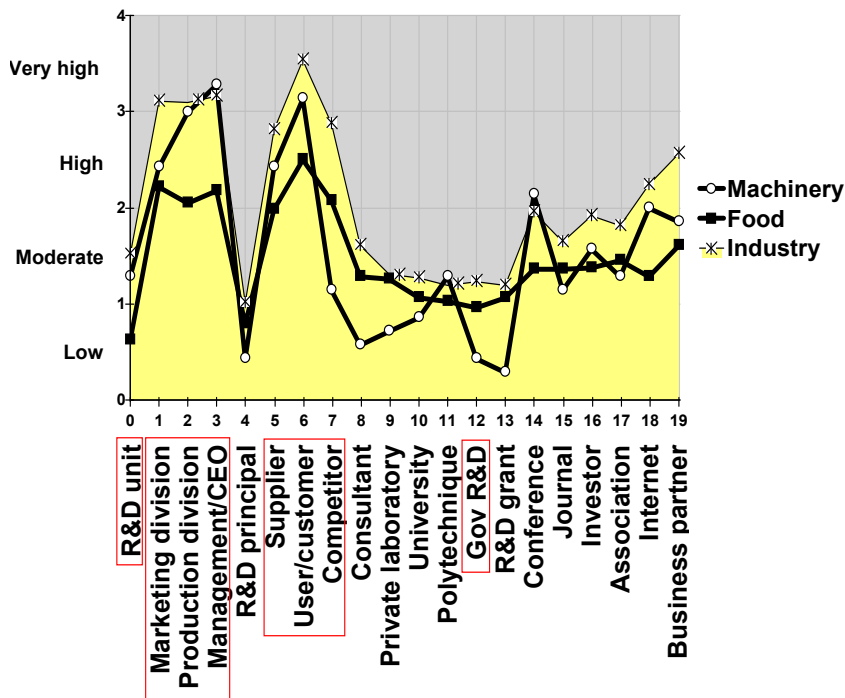


Data and estimation based on Indonesian science and technology indicators.
Source: Pappiptek-LIPI (2013, 2011, 2007, 2006)

Based on data from the 2011 Indonesian innovation survey, 61% of industries engaged in product, process, organizational, and marketing innovations (Pappiptek-LIPI, 2013). Among those firms, around 52% innovate not by engaging in formal R&D activities but through non-R&D and informal R&D innovations. The sources of information frequently used by the industry to innovate are: external sources (supplier, user/customer, and competitor); internal sources (marketing, production, management, business partner);

and other sources such as the Internet and conferences. The role of R&D units is less influential as a source of innovation for the industry. Similar patterns of innovation without formal R&D playing a dominant role also exist in the machinery and food sectors. The machinery sector is a more frequent user of the aforementioned sources of innovation, whereas the food sector is likelier to utilize institutional sources of innovation such as private laboratories, universities, and government R&D institutions (Figure 2-3).

[Figure 2-3] Sources of Innovation in the Indonesian Manufacturing Industry (2011)



Processed using the data from 2011 Indonesian innovation survey 2011, with permission
Source: Pappiptek-LIPI

The private industrial sector mostly generated innovation through learning without formal R&D. This is related to the low-skilled labor force in the economy, especially in the industrial sector. The labor force in the industrial sector consists mostly of graduates of junior high school or lower, followed by general school and vocational school graduates. Very few university graduates work in the industrial sector. Aminullah, E., Fizzanty, T., and Soesanto, Q.M.B. (2018) explained that the low capacity of human resources in the

industrial sector is in line with the dominant industrial output, i.e., low technology products, such as recycling, wood, pulp, paper and paper products, printing and publishing, food products, beverages and tobacco, textiles, textile products, leather, and footwear. With the exception of the period 2009–2012, there was a decrease in the share of low-tech industries as they were substituted by the increase of medium- to low-tech industries, such as building and repairing of ships and boats, rubber and plastic products, coke, refined petroleum products and nuclear fuel, and other non-metallic mineral products.

3.2.2 Evidentiary Case Studies

(a) Innovation in capital goods firms

Product innovation in Indonesian capital goods firms is conducted with the aim of improving existing machinery products. Informal learning without conducting formal R&D activity generally results in the creation of minor innovations, which are usually “new to the firm” innovations rather than “new to the market.” Innovation is stimulated by the limited cooperative arrangements made between producer, supplier, and consultant. The user–producer interactions are in the form of exchanges in goods and information as well as cooperation to stimulate innovation. Some cases of upgrading machines and related technologies in capital goods firms are as follows: (i.) co-development of new machines through user–producer interaction, i.e., modification of Fluidized Bed Combustion technique; (ii.) collaboration between users and producers in upgrading or customizing standard machines, i.e., supplying the control panel; and (iii.) development of new machines, equipment and tools, i.e., development of Nozzle Air Distributor, power plant, passenger boarding bridge, and medical device (USG). The main driver of technological learning in capital goods firms is vision as determined by the top management (Aminullah, E., Dian, P., Irene, M. N. and Laksani, C. S., 2014).

(b) Innovation in natural resource-based firms

Some natural resource-based firms have conducted product innovation as driven by the internationalization of production network through Outward Foreign Direct Investment (OFDI). Natural resource-based multinational enterprises (MNEs) operate in the areas of foods and natural cosmetics. MNEs have developed their own knowledge capability, and then transferred and adapted the knowledge to host countries. The firms conduct product innovation with R&D activity, either with intramural or extramural R&D. One large food firm does not have its own R&D facilities, generally conducting product innovation through extramural R&D by collaborating with other innovation sources, such as universities and research institutions. The food firm provides full R&D financing and buys the intellectual property for the product innovation conducted by universities. Another food firm has invested in an R&D facility and allocated intramural R&D financing for developing new product innovations. Meanwhile, a natural cosmetic firm grew from a small firm to a large one, and then expanded into an MNE by developing various product innovations through internal R&D activity (Aminullah, E., Fizzanty, T., Kusnandar, K., and Wijayanti, R., 2013).

(c) Innovation in bio-based chemical firms

Bio-based chemical firms acquire, develop, and accumulate knowledge and technology through interactive learning among R&D, production, and marketing units within the firms as well as through interactions with external sources of knowledge (universities, research institution, suppliers, and users). Large-scale firms acquire knowledge through R&D activities, monitoring of scientific advancement, and technology spillover, but the small firms gain knowledge and technology through learning by DUI (doing, using, and interacting) in their small laboratory. The bio-based chemical firms develop technology by conducting R&D activities to solve the negative environmental impact, find effective pest control, and improve soil fertility as well as meet the standards for the safety and security of bio-based chemical products. Encouraging the chemical firms to interact with public R&D institutions and universities is important to promote technological innovation through R&D. Enabling factors for bio-based chemical products innovation are the availability of scientists and researchers in universities to collaborate in research with the firms' R&D units (Aminullah, E., Fizzanty, T., Indraprahasta, G. S., and Asmara, I. J., 2015a).

4.

Explaining Indonesian STI Capacity and R&D Intensity

Capacity building in STI is the process of developing STI capacity, which involved some interactive components of STI capacity. Viewed from four important STI capacity components, those are: intellectual capital formation, knowledge mastery accumulation, technical capabilities acquisition, and STI policy governance. Indonesia has been less successful in developing STI capacity building in the past. Although all components are important in STI capacity building, the component of STI policy governance plays a decisive role in STI capacity building.

4.1 Cause of weak STI capacity

Intellectual capital. STI capacity building in terms of intellectual capital formation can be seen from some education indicators: (i) number of doctoral degrees especially in science and engineering; (ii) government expenditure on education especially in tertiary education as a percentage of GDP; and (iii) government expenditure on tertiary education as a percentage of government expenditure on education.

The formation of intellectual capital in Indonesia was described in terms of doctoral degrees produced by tertiary education, where Indonesia's position was lowest among selected Asian countries in 2014. Even though the total doctoral graduates in Indonesia reached 170 thousand in 2014, it ranks among the 15 countries with the biggest number of doctoral graduates, ranking 13th after Turkey (12th) and ahead of Russia (14th). Nonetheless, data on total doctoral graduates in science and engineering was not available. Based on the assumption that around 50% of the total doctoral graduates are doctorate degree holders in science engineering, the doctoral degrees in science and engineering were estimated to be around 1745 in 2014 (Table 2-1).

[Table 2-1] Doctoral Degrees by Selected Asian Countries

Doctoral Degrees by Selected Asian Country (Science and Engineering Degrees)

Country	Year				
	2010	2011	2012	2013	2014
China	48 987 (31 410)	50 289 (32 208)	51 713 (32 331)	51 139 (33 490)	53 653 (34 103)
India	-	21 544 (13 782)	23 630 (14 658)	-	21 830 (13 144)
Japan	16 760 (7 470)	15 911 (7 100)	15 902 (7 100)	6 791 (15 427)	15 045 (6 743)
South Korea	-	11 645 (5 558)	12 243 (5 782)	12 625 (5 933)	12931 (6 032)
Taiwan	3 705 (2 509)	3 846 (2 472)	3 861 (2 478)	4 241 (2 678)	4 048 (2 487)
Indonesia	-	-	-	-	3 591*) -

Sources: NSF, *) Weforum.org

The Indonesian government spent 3.36% of GDP on education in 2013 and 3.58% in 2015. Indonesia's position was similar to India, surpassing Singapore, but it lagged behind other ASEAN countries like Malaysia, Thailand, and Vietnam whose education spending was more than 4% of GDP in 2013. Furthermore, Indonesia's expenditure on tertiary education remained stable at around 0.5% of GDP; this figure was similar to that of Thailand and Vietnam but far below Malaysia and Korea whose expenditure on tertiary education was more than 1% of GDP. Indonesia's low expenditure on tertiary education was related to the priority of education expenditure of around 85% for primary and secondary education, with the remaining 15% as expenditure on tertiary education (See Tables 2-2 and 2-3.).

[Table 2-2] Government Expenditure on Education per GDP (%)

(Expenditure on Tertiary Education per GDP in %)

Country	Year				
	2013	2014	2015	2016	2017
Indonesia	3.36 (0.55)	3.29 (0.49)	3.58 (0.57)		
Malaysia	5.48 (1.67)	5.21 -	4.98 -	4.48 (1.13)	4.03 (1.00)
Thailand	4.12 (0.64)	-	-	-	-
Vietnam	5.65 (0.64)	-	-	-	-
Singapore	2.90 (1.02)	-	-	-	-
South Korea	5.25 (1.02)	5.27 (1.09)	5.25 (1.00)	-	-
India	3.84 (1.1)	-	-	-	-

Sources: UIS-Unesco

[Table 2-3] Expenditure on Tertiary Education per Government Expenditure on Education (%)

Country	Year				
	2013	2014	2015	2016	2017
Indonesia	16.41	15.05	15.80	-	-
Malaysia	30.51	-	-	23.39	20.99
Thailand	15.55	-	-	-	-
Vietnam	15.01	-	-	-	-
Singapore	35.28	-	-	-	-
South Korea	19.38	20.78	18.95	-	-
India	28.53	-	-	-	-

Sources: UIS-Unesco

Knowledge Mastery. The accumulation of knowledge mastery is obtained by continuously fertilizing intellectual capital through education (mainly tertiary education), learning without R&D, and conducting R&D activities. Fertilization of scientific knowledge generally takes place through institutionalized R&D activities primarily in the business sector and academic world (R&D ` R&D institution; thus Indonesia's R&D expenditure, which mostly came from government budget, reached 0.24% of GDP in 2016 and 2017. Among the five ASEAN countries (Singapore, Malaysia, Thailand, Vietnam, Indonesia), the neighboring countries of Indonesia have experienced an increase in mastery of knowledge through R&D. Indonesian R&D expenditure continues to decline, occupying the lowest position since the 2000s (Table 2-4).

[Table 2-4] Gross Expenditure on R&D per GDP (%)

Country	Year				
	2013	2014	2015	2016	2017
Indonesia	0.08	-	-	0.24	0.24
Malaysia	-	1.26	1.30	1.44	-
Thailand	0.44	0.48	0.62	0.78	-
Vietnam	0.37	-	0.44	-	0.53
Singapore	1.99	2.16	2.28	2.22	
South Korea	4.15	4.29	4.22	4.23	4.55
China	1.99	2.02	2.06	2.11	2.13

Sources: UIS-Unesco

STI capacity development through strengthening of knowledge mastery by increasing R&D expenditure is limited. The limitation is subject to the availability of R&D personnel and researcher to conduct R&D activities. Indonesia's R&D personnel per million population was 228 in 2016 and 244 in 2017 in FTE³. The figure is only one-tenth of that of Malaysia

3 FTE is considered to be a true measure of the volume of R&D and also the main R&D personnel statistic for international comparisons. The Full-time equivalent (FTE) of R&D personnel is defined as the ratio of working hours actually spent on R&D during a specific reference period (usually a calendar year) divided by the total number of hours conventionally worked in the same period by an individual or by a group. In other words, one full-time equivalent may be thought of as one person-year. Thus, a person who normally spends 30% of his/her time on R&D and the rest on other activities (such as teaching, university

and is the lowest among five ASEAN Countries (Singapore, Malaysia, Thailand, Vietnam, Indonesia). In terms of the actual number of total R&D personnel, Indonesia only had 59 thousand in 2016 and 64 thousand in 2017. The number was still smaller than that in three ASEAN countries (Malaysia, Thailand, and Vietnam) with smaller population, surpassing only Singapore, which had the smallest population. The slow increase in the number of R&D personnel could be the limit to absorb the intended rapid increase in government R&D budget in Indonesia in the coming years (Table 2-5).

[Table 2-5] Total R&D Personnel per Million Population (%) – In FTE

(Nominal)

Country	Year				
	2013	2014	2015	2016	2017
Indonesia	-	-	-	228 (59 657)	244 (64 635)
Malaysia	-	2 483 (75 062)	2 680 (82 360)	2 859 (89 178)	-
Thailand	-	1 230 (84 216)	1 305 (89 617)	1 632 (112 386)	-
Vietnam	877 (80 310)	-	868 (81 222)	-	886 (84 733)
Singapore	7 756 (41 582)	7 808 (45 542)	-	-	-
South Korea	8 001 (401 443)	8 551 (430 868)	8 726 (442 027)	8 808 (447 408)	9 242 (447 201)
China	2 554 (3 532 816)	2 669 (3 710 580)	2 690 (3 758 847)	2 763 (3 878 056)	2 861 (4 033 597)

Sources: UIS-Unesco

The limited capacity to conduct R&D is visible in the private sector. The ratio of researchers in business enterprises to the total number of researchers is less than 10%, the lowest figure among five ASEAN countries (Singapore, Malaysia, Thailand, Vietnam, Indonesia). The important role of the private sector in driving innovation had been shown by Thailand, which was approaching the figure of Singapore where the ratio of researchers in business enterprises to the total number of researchers is around 50%. This figure indicates that STI policy in Thailand has been more supportive than Indonesia in terms of promoting R&D activities in business enterprises (Table 2-6.).

administration, and student counseling) should be considered as 0.3 FTE. Similarly, if a full-time R&D worker is employed at an R&D unit for only 6 months, this results in an FTE of 0.5. (Source: Uis-Unesco)

[Table 2-6] Ratio of Researchers in Business Enterprises to the Total Number of Researchers (%) – In FTE

(Nominal)

Country	Year				
	2013	2014	2015	2016	2017
Indonesia	-	-	-	7.6 (3 534)	7.5 (4 246)
Malaysia	-	10.3 (6 289)	12.3 (8 613)	21.9 (16 134)	-
Thailand	-	43.1 (28 440)	50.9 (45 067)	56.8 (47 363)	-
Vietnam	21.1 (12 987)	-	21.7 (13 623)	-	24.1 (16 110)
Singapore	50.9 (18 328)	50.5 (18 520)	-	-	-
South Korea	78.7 (253 446)	79.5 (274 637)	79.7 (284 136)	79.7 (287 869)	81.3 (311 565)
China	62.2 (922 681)	62.1 (946 076)	62.7 (1 014 614)	61.9 (1 048 146)	60.7 (1 055 746)

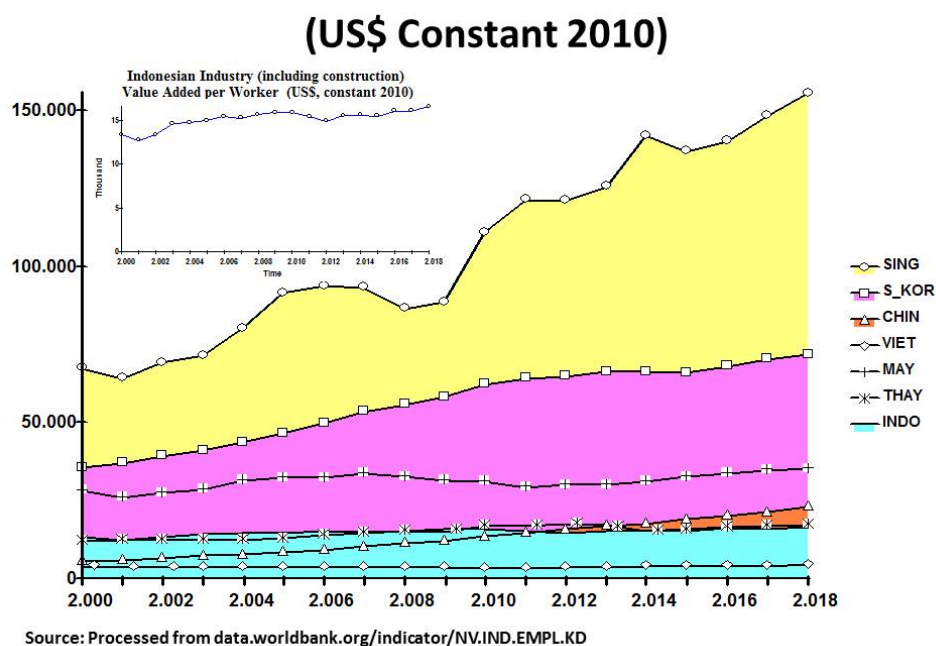
Sources: UIS-Unesco

Technical capability. The acquisition of technical capability is indicated by a shift of technical capabilities from the ability to buy, combine (buy some and make some), and make almost fully, both in terms of production tools and goods produced. High technical capability is related to the accumulation of knowledge mastery through R&D, availability of intellectual capital, and innovative STI governance. High technical capability gained from faster processing lowers input, and more output produces high business efficiency. All of these things create value added in business enterprises, especially in the industrial sector. The Indonesian value-added per worker in the industrial sector (2000-2018) was stagnant at around US\$ 15 thousand and is only higher than Vietnam. Countries generally have succeeded in increasing value-added per worker except Indonesia and Vietnam, which means more technical capability to produce low-technology products (Figure 4).

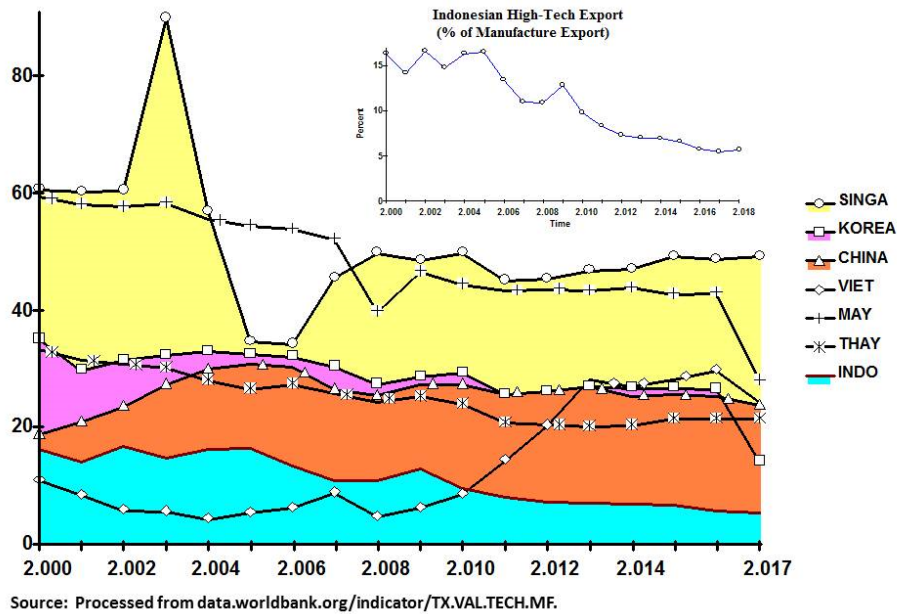
Such low technical capability in Indonesia was inter-related with the previous STI capacity's components (intellectual capital formation and knowledge mastery accumulation). It was demonstrated by Singapore, South Korea, and Malaysia with their high level of intellectual

capital formation and knowledge mastery, showing greater technical ability to master high technology and produce and export hi-tech products as well. In terms of hi-tech product export as a percentage of manufacture export, Indonesia is at the lowest position among five ASEAN countries (Singapore, Malaysia, Thailand, Vietnam, Indonesia). The hi-tech export of Indonesia has continuously declined since 2000, whereas Vietnam has shown rapid increase in hi-tech export since 2010 (Figure 2-5). This trend clearly denotes that the technical capability of Indonesian business enterprises is dominated by producing and exporting low- and medium- tech products such as textile, footwear, horticulture products (palm oil, cocoa, coffee), rubber products, forest products, and automotive.

[Figure 2-4] Value-Added per Worker (including construction)



[Figure 2-5] Hi-Tech Export (% of manufacture export)



STI policy governance. The capacity to govern STI policy contributes to social development, economic competitiveness, and long-term sustainable development. As described earlier, in terms of intellectual capital formation, knowledge mastery accumulation, and technical capability acquisition, Indonesia is lagging behind four ASEAN countries (Singapore, Malaysia, Thailand, Vietnam). Such is attributable to Indonesia's STI policy governance, which is still underdeveloped in comparison with the better STI policy governance and performance of South Korea and China, for example (Table 2-7).

[Table 2-7] STI Policy Governance and Industrial Technology

Korea	China	Indonesia
• Use industrial policy to accelerate innovation in large chaebols • Utilising domestic resource and digesting foreign technology to develop Korean technology. • Policy coordination (economic, industry and STI), focusing on specific industrial technology for export-led economic growth • With private sectors largely self-sufficient in industrially oriented R&D, GRIs moving on towards public and welfare R&D.	• Strong support to form industrial cluster for global export market global • Dedicated support system (regulatory reforms, monetary incentives for R&D, financial assistance to SMEs, and innovation research centres in collaboration with local universities and research institutes)	• Ever used industrial policy to deepen industrial structure and to transform POCs' high-technology industries → policy discontinuity after hitting by economic crisis 1998. • Broad base industrial policy covering upstream and downstream industries to strengthen industrial structure with the support of infrastructure catch up, reform packages, and incentives. • Lack of STI policies for industrial technology development

Source: OECD (2009), Jang (2015), Li and wang (2015), Sirgudson and Polanka (2002), Naude (2013) and ADB (2019)

The success Korean's technology development was driven by the successful digesting of foreign technology. The acquisition of foreign knowledge is done by utilizing domestic R&D supported by the high quality of intellectual capital. Korean STI policy on intellectual capital accumulation and R&D investment is linked with industrial policy to accelerate innovation in large chaebols. Therefore, Korean STI policy emphasizes building Korean technology and relying less on foreign technology through FDI; thus, attracting FDI was designed to create technology transfer and drive economic growth. In Korea, inter-ministerial R&D coordination is done by a powerful NSTC (National Science and Technology council) that has played an important role in securing significant R&D budget increases for STI from the national budget. The role of the main Korean research funding

agencies⁴ is important for financing and managing their R&D projects (OECD, 2009). In short, the main characteristic of the Korean STI policy was policy coordination (economic, industry, and STI) by focusing on specific industrial technology for export-led economic growth, where STI policy implementation is supported by all necessary actors and institutions (Jang, Y. 2015).

Furthermore, the success of Chinese technology innovation catching-up was driven by strong support to create an industrial cluster for the global export market. The industrial cluster has transformed China into an R&D-intensive, world-class manufacturing country. Basic research enjoyed higher priorities, being dominated by the state sector to meet societal needs and problems (e.g., personal health, environmental protection, sustainable urbanism, water, pollution, natural catastrophes, food supply, employment), whereas technical dissemination and adoption of innovation were largely left for enterprises that are R&D-intensive and world-class manufacturers. In China, focus was more on market-driven research for economic growth with the support of policy system, including infrastructure support, tax incentives, administrative support, and industry-HE Institution-R&D institution collaboration. Although STI policy instruments generally provide fiscal support for medium, small, and micro firms, China showed more interest in international collaboration and public procurement (Li, F. and Wang, P., 2015).

Indonesia experienced failings in technology transfer and domestic R&D in the past. In the period of high economic growth (7-8 % of GDP) in the 1990s, Indonesia tended to ignore reinvesting the results of economic growth in mastering knowledge by investing in R&D activities and intellectual capital formation (Sigurdson, J. and Polanka, K., 2012). Indonesia implemented an industrial policy to deepen the industrial structure and to transform POCs into high-technology industries in 1990s. Nonetheless, the industrial policy experienced policy discontinuity after being hit by the economic crisis 1998. Since 2000s, Indonesia has

⁴ Some ministries have established their own funding agencies for financing and managing their R&D projects, such as: (i) Korea Science and Engineering Foundation (KOSEF) for bridging the government, public R&D institutes, universities, and some private firms; (ii) Korea Industrial Technology Foundation (KOTEF) for nurturing industry-academe cooperation in research and innovation (e.g., through the funding of industry-academe cooperation centers in universities, improving corporate technology management, especially in SMEs, through education); and (iii) Korea Research Foundation (KRF) for promoting and supporting academic activities and on upgrading research quality through its support for academic research foundations and new researchers. (OECD, 2009)

implemented a broad-based industrial policy covering upstream and downstream industries to strengthen industrial structure with the support of infrastructure catchup, reform packages, and incentives (Tijaja, J. and Faisal M., 2014). Due to weak inter-ministerial policy coordination on STI and lack of linkage between STI policies and industrial technology development, however, Indonesia experienced failings in realizing industrial technology development based on R&D in the past. In addition, Indonesia needs an innovative industrial policy based on recognition of the entrepreneurial economy and promotion of knowledge acquisition and commercialization through investments in R&D, human capital, and information and communication technology (Naude, W. 2013).

4.2 Condition and consequence of low R&D intensity

R&D intensity is a main indicator for judging a country's STI governance. Low R&D intensity is generally related to low or weak STI indicators including intellectual capital formation, knowledge mastery accumulation, technical capabilities acquisition, and STI policy governance. The following section explains the condition and consequence of low R&D intensity in Indonesia.

4.2.1 Condition

As explained in section 2, the past trend of Indonesian R&D intensity was initially surging during the period of the 1980s but has been continuously on a downward spiral since the 1990s. There are three conditions related to the low R&D intensity in Indonesia: (i) large portion of low and medium technology (LMT) industry in industrial sectors; (ii) lack of government attention on R&D; and (iii) continuously low share of private R&D intensity. Such low R&D intensity has kept Indonesia "on the defensive," producing low-end products in the economy as shown by the large portion taken up by the LMT industry in industrial sectors (Aminullah, 2011; 2012).

First, the Indonesian industrial sector is generally dominated by the LMT industry. This industry produces low-technology products that do not require R&D facilities. Manufacturing firms in Indonesia are micro and small firms (99.3%), with medium and large firms accounting for only a small percentage (less than 1%). This is very important in

understanding what constrains the manufacturing sector and in thinking about the future of the sector. The reality is that it is very difficult for micro and small firms to grow to a size that makes an impact in terms of productivity, product upgrading, or R&D (ADB, 2019). These industries generally produced low-technology products, such as food products, beverages, shoes, textiles and textile products, leather and leather articles, wood and paper products, rubber and plastic articles, and metal and metal goods. Around 20–25% of the industries produced medium-technology products, with only a small portion performing R&D activity with low intensity. Parts of the medium-technology industries, such as some in the automotive industry, did not even perform their own R&D to source innovation because their R&D activities were generally performed by their parent companies abroad.

Second, government attention toward the development of S&T has consistently decreased since the 1990s. In the 1980s, the S&T budget was about 2.5% of the total government budget, dropping to around 0.5% in the 1990s; the S&T budget then remained at approximately 0.5% of the total government budget in the 2000s. The continuous decline in the perennially low S&T budget, parallel to positive economic growth, seems to be a typical phenomenon in Indonesia. For comparison, in China, the high attention given to S&T was paired with high economic growth, with the S&T budget amounting to 4–5% of the total government budget over the last 30 years. (Kou, Z., 2010)

Third, the private sector is generally less interested in reinvesting their production yields in R&D activity. Aminullah (2012) showed that, during the period 1990–2013, the increase of production did not affect the increase of R&D intensity. The objective of investments in the private sector is normally productivity improvement through capacity enhancement, business expansion, distribution network, and sales promotion, which have less or no need for the support of an R&D facility. In 2010, industrial R&D expenditure was very small at only 0.013 % of the Indonesian GDP, or about US\$90 million compared to US\$700 billion GDP (Pappiptek-LIPI, 2011).

4.2.2 Consequence

As a consequence of low R&D intensity, Indonesia has experienced delays in building

innovative capability in the industrial sector. Indonesia's industries gained capability in plant operations and basic industrial skills in the 1980s. Furthermore, in the 1990s, the Indonesian industry developed capabilities in redesigning old products and processing. The increase in technological capability was facilitated by the use of imported technology without digesting the technology. The capability in technology use was gained through learning activities that were generally performed without R&D or with less R&D in selected sectors. Additionally, since the 2000s, some manufacturing firms (sections 3.2.ii) have focused on building capability in designing and redesigning system engineering as well as in competing with innovative products in the global market.

By engaging in innovation with less or no R&D, the industries have successfully mastered the basic skills needed in operating plants as well as in redesigning existing products and processes toward better quality and more competitive prices. With such capabilities, the industrial sector could emerge as a potentially fast runner in industrialization. Note, however, that innovation with less or no R&D has resulted in a weak absorptive capacity for foreign technology; consequently, the Indonesian industrial sector cannot raise its technology capability level to the class needed to redesign innovative products and processes based on R&D. In other words, Indonesia maintains its position as a latecomer not as a fast runner in industrialization such as China and India.

Another consequence is the undercapacity to absorb the rapid increase of R&D intensity in the public sector. As shown in Table 4, R&D intensity increased rapidly from around 0.08 to 0.24 during the period 2013-2016. Such tripling in R&D intensity was absorbed by a very small number of researchers engaging in R&D. The number of researchers in R&D per million population (RRD) in Indonesia decreased from 210 to 90 during the period 2000-2010. Then RRD increased to 228 (2016) and 244 (2017). Such small figure of RRD is far below that of Malaysia, Thailand, and Vietnam and is worse for Indonesia as a member of G20. If Indonesia maintains such low RRD value or no policy change for giving preferential treatment in the handling of R&D institutions and enhancement of human capital for innovative activities, then it will consistently remain a low R&D country in the future caused by lack of researchers in R&D (Aminullah, E., 2015).

The institutional setting for intellectual capital, i.e., RS&E (researchers, scientists, and

engineers) engaging in R&D in the public sector, is subject to government employee regulations. Indonesia has, for a long time, faced a severely limited quantity of RS&E engaging in R&D; likewise, diminishing quality of such small numbers of RS&E engaging in R&D is likely to happen in the future. The government urgently needs to increase the quantity and quality of RS&E engaging in R&D by establishing conducive regulations on handling intellectual capital in the public sector, especially for national research institutions and universities. The regulation under the existing State Civil Apparatus Act would diminish the quality of intellectual capital, because innovative and productive intellectual capital is hardly cultivated under the working climate of rigid bureaucratic procedures as commonly fit for administrative employees or administrators (Bozeman, B., 2015 and Coccia, M., 2009). The universal ethos for RS&E is that science discovery, technology invention, and process/product innovations are produced by continuously building scientific competence under the climate of scientific freedom while guided by science and research ethics and appropriate levels of funding.

5.

Promoting STI and R&D Governance

5.1 STI policy challenge

Indonesia applied various STI policy instruments in the past. The policy implementation failed to give feedback and strengthen the national innovation system, because the system was not well-established due to some policy problems such as policy discontinuity, integration, and instrument. Such categories of policy problems were based on reflections from the STI policy implementation. First, STI policy implementation was carried out with strong support from the government; due to the weak public support, however, its continuity was not sustained, e.g., program of national car in the automotive industry. Second, building technological capability from downstream (production engineering) to upstream (basic research) was successful, thanks to the strong STI leadership that pursued industrial transformation through mastery of high technology. Nonetheless, the policy failed to master high technology due to lack of integration with building technological capability from upstream (basic research) to downstream (production engineering). Third, policy implementation using some policy instruments (i.e., financial support) to stimulate R&D in the private sector was applied in the past. As demonstrated in the previous statistical evidence and case studies, however, such policy instruments appeared to have not increased R&D activities significantly in the private industrial sectors. The lack of R&D promotion to develop industrial technology has significantly contributed to the slowdown of R&D intensity since the 1990s.

Policy failure in spurring R&D-based innovation in industrial sectors was a matter of fact in the past. Despite the lack of attention and confidence in the private sector to innovate, based on R&D, this can be considered not to pose a problem for some industries that have experienced growth without R&D. By all means, it is not a justification that all industries can maintain growth without R&D in the future. The competitively global environment, including the ASEAN Economic Community (AEC), has now challenged the Indonesian industry to compete in innovation based on R&D. The losers and winners in the race of R&D-based innovation are determined not only by the speed of innovative product

creation but also by competitive prices in the global market. In the global market competition, Japan's consumer electrical industry was defeated by Korea, and then the less competitive subsidiaries of the Japanese consumer electrical industry were taken over by their more innovative, highly efficient Chinese counterparts (Aminullah, 2017).

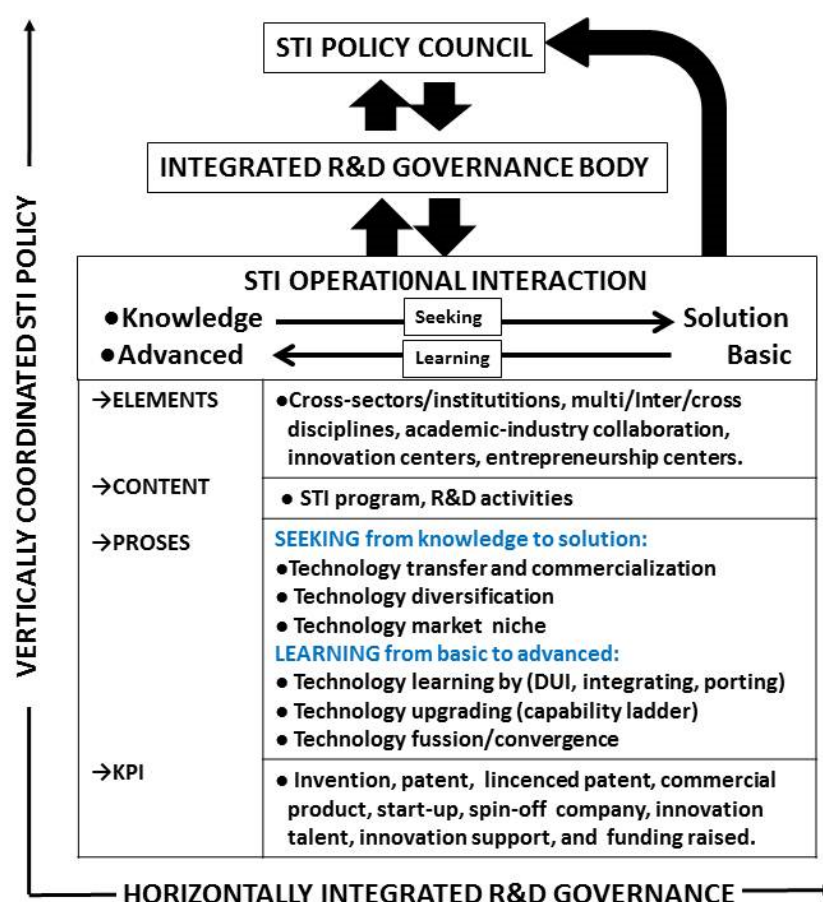
Referring to the framework (see Figure 1, section 2), the STI policy challenge for Indonesia is that the high level of STI policymaking institutions should be redesigned under the perspective of coordinated, integrated institutions. Even though the STI policies of each country apply different policy solutions to specific economic structures, strategies, and local conditions, a coordinated, integrated institution is urgently needed for catching up with an innovative institution such as that in Japan (CSTI), Korea (KSTC), and Malaysia (MIGHT). STI policy coordination needs to be maintained continually by the function of the STI policy council, which is led by the President and capable of coordinating STI resources in an integrated manner.

Putting STI policy coordination to work, Indonesia needs to set up an integrated R&D governance body applied as the instrument of STI policy under the direction of the president. The integrated R&D governance body is needed to implement a coordination mechanism and to integrate sectors and institutions at the levels of policy implementation, program budget planning, and evaluation in a clear, straightforward manner. R&D programs should be governed relevant to societal needs and economic development. An R&D program needs to solve the real problems faced by Indonesian people, such as S&T solutions and innovations in the field of basic needs (adequacy of food, health, sanitation, nutrition, medicine, energy, etc.), to meet the Sustainable Development Goals (SDGs).

For that purpose, the R&D governance program needs to be redesigned and assessed continuously. Integrated R&D governance brings all STI actors (scientist, technologist, innovator, and entrepreneur) to interact with each other to realize the final goal, i.e., creating innovative products and services. Indonesia needs to shift from the capability to buy toward combine (buy some and make some) and move on to make almost fully innovative products and services. To realize it, high technical capability in terms of knowledge mastery should be accumulated through of R&D activities, supported by the availability of intellectual capital in business enterprise. Therefore, the STI operation

system should focus on interaction between academic and industry collaboration, which stimulates business enterprises to hire scientists and engineers to conduct R&D and innovate with measurable key performance index (KPI) as well. The whole process in a vertically and horizontally integrated system is depicted in the operational model of STI policy and R&D governance system (Figure 2-6).

[Figure 2-6] Operational STI Policy and R&D Governance System



5.2 Way forward in promoting STI and R&D governance

The way forward in promoting STI and R&D governance is putting the STI operational interaction to work within detailed process and measurable performance. The couple elements — vertically coordinated STI policy and horizontal R&D governance — are

implemented through the STI program and R&D activities. The detailed program and activities are as follows:

5.2.1 Vertically coordinated STI policy

Vertically coordinated STI policy content needs to be expressed by picking the strategic, technology-intensive sector for future competitiveness. Indonesia should focus its limited R&D financing on the country's potential for technological leadership through a "make some/buy some" strategy. The future identity of Indonesian technological leadership would be embedded in Indonesia's strength of R&D capacity in new materials and life science-based industry (Aminullah, E. 2017). The important position of life science R&D in Indonesia has been shown by the internationally recognized scientific reputation of Indonesian researchers who engage in the area of life science R&D. The most developed data from the scientific activities of Indonesian researchers, in terms of international collaboration and international publication, are in health and medicine, biology, plant science, ecology, and environmental science. Therefore, the life science-based industrial innovation could be explored from the opportunity of international collaboration on R&D in the health, agricultural, and marine sectors (Akil, H.A., et al, 2014).

The technology-intensive sector for future competitiveness could be based on the unique mineral and natural wealth of Indonesia, utilizing the country's comparative advantage of vast minerals and biological diversity in both land and sea. The development of the natural resource-based industry driven by new materials and life science-based technologies for the future should not create a totally new industry but should be channeled into developing the existing industries with strong potential. Therefore, new materials, life science R&D programs, and its implementations need to be facilitated to cooperate with existing related industries such as steel, bio-based chemicals, biomedical and biotechnology industries, etc. The Indonesian government needs to encourage public-private partnerships or a consortium of joint research. Long-term research grants (for 5–10 years) in sufficient quantity (rather than small R&D grants) are required to achieve significant results. These efforts can be realized by implementing conducive legal and institutional support for conducting R&D domestically.

Initiatives to enhance innovation capacity by intensifying R&D in life sciences have long existed in Indonesia. The availability of life science parks, such as the so-called LIPI Cibinong Science Centers (CSC), has also been driving life science-based technology development in Indonesia. Upgrading of CSC, which is currently managed by the Indonesian Institute of Sciences (LIPI), can be developed toward progressive S&T infrastructure to support the development of leading industries in the future. Natural resource-based industries, which are driven by life science-based technologies, will further develop faster, including the bio-based chemical, biomedical, and biotechnology industries. The existing life sciences park for nurturing industries should be the core of developing a national STI policy for industrial technology development.

5.2.2 Horizontally integrated R&D governance

The horizontally integrated R&D governance body is the instrument of STI policy to execute the coordinated, synergized national R&D program and activities, which must resolve the constraints of cross-sectoral ministries/institutions. It can be in the form of “holding research institution” consisting of multi-/inter-/cross-disciplines (from science, technology, innovation, startup operations, and spin-off company), encompassing academe–industry collaboration. The scale of R&D activities under the control of an integrated R&D governance body should focus on large-scale R&D activities or strategic innovation consortium, which have the following requirements: (i) pooling STI resource inputs (scientists, engineers/technologist, innovator/entrepreneur) to transform R&D investment into commercial innovation; (ii) integration of processes from research, development, design/engineering and commercialization to create convergence technology and innovation; (iii) downstreaming of outputs from science, technology, innovation, and commercial products to seek knowledge-based solution, and; (iv) integrative learning from basic operation to advance knowledge in acquiring science-based innovation in the strategic sector.

The process of integration in R&D governance basically has two directions: (i) seeking solution from science; and (ii) learning from basic operation toward advanced knowledge. Under the direction of seeking solution from science are: (i) Technology transfer and

commercialization, which can be the transfer of technology from a public research institute or a university to the industry or the commercialization of a public research institute or a university's patents by industry; (ii) Technology diversification, which involves developing new technology from an existing one, driven by the needs for better technology to support industrial development, e.g., developing high-precision tools and machinery from forest machinery technology based on the needs for forest equipment, and the high-precision tools diversified into ICT equipment industry by Nokia; (iii) Seeking a technology market niche driven by technology innovation and developing the networks of commercialization for innovative products, for the creation of a cluster of demand/market niches as an integrated industrial ecosystem, e.g., seeking and creating the market networks of unique (halal) technology products to meet the cluster demand of firms or industries that put obligatory halal in their material sources, production process, and commercial products.

The direction of learning from basic operation toward advance knowledge is detailed as follows. (i) Technology learning starts from learning by doing, moves on to learning by using new technology, upgrades to learning by integrating the advance technology into the existing technology and learning by porting the latest technology into the existing technology. For example, steel making technology started from learning by doing efficient steelmaking, and then it was made very efficient by using computerization in steelmaking. The case of learning by integrating the latest technology involves the integration of LCD technology in the computer, whereas the case of learning by porting involves putting the computer on numerical control machinery. (ii) Technology upgrading (in capability ladder) emerges from "operational capability" in basic industrial skill and plant, moves to "technical capability" to redesign old products/processes, upgrades to "design and engineering capability" to design and redesign system engineering, and competes with an innovative product, and then achieves "technology development capability" to create advance/frontier industrial goods through the design and redesign of high system linkages. An example of moving up the technology capability ladder is moving up from making traditional herbal medicine to producing standardized herbal medicine, and then creating phyto-pharmaceutical herbal medicine based on R&D activity. (iii) Technology convergence is a systemic solution to the limits of existing technology in solving a complex

problem that needs the application of science convergence to the so-called technology convergence. For example, in a biological-environmental system perspective to solve the degradation of soil fertility due to non-degradable waste, the perspective drives the application of bio-based chemical science to create convergence technologies in bio-fertilizer and bioplastic products.

5.3 Envisioning the STI policy outcome

Various forms of R&D support and incentives have been launched; nonetheless, the outcomes of past instruments were still not appealing enough for the industrial sector in Indonesia. Therefore, the future challenges are to design a “breakthrough formula” that is of interest to the industrial sector. The design of policy instruments should deal with markets, funding, and intellectual capital as well as a consortium of strategic innovation. STI policy outcomes by implementing integrated R&D governance must be measured by key performance indicators (KPI), i.e., invention, patent, licensed patent, commercial product, startup, spin-off company, innovation talent, innovation support, and funding raised.

Under a new institutional setting, as explained above, industrial policy for the promotion of R&D in a specific technology-intensive sector can be applied. As for the expected outcome of driving R&D activities in the private industrial sector, considering the fact that the routine increase in R&D investment in the public sector is around 10–15% per year, it can be deduced that a non-routine increase in R&D investment is an urgent matter for Indonesia. In order to realize such, the Indonesian private sector should drive rapid growth in R&D investment of around 20–30% per year. To achieve this, the industrial sector must shift from innovation without R&D toward innovation with R&D. For the government, the policy of super-tax deduction for R&D expenditure needs to be implemented without delay. By targeting an increasing rate of R&D investment in the industrial sector, which is two times that in the government sector, problems of stagnation in R&D activities, low absorptive capacity of technology, and improvement of low technological capability in the private industrial sector would be solved in the future.

If a rapid increase in the rate of R&D investment in the industrial sector can be implemented, it would gradually change the pattern and trend of R&D intensity. Indonesia's R&D intensity would increase from around 0.11% of GDP (2015) to 0.42% (2025), wherein the contribution of the industrial sector would increase from approximately 22% (2015) to 62% (2025). The amount of R&D spending in the industrial sector is expected to increase exponentially in the next 10 years from an estimated Rp 3.2 trillion in 2015 to 77.2 trillion in 2025. In the period 2025–2045, industrial R&D spending would be expected to grow in a parabolic curve toward a stable point. In the year 2045, Indonesia's R&D intensity would reach 0.71% of GDP, with the contribution of the industrial sector amounting to about 85%. Meanwhile, the public sector would routinely increase its R&D spending from an estimated Rp 11.5 trillion in 2015 to Rp 46 trillion in 2025 and further to Rp 115 trillion in 2045. Indonesia would become a developed country with GDP per capita of US\$ 24,250 and mid-level R&D intensity (between 0.5-1%) in 2045 (Aminullah, E., 2015). (Table 2-8)

[Table 2-8] Estimation of Indonesian R&D Intensity (1990–2045)

TIME	1990	1995	2000	2005	2010	2015	2020	2025	2030	2035	2040	2045
RD_exp (billion)	274,55	597,66	1.378,4	2.743,8	5.843,5	14.771	40.323	123.207	227.267	353.958	546.942	769.682
Gov_RD (billion)	247,10	537,89	1.171,6	2.332,2	4.674,8	11.481	28.702	45.924	63.146	80.367	97.589	114.810
Prv_RD (billion)	27,46	59,77	206,76	411,56	1.168,7	3.289,7	11.621	77.283	164.121	273.591	449.354	654.872
RDpGDP (%)	0,13	0,128	0,138	0,114	0,0915	0,111	0,217	0,418	0,523	0,577	0,656	0,711
GOVRDsh (%)	90,00	90,00	85,00	85,00	80,00	77,73	71,18	37,27	27,78	22,71	17,84	14,92
PRVRDsh (%)	10,00	10,00	15,00	15,00	20,00	22,27	28,82	62,73	72,22	77,29	82,16	85,08
rGOVRD (%)	15,01	9,27	16,05	11,33	5,71	13,94	12,00	7,50	5,45	4,29	3,53	3,00
rPRVRD (%)	27,15	9,27	17,15	11,33	33,44	34,60	4,88	23,59	15,36	8,70	8,27	6,55
GDPpCAP_s	0,639	1,06	0,779	1,24	3,04	3,76	5,65	8,39	11,66	15,50	19,83	24,25

RD=R&D expenditure, Gov=Government, Prv=Private, sh=share, GDP= Gross domestic product, CAP= Capita, p=per, s=US\$
Source: Aminullah, 2015

By implementing the future focus on industrial technology development, where the amount of R&D spending in the industrial sector is expected to increase exponentially from an estimated Rp 3.2 trillion (2015) to Rp 164.1 trillion (2030), the share contribution of the industrial sector would increase from approximately 22% (2015) to 72% (2030). By driving force of industrial technology development, the GDP per capita of Indonesia is expected to increase from US\$ 3,760 (2015) to US\$ 11,660 (2030). In short, by applying industrial policy for the promotion of R&D in a specific technology-intensive sector, Indonesia can be expected to escape the trap of being a middle-income country in 2030.



CHAPTER 3

Examining the Importance of Science and Technology Governance in Indonesia

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Chapter 3. Examining the Importance of Science and Technology Governance in Indonesia

1. Conceptual Framework

Any policy, including S&T policy, is government's response to unwanted or changing condition of their environment. Some scholars may recognize the seriousness of a policy problem, but ultimately, government's commitment to policy is generally determined by the existence and intensity of public demand to address and solve that problems. The understanding and quality of discussion among various actors involved in problem identification and policy formulation will determine the quality of policy. And public demand and political pressure will determine the amount of resources that can be mobilized for that policy. Such mobilization requires good governance to make the establishment and implementation of policy accountable and achieved the goals it has been assigned. The importance of S&T development in the eyes of a nation will manifest in the intensity of its governance. The more important a matter for society, the more control they seek to exert over it.

To analyze S&T governance, the following stages of policy will be evaluated based on various actors that are involved, their various roles, their power and influences, how they organize themselves and their works, incentives if they do the work, and constraints that limit what they can do. Governance of each stage will be analyzed by exploring the answers of the following questions.

- 1) Problems identification
 - a. What problems are being raised?
 - b. Who raise the problems related to S&T development?

- c. How is their authority related to the problems?
- d. How influential are they? Do they already include all necessary actors/parties that are needed to identify and solve the problems?
- e. What incentives they can get if they can solve the problems?
- f. What limits the identification and solving the problems?

2) Policy formulation and planning

- a. Do the boundaries of policy interventions sufficiently address the boundary of the problems?
- b. If the problems are intersectorally dependent, is S&T policy interconnected with the policies in other sectors?
- c. Actors involved in policy formulation correspondence with actors involved in the problem.
- d. What limit the boundary of policy?

3) The Implementation of S&T development

- a. Who should make sure that the policy is being implemented?
- b. What limit the implementations of S&T policy?

4) Evaluation

- a. Who do the evaluation?
- b. How the evaluation affects the changing policy?
- c. What limits the evaluation?

5) Institutions and Mobility of Human Resources

2. Problem Identification

This report will discuss problems and actors who propose them indirectly from secondary documents: Long-Term National Development Plan (RPJPN) 2005-2025, Mid-Term National Development Plan (RPJMN) 2015-2019, and National Research Masterplan (RIRN) 2017-2045.

Long-Term National Development Plan (RPJPN) 2005-2025 states that the national development that has been conducted shows progress in various areas, such as economy, politics and also science and technology (S&T), but it also recognizes that Indonesia has low competitive advantage, which is caused by "low productivity of human resources and low mastery and implementation of technology in production process"¹ (RPJP, p. 8).

Even though S&T have come rarely into public discourse, they are beautifully formulated and seem rethorically important. For example RPJP also mention that "The vision of socio-cultural development is the realization of a strong, competitive, and high moral character of the nation through the development of a national culture characterized by the character and behavior Indonesian people who believe in God Almighty, intelligent, innovative, high work ethic, virtuous noble, tolerant, collaborative, patriotic, dynamic, and oriented to science and technology"² (RPJP, p. 26, emphasis is mine).

The development of innovative culture oriented toward S&T is attempted to make Indonesia competitive in global competition. S&T, competitive advantage, competition, productivity and innovation get significant attention in RPJP. For example, industry policy is carried out partly through "providing physical infrastructure, economic, and technology that are responsive to the need and competences of each industry clusters"³ (RPJP, p. 31). In some parts of the document, technology is treated as resource that can be acquire and is separated from human resource development. It seems to assume that technology equals to machinery that can be bought and utilized when human resources are capable. The low technological capability is regarded as caused by low education.

Mid-Term National Development Plan (RPJMN) 2015-2019 identifies three main challenges: low contribution of national S&T on economic development, national resources is continuously decreasing, increasing globalization of socio-cultural life. This

report will only discuss the first challenge.

Despite recognizing that S&T development has been long attempted in Indonesia, and government has paid attention and invested in laboratories and facilities, training and educating researchers, and the results can be seen in publications, patents, and technological services for society, but the contribution of national S&T to national economy is still limited.

National Research Masterplan (RIRN) 2017-2045 which was released in 2017 by Ministry of Research, Technology and Higher Education (MoRTHE) acknowledge that the low of S&T utilization and development is caused by the weakness in 1) research capacities and competences, 2) capabilities of development for S&T based creation; 3) local, regional and global network of institutions and researchers; 4) national R&D productivity and relevance to answer societal demand for technology; and 5) utilization of R&D to create added value for natural resources and product innovation to improve economic competitive advantage.

RIRN regards the above conditions are implicated by low scientific and technological culture and literacy of the nation. This problem also manifests into various phenomena, such as disconnection of research results from industries' needs; disconnection of university research from those conducted by research institutions; as well as the underutilization of research resources (human resources, which include researchers, engineers and academics; funding, and research facilities).

The above three documents acknowledge the necessary of S&T development for improving competitive advantages and the low contributions of S&T for economic development. RPJPN and RPJMN do not explicitly admit that the low contribution of S&T also implies the low contribution of government S&T sectors. RPJMN even states that there is progress in previous S&T as can be seen from scientific publications, patents, etc. Even though this document does not explain why the contribution of S&T on economic development is still low despite of such progress in S&T development. On the contrary, RIRN acknowledges the weakness in R&D capacity as well as the networking of S&T community.

RPJPN and RPJMN were published by the National Development Planning Agency (BAPPENAS). In relation with S&T policy, they have consulted various government R&D institutions, but it is ultimately up to them to write the final version of the document.

3. Policy Formulation and Planning

The importance of science and technology (S&T) for Indonesian development has been formally recognized. Book I of Medium-Term National Development Plan (RPJMN) 2015 – 2019 recognizes Term National Development Plan (RPJMN) 2015 – 2019 recognizes S&T as one among six national development area which become the target of policies for taking the advantages of demographic bonus. One of the listed policies is "S&T for improving productivity of works." Book II of the RPJMN even has special chapter (Chapter 4) allocated for S&T development.

Table 3-1 shows the number of various terms related to S&T, such as "science," "technology," "R&D" and "innovation," which are spread over many parts of the two volumes of Medium-Term National Development Plan (RPJMN) 2015 – 2019 recognizes Term National Development Plan (RPJMN) 2015 – 2019. Apart from "Industri", "Teknologi" is the most mentioned word in the RPJMN, and then "daya saing" (competitive advantage). In some parts of RPJMN, technology is partly related or needed to improve the competitive advantage, or that the low competitive advantage is caused by the low utilization of technology. Sometimes the term "teknologi" is mentioned together with "ilmu pengetahuan" (science), and more often mentioned in their abbreviation, "iptek" (S&T).

In Chapter 6, National Development Agenda, of RPJMN Book I, it is stated that the improvement of competitive advantage and productivity is achieved by the improvement of mastery of S&T and innovation. Hence, competitive advantage and productivity are mentioned many times in RPJMN. The notion of "competitive advantage" cannot be separated from innovation. Innovation has been recognized as a source of competitive advantage, and technology is one element of innovation. Productivity, especially in industry, is heavily dependent on the utilization of technology. However, Table 4.1 shows that R&D and research are much more mentioned than engineering. And "perekayasa" (engineer) is not mentioned at all. It is in contradiction with the closeness of the concepts

"competitive advantage" and "engineering" rather than "competitive advantage" and "research".

The notions of "perekayasa" and "perekayasaan", which have been formally adopted and even there is functional position called "perekayasa" which is comparable to researcher, are less popular among public or outside government S&T community. Public are more familiar with the English word "engineer" and "engineering" or Indonesian word derived from the Dutch, "Insinyur".

[Table 3-1] Number of words related to S&T in Medium-Term National Development Plan (RPJMN) 2015 – 2019 recognizes Term National Development Plan (RPJMN) 2015 – 2019

Words	Number of Words in RPJMN 2015 – 2019		Total
	Book I	Book II	
Ilmu pengetahuan (science)	64	16	22
Teknologi	107	322	429
Iptek (S&T)	36	102	138
Litbang (R&D)	7	20	27
Peneliti (Researchers)	1	17	18
Penelitian (Research)	23	94	117
Perekaya (Engineer)	0	0	0
Perekayasaan (Engineering)	2	13	15
Inovasi	36	74	110
Daya saing (competitive advantage)	70	167	237
Produktivitas	40	87	127
Industri	239	546	785

[Table 3-2] The number of Web pages in the Web sites of some institutions containing particular words (Source: google.com, accessed in October 5, 2019)

Words	LIPI	BPPT	MoRTHE	Mol
Peneliti (Researcher)	103,000	4,140	60,100	31,900
Penelitian (Research)	153,000	8,580	781,000	31,300
Litbang (R&D)	8,410	1,640	18,800	9,850
Perekayasa (Engineer)	1,730	13,400	1,950	540
Perekayasaan (Engineering)	633	1,710	1,710	2,490

Table 3-2 shows the occurrences of the words peneliti (researcher), penelitian (research), litbang (R&D), perekayasa (engineer), and perekayasaan (engineering) in the Web sites of four government institutions: Indonesian Institute of Sciences (LIPI), The Agency for the Assessment and Implementation of Technology (BPPT), Ministry of Research, Technology and Higher Education (MoRTHE), and Ministry of Industry (Mol). The core business of LIPI is the development of science through research, so it is understandable that their Web site mentions research more than engineering. While the Web site of BPPT mentions engineer more than it mentions researcher, but research occurs more than engineering, even though their name gives the indication that they should give more emphasis on assessment and application of technology, which is closer to engineering than to research. The occurrence of researcher, research and R&D is also higher than engineer and engineering in the Web sites of MoRTHE and Mol. This brief data may indicate the assumption behind the S&T development in Indonesia that discursively gives more emphasis on R&D activities as the main contributor of the improvement of economic competitive advantage in Indonesia. Discursively it has been the norm for a long time in S&T policy, however it contradicts with the fact that for developing countries, "mastering existing technologies is more significant than innovation as the focus of technological activity. In developing countries, by contrast, not only is the internal base of knowledge for

mastering technologies relatively weak, the supporting network of other enterprises, institutions and human capital is also underdeveloped" (Lall & Teubal, 1998, p. 1371).

According to the chapter 4 of the Book II of RPJMN 2015-2019 (p. 4-3), the target of S&T development is to improve S&T capacity, which include the following targets. First, improving research, development and application of S&T to support: competitive advantage in the sectors of production of goods and services; sustainability and exploitation of bio- and non-bio-resources; and preparing society to face globalization. Second, improving the support for S&T activities, which include providing human resources, facilities and infrastructures, institutions, and networks. Third, which is very specific, the development of 100 Techno Park in regencies/cities and Science Park in every Province.

While all the above targets are very challenging, only the latest target that has the very clear and quantitative target, because it states the exact number of techno parks and science parks. Only within a year this target has been revised, as if it could not be assessed in advance that this target is impossible to achieve. The correction of this target also indicates that there is serious problem in the epistemic governance of S&T planning.

Further, RPJMN 2015-2019 directs the S&T development toward the five following activities:

- 1) R&D that deliver new technology/product diffused to production sectors;
- 2) Engineering services, providing facilities for engineering, design and testing;
- 3) Quality assurance services: standardization, metrology, calibration, and testing of quality;
- 4) Nuclear energy monitoring services: monitoring the utilization of nuclear technology in industry, agriculture, health, and energy;
- 5) Strengthening academics-private sectors-government partnership through the facilities of science and technology parks, incubators, and venture capital.

The direction into the above five activities may not be the right word, because those

activities have been done, not something to be pursued. Points 1 to 4 are the tasks of the main government R&D organizations that are mentioned in Chapter 4, namely, Indonesian Institute of Sciences (LIPI) for R&D, the Agency for the Assessment and Application of Technology (BPPT), National Nuclear Energy Agency of Indonesia (BATAN), and National Standardization Agency (BSN). The last point applies for all the above organizations.

Chapter 4 of RPJMN states that national researches are focused on (1) food and agriculture; (2) energy, new and renewable energy; (3) health and medicine; (4) transportation; (5) information, communication, and telecommunication (ICT); (6) defense technology; and (7) advanced material. These areas will be called as Main National Program (PUNAS) of research. These programs almost cover all areas, which are currently becoming the area of government R&D organizations. Among those areas only number 7 address specific research areas, namely, research on rare earths and silicon-based material. The others will cover almost all existing research that have been conducted in government research organizations. In PUNAS of food and agriculture research, the organizations involved are BATAN, LIPI and BPPT, whereas Ministry of Research, Technology and Higher Education (MRTHE) is responsible to organize research consortium for the Indonesian competitive fruits and vegetables to compete in global markets.

In PUNAS of energy research, the involved organizations are BATAN, BPPT and LIPI. In PUNAS of health technology and medicine research, the organizations involved are BPPT, LBM- Eijkmann, and LIPI. MRTHE is responsible to establish two consortia: for research in tropical diseases to develop national vaccine industry to deal with HIV, "demam berdarah", and "TBC" for research of stem cell.

Organizations that involved in transportation technology research are BPPT and National Institute of Aeronautics and Space (LAPAN). And for those in PUNAS of ICT research are BPPT and LIPI. Those involved in PUNAS of defend technology research are not mentioned at all. And in advanced material are BATAN, BPPT, Tekmira belonging to Ministry of Energy and Mineral Resources, and Metal Industry Development Center (MIDC) of Ministry of Industry.

So, the most frequently mentioned R&D organizations are LIPI, BPPT and BATAN. And two research centers belong two ministries that have been mentioned are only Tekmira of Ministry of Energy and Mineral Resources and MIDC of Ministry of Industry. Research centers belonging to Ministry of Agriculture, which in total have the largest number of researchers, are not mentioned at all. While PUNAS of food and agriculture research is the first PUNAS, and is supposed to be the most important area in Indonesia.

Two comments can be given here. First, the S&T content of Chapter 4 of Book II of RPJMN seems to present what have been done and also will continue to be done. It gives the legitimacy of what have been done by government R&D organizations rather than giving policy directions to them. Second, even though in the beginning of the Chapter 4, it is stated that S&T development is attempted to improve the competitive advantages of sectors of goods and services production, but it does not clarify how that attempt can be achieved through the activities of government R&D organizations.

Not only Chapter 4 of RPJMN does not address the integration of S&T development and industry development, even related R&D government organizations are not fully integrated. R&D organizations belonging to the ministries, except Tekmira and MIDC, have been excluded. It seems that Chapter 4 of RPJMN only provides legitimacy to what public R&D organizations, which do not belong to ministries, will do. In this case the function of public R&D organizations as supporting structure cannot be fulfilled.

R&D organizations are supposed to be the supporting structure of industrial development (Teubal, 2002). They cannot develop national technological capabilities on their own. Ultimately, technology-in-use is what is developed and produced in industries. R&D organizations can play role in enriching the stock of knowledge, including exploring various technological opportunities, but they are not designed to commercialize the technology to society by themselves. Technological development is the result of negotiations, interactions and transactions of various heterogeneous actors. Structural transformation from one phase of industrial development to another requires collective learning, coadaptation or coevolution among those actors.

Often S&T communities in government R&D organizations see themselves as the center of technology development rather than the supporting structure. Hence they often determine their research agenda without consulting with industries. Industries need only to wait the results of R&D and then apply them in their productions. This approach, which is currently popular in S&T policy discourse, is called in Indonesia language as “hilirisasi”, the end of which is “commercialization” of R&D results. It is based on the linear model of research-to-innovation development. This is the opposite of what was employed in the era when B.J. Habibie was minister of research and technology, back in 1980-1990s, which started the development of technology from the end (manufacturing/production) and end at the beginning (R&D). The “hilirisasi” approach is based on the technocratic assumptions that scientists, researchers and engineers working in R&D organizations know everything what to research and develop, and the result can be readily adopted by industries.

RPJMN was prepared by National Development Planning Agency (BAPPENAS). And Chapter 4 of Book II of RPJMN 2015-2019 was written by a unit at the level of echelon II in consultation with various people from planning bureau of R&D organizations. But, ultimate they are responsible to do the writing. But research areas are too broad to be comprehended by either a small group of people in BAPPENAS or planning bureaus of R&D organizations. Hence, they rely on the input given by each research centers. That is why RPJMN is more like legitimacy of what have been done by the organizations. Given this kind of RPJMN, it gives flexibility or does not constrain the planning carried out by each R&D organizations as well as MoRTHE. It is quite easy for public R&D organization to claim that their strategic plans have already been aligned with RPJMN. MoRTHE can derive strategic planning, which including funding schemes for research activities in the aread determined by RPJMN.

The constraints that limit the scope of S&T policy is because each part of RPJMN is the responsibility of particular unit of BAPPENAS. The structure of BAPPENAS implies the boundary of development area as well as the sovereignty of each unit over its area, which cannot be violated by other same-level units. That is why research centers under ministry of agriculture are not included in S&T development chapter, because agriculture research belongs to agriculture sector, which is the responsibility of other unit.

The way RPJMN was constituted allows government R&D organizations to do what they have usually been doing. It does not impose any constrain or direction to them. They can develop their strategic plan in line with the nature of their research areas. There are challenges because they also

have to set their own priority and integration among their own research centers. For specific R&D organizations, such as National Nuclear Energy Agency (BATAN) or National Institute of Aeronautics and Space (LAPAN), integration is relatively easier to do. But, for the organizations such as BPPT and LIPI, which consist of heterogeneous units, the integration and priority setting is quite challenging.

4. The Implementation of S&T development

RPJMN 2015-2019 assigned the research institutions to do what have already been established in it, such to do research in main priority areas, or to establish national science and technology parks, science parks and technology parks. MoRTHE, BPPT and LIPI have been assigned to establish national science and technology parks. And Ministry of Marine Affairs and Fisheries Republic was assigned to establish 24 techno-parks in various parts of Indonesia, and ministry of agriculture 17 science parks and 26 techno-parks. But, the needed resources for those parks were not only funding, but more importantly human resources that should manage and do the R&D in those parks. All those targets were too ambitious and actually could be assessed in advance based on the available resources, financial as well as human resources. But, it was waiting until the implementation could not be done. All those targets were significantly reduced in the following years without explanation that allow people to learn about previous policy making.

The most important instrument of S&T policy is sufficient funding that can be controlled. If it cannot be controlled, it is not policy instrument. Central government allocates certain amount of budget to government R&D organizations annually. But, the latter can use their budget relatively autonomously. Practically, they can operate independently of other organizations even though they can still obtain additional sources of funding. Their annual budget is guaranteed. Formally the tasks assigned to them (such as those of the RPJMN) or what they have planned (those of their own strategic plans) usually can be carried out and achieved.

Government R&D institutions have their own strategic plans as guidance to direct and coordinate their programs and projects. They can claim that their plans are derived from RPJMN, because RPJMN is general enough to accommodate any research activity. Generally, they have already set their priorities in their strategic plans. Having priorities implies that they have to focus the spending of budget on certain activities which require certain competences. However, implementing such priorities can be very challenging and

even in many cases almost impossible. Their researchers and engineers are usually permanent staffs whom they have to utilize even though their competences and research may not be relevant to the established priorities. So, the organizations have to allocate budget to fund their research interests. Ultimately and usually the budget will be distributed evenly to fund the activities of all researchers, and priority setting becomes meaningless.

MoRTHE has the flexibility to fund any research which accord with established priorities, because this ministry does not have their own researchers and engineers. They give funding to researchers in other institutions. But, not all researchers and research centers under R&D organizations submit their proposal for the funding. Some people may see application for external funding just gives them additional burden. They can only spend the funding for research purposes, so no additional incentive other than to do more research. Hence, the funding scheme of MoRTHE is not strong and attractive enough to direct R&D in Indonesia. However, formally MoRTHE might set ambitious targets for their funding programs, such as stimulating the diffusion of R&D results into industries. However, when the number of applications for the scheme is lower than the number that they can fund, they will lower the requirements so that the available budget can be fully utilized. Because the ministry also obliges to spent all the budget, otherwise they will be considered as underperforming.

5. Assessment

Monitoring and evaluation have been the usual bureaucratic procedures ensuring that government programs and projects are being carried out according to established plan. When the evaluator and the evaluatee belong to the same organization, the former will exert pressure on the latter because the failure of the latter will result to the failure of the former. But, the problem is when the failure of the latter cannot be prevented by the former, consequently it will become the failure of the former. While the failure is unacceptable. Hence, whatever the content of intermediate evaluation of a project or program, the final results can only show success.

That S&T development, especially R&D, have to deliver successful results is actually against the nature of research and innovation. The activities of research and innovation are always hampered by uncertainties. That the bureaucracy cannot accept such failures indicates a foundational problem in Indonesian S&T policy. The problem with the evaluation can be seen from the fact that even though every year government's programs considered success, but over a long period of time, S&T development does not make significant progress.

This problem has been realized by bureaucrats of R&D organizations. Even though they should play within the existing norms, that research is not allowed to fail, they understand not only the uncertainty of research but also the still low capacity of research in Indonesia. This problem, the low research capacity and competence, has been acknowledged in National Research Master Plan (RIRN).

MoRTHE have established programs having high targets, for example to deliver results that can be applied by industries, or to stimulate the collaboration between R&D institutions, universities and private sectors. Introducing technology to private sectors that can buy mature technology from overseas technology vendors is not easy, and also government R&D institutions and universities work as different pace and pursuing different goals from those of private sectors. That it was difficult to achieve was realized by

a former program manager of MoRTHE. He said that the funding program was to educate government R&D institutions and universities to collaborate with each other as well as with private sectors. Dr, Teguh Rahardjo⁵, former deputy of MoRTHE, who have designed and implemented funding schemes with his team, admitted that they had to accept such limitation, but still believed that they could make improvement. For example, most proposals for funding were not feasible, but they accepted them, and along the way they monitored and directed the participants to improve their research capabilities and cooperation. It was a learning process. However, learning could not be made explicit and reported. It is only known by people who were involved in managing the program. When the program was continued by other people, the lesson learned was lost, because it could only be transferred through informal conversations not reports.

⁵ Personal conversation in his office on October 3, 2019.

6. The role of Higher Education in S&T Development

That the role of universities is not included in Chapter 4 of Book II of RPJMN 2015-2019 needs clarification. The role of universities in S&T development cannot be ignored. Not only they provide highly educated human resources, some of them also produce more scientific publications than do R&D organizations. Indeed, academics in universities have teaching responsibility that reduce their research time, but some of them, especially universities having status of Perguruan Tinggi Negeri Badan Hukum (PTN BH) or Legal Entity State University, have graduate students who have capability and especially high motivation to do research. Assisting by their graduate students, academics can be very productive in publishing scientific papers.

In terms of commercialization of research, these Legal Entity State Universities (PTN BH) also have more flexibility than government R&D organizations, because they can manage their acquired resources independently. Hence, they can establish their own business entities to commercialize the results of their results or sell their scientific services. This kind of flexibility, which is very much needed in commercializing research results, is not possessed by government R&D institutions. Besides that, universities also have better network with private sectors through their alumni who work there.

7. Discussion on R&D Governance

This chapter will discuss Indonesian R&D governance at national level and organizational level. Governance at national level is that is carried out by central government, either through Ministry of Research, Technology and Higher Education (MoRT&HE) and National Development Planning Agency (BAPPENAS). While governance at organizational level occurs at each R&D organization. The characters of governance are listed in Table 3-3.

[Table 3-3] Arenas of governance at organizational and national level

	Organizational	National
Policy arena (overall goals)	Incoherence between: - The goals of organization to support industry vs the incentive system of S&T staffs.	Incoherence between: - The goal of R&D to promote production sectors vs the detachment of R&D activities from industrial policy.
Selection arena (priority setting)	- There is no measure to ensure that the priorities comply with the aspirations of stakeholders. - The priority setting can be countered or cancelled by the usual business of dividing budget equally across programs.	- Information asymmetries between actors have not been bridged. There is no systematic effort to engage industries in priority setting. - Priorities seem to accommodate what have been or will be done by public R&D organizations separately and independently from one another.
Control arena (progress monitoring)	- No interactions between S&T personnels and industries. - The emphasis of control is on the time and cost management of the project rather than on the quality and relevance of R&D to satisfy industries' needs. - There is no public or stakeholders' assessment of outputs.	- Priority is not strongly reflected in budget allocation. There is no financial control by central government to assure that the goals will be achieved. - R&D results are demanded too soon to contribute to industries, which are impossible to achieve. The impossibility is realized by all parties, including monitoring and evaluation teams.

7.1 Governance at National Level

The first goal of national S&T development is to improve the competitiveness of goods and services production sectors. The targeted S&T development areas cover seven national main research program (PUNAS), from food and agriculture to ICT to advanced material, which are very broad for a country that has limited financial resources and knowledgeable human resources. Some industry sectors are even too advanced for any developing country to make significant R&D contribution. While the goal seems very ambitious, the S&T policy is detached from industrial policy. Every area of PUNAS consists of random set of research belonging to the same category, rather than systematic set of complementing technologies needed by particular industry.

Industry is a complex and networked enterprise involving heterogeneous actors. There is information asymmetry between different actors that cannot be easily overcome. However, there is no systematic measure to engage industries as well as ministry of industry in developing S&T policy. The team that wrote up S&T policy in RPJMN is quite small in number to deal with the complexity of S&T development. Given that the Chapter 4 of RPJMN Volume II consists of what have been done and will be done by government R&D organizations, this chapter seems to give the legitimization to what will be done by them rather than derive a systematic plan to pursue scientific and technological targets.

What have been planned are supposed to be executed by R&D organizations. However, what have been mentioned in the chapter 4 are only small parts of what have to be done by R&D organizations. Previously there was no strong financial control over the achievement of the plan. But, recently the funding of national research is conducted through the so called national priorities which are established together by the R&D organizations and a small team from BAPPENAS.

In addition to the funding through national priority programs, MoRT&HE also provides funding to which R&D organizations and universities can apply. This program provides funding for a large number of proposals but managed by a team of less than ten people. Disbursement of funding is often late, but the research is demanded to deliver result very soon.

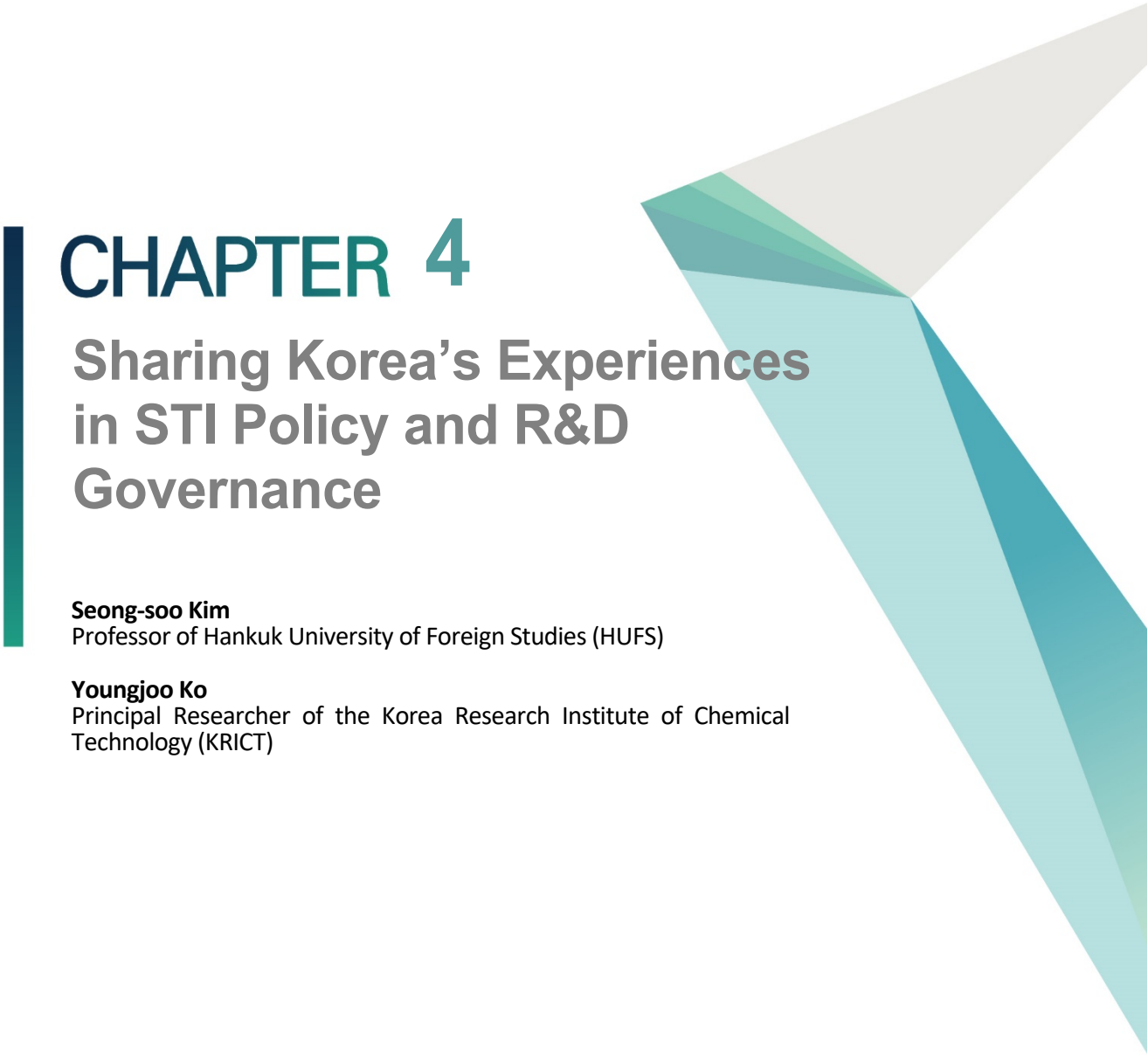
7.2 Governance at Organizational Level

Technologies that are utilized by large corporations in manufacturing industries are often very advanced and the latest. In order to improve in production process or product development, R&D organizations need to get familiar with what happen in the industries. They have to work with the same rhythm with their counterparts in industries. However, the goal to support industries may not be compatible with the incentive systems of, for example, researchers. Researchers will be assessed based on their scientific publications, whereas working for industry requires engineering works that consists of optimization that takes time and a lot of efforts. This kind of work is necessary and important for industry but may not give sufficient material for scientific publications.

Priority setting is very important to deal with the limited resources. The priority setting should be done in cooperation with industries, otherwise R&D organizations will not know the problems and needs of industries. However, there is no scientific measure to ensure that the priorities are in accord with the aspirations of stakeholders.

The other problems are that the priority setting can be countered or cancelled by the usual business of dividing budget equally across programs. R&D organizations have research or engineering staffs that they have to use from the time when the latter were recruited to the time they reach retirement. They have to keep their staffs even though their competences are beyond the scope of established priorities. Hence, there will always be programs which are not included in the priority list and the programs also get funding. This causes the priority setting is difficult to execute because the funding ultimately is divided across programs including the ones beyond the priority list.

When consultants work for private sectors, reporting and monitoring do occur periodically. Constant interactions are needed between two parties to ensure the achievement of the program. But, in the case of R&D activities, even though they are intended to support industries, there is no or little interactions between S&T personnels and industries. Control is conducted internally and especially more on time and cost management of the project rather than on the quality and relevance of R&D to satisfy industries' needs. There is neither public or stakeholders' assessment of R&D outputs.

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are several overlapping, semi-transparent geometric shapes in shades of teal, light blue, and grey, creating a modern, abstract design.

CHAPTER 4

Sharing Korea's Experiences in STI Policy and R&D Governance

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Chapter 4. Sharing Korea's Experiences in STI Policy and R&D Governance

1. Korean Experience in Nurturing Public Research Institutes & NST

1.1 Policy change in Public Research Institutes in Korea until the establishment of National Research Councils (Ko, Y.J., 2005)

As a third type of public research institutes, non-profit research institutes are usually sponsored by government-owned public companies, accounting for less than 10% of R&D by value within all public sector research organizations. NRIs are part of central government departments established by law. Public Research Institutes (PRIs) are classified as research organizations created by the regulations and rules of local governments. Researchers in both NRIs and PRIs are civil servants. They conduct mainly testing, investigation, and research in the fields of agriculture, engineering, medicine, and physical science (Ko, Y.J., 2005).

N/PRIs were first set up in the period of Japanese rule between 1910 and 1945. After the Korean War, which lasted for three years from 1950 to 1953, a research institute for defense science and technology was established under the Department of Defense. The Minnesota project with the US was launched in the mid-1950s, providing the opportunity for Korean professors to undergo scientific training and conduct research in the US. The highly educated researchers became the basis of scientific education after their return. In 1959, an atomic energy research institute - which was the beginning of modern research in Korea - was created in the government based on the Atomic Energy Law in 1958. Other N/PRIs continued to be formed, playing the role of main public sector research organizations until the mid-1970s (Ko, Y.J., 2005).

Of the government expenditure on R&D, N/PRIIs accounted for 82.1% in 1967 and 59.3% in 1975 (MOST, 1997: 258-264), even though the volume of research conducted by them was very small (MOST, 1997: 16-19). The role and number of N/PRIIs have recorded a decrease that is inversely proportional to the creation and development of Government-Supported Research Institutes (GSRIIs) since 1966. GSRIIs have been carrying out R&D in order to achieve the national goals for S&T, and they have been a crucial vehicle for the implementation of the national science and technology policy. GSRIIs have changed with changes in industrial or economic policy as well as science and technology policy and, to some degree, with partially public sector management or reform. All of the policies and reform approaches are rapidly integrated into innovation agenda and policy (Ko, Y.J., 2005).

1.1.1 Establishment and Proliferation of Korean-Model GSRIIs in the 1960s and 1970s

a) Objectives and Strategies

The genuine development of Korean S&T began with a military government taking power by coup d'état on May 16, 1961. Shortly after seizing power, General Park Chung Hee, who soon became president and remained in that position for 18 years until he was assassinated by one of his subordinates in 1979, announced the First Five-Year Economic Development Plan from 1962 to 1966. The government initially tried to pursue inward-looking industrialization involving the so-called "import-substituting industrialization strategy." Nonetheless, the strategy was revised to an "outward-oriented export promotion strategy" in December 1962 in response to US demands (Lim, 2000). According to an OECD report, Korea's pattern of development from 1958 to 1964 involved a strategy of import substitution designed to revitalize the country's light manufacturing industries. This was achieved by the use of stringent import controls, multiple exchange rates, and foreign aid to pay for imports (OECD, 1988:34).

The Park government decided to rely heavily on foreign loans from the private sector by allowing state-owned banks to guarantee foreign borrowing especially by large private

firms. This represented a so-called “risk partnership” with large firms instead of relying on foreign direct investment (FDI) and control by state-owned enterprises. The Korean government’s strategy for development in the 1960s was export-led industrialization based on “government-business risk partnership,” “high degree of control over the financial sector,” “labor-intensive light industry” with “zero tolerance for trade union movements,” and “foreign technology imports” mainly in the form of capital goods that enabled business to learn from reverse-engineering.

Accordingly, the major objectives and strategies of S&T policies in the 1960s were to build a technological infrastructure to support the government strategy including the creation of a government department responsible for S&T matters, enactment of laws to support S&T development, building of a new R&D performing capability, and development of human resources. In particular, during the Five-year Plan, imports of foreign technology were crucial components of the Plan due to the lack of technological capabilities necessary for industrialization.

The rapid progress of industrialization forced Korean industries to ask the government develop and provide domestic R&D capabilities. N/PRIs, which were at that time part of government departments, were considered to be inappropriate as suitable national R&D resources because their role focused mainly on testing and investigation and not R&D, and their R&D capability was small and weak. Their capability was insufficient to attract scientists and engineers who studied and worked abroad. The government decided to build a new research institute, the Korea Institute of Science and Technology (KIST), which offers excellent environment and high salaries, conducts a wide range of industrial R&D including technological feasibility studies, provides technological services for small and medium companies, and conducts engineering studies on a pilot plant scale. The main focus was on solutions for simple and practical problems arising from the application of transferred technology from abroad.

KIST led Korea’s early stage of technological innovation and contributed to solving many problems during the process of adaptation and application of imported technology through the contractual R&D system with business. Government-business risk partnership

and rapid economic development fueled what may be characterized as an investment explosion in the second half of the 1960s. The reckless investment boom inevitably produced a number of firms that could not meet their foreign debt obligations and went bankrupt (Lim, 2000). As the Korean economy showed signs of overheating near the end of the 1960s, IMF stepped in to prescribe a stabilization package. The government initially resisted but had to accept the package in 1970 when the US offered additional development loans contingent upon the acceptance of the IMF program (Cho and Kim, 1997: 81).

In the 1970s, as labor-intensive industries were gradually replaced by heavy industry and chemicals, S&T infrastructure capable of modifying and internalizing foreign technology was essential in order to meet the demand for the greater technological sophistication needed for producing machinery, steel, chemical engineering, shipbuilding, and electronics (OECD, 1996). In order to meet the increasing demand for qualified scientists and engineers, the S&T policy focused on expanding technical and engineering education in related fields. Furthermore, a number of GSRI were established in the 1970s in fields such as machinery, shipbuilding, chemicals, marine science, electronics, and electricity in order to satisfy various industrial needs and to improve the institutional mechanism for adopting and internalizing imported technology.

b) Governance Structure

The government created the Economic Planning Board (EPB) in July 1961. This was responsible for formulating and implementing economic development plans. In addition, the commercial banks were re-nationalized, and an amendment to the Bank of Korea (BOK) Act in 1962 transferred authority for monetary policy, including control of foreign exchange, to the Ministry of Finance. This step turned out to be crucial to the government's strategy for financing development projects (Cho and Kim, 1997: 34). As a government unit responsible for S&T policy, the "Technology Development Bureau" was also created within EPB in 1962. The first and second five-year plans for the promotion of S&T were formulated in line with two Five-Year Economic Development Plans from the

periods 1962-1966 and 1966-1970.

To meet industrial needs, KIST was created in 1966 in Seoul as an independent research institute covering all areas needed by the industry, based on a recommendation of the US model of contractual research institute. After the summit talks between Park and President Johnson of the US in May 1965, the US scientific adviser to the president, Dr. Honic, visited Korea in July 1965 to discuss the possibility of building a new modern research institute to support the industry. In September 1965, investigators from the Battelle Memorial Institute (BMI) in the US visited Korea again and recommended the establishment of an integrated comprehensive R&D center to respond to the demand of the industry based on a detailed plan from Battelle and the support from the US. Even though there were about 50 N/PRIs and 30 laboratories in companies and universities at that time, they thought N/PRIs were inappropriate for meeting industrial needs owing to their weak R&D capability, small size, poor facilities, and bureaucratic culture. The 1966 KIST Assistance Act was promulgated to change KIST into an independent legal entity. It guaranteed for the institute an annual endowment from the government to be used for basic operations and capital expenditure for construction.

In addition, the S&T Promotion Law was enacted in 1967, and MOST (now MSIP) was established as a lower-level government department in 1967 in order to put science and technology higher on the agenda. Such establishment and the functions of MOST within the policy mechanism of the central government meant that S&T would be one of the national tools from the outset of national development, unlike in other less developed countries (MOST, 1997: 43). Nonetheless, MOST was not able to manage and control the entire process and issues of S&T policy for two reasons. First was the existence of other much more powerful departments developing economic policy and controlling budgets resulting from the national goals of rapid economic growth. Second was its ignorance on how to establish a long-term S&T policy (Kim, L.S., 1997: 136).

In 1971, the government, anticipating increased demand for high-caliber S&T manpower, established the Korea Advanced Institute of Science (KAIS) as a graduate school for applied science and engineering, later renaming it the Korea Advanced Institute of Science and

Technology (KAIST). This research-oriented university was unique since its faculty and students were fully supported by the government under the jurisdiction of MOST, unlike other universities that were under the Ministry of Education (OECD, 1996: 66).

In the 1970s, the expansion of the economy and the more complex industrial structure demanded more R&D capability. Note, however, that R&D capability in the private sector was still weak. KIST could not meet the increasing demand alone. Based on the success of KIST, other GSRI were able to expand over time mainly through spin-offs from KIST into specialized research institutes in fields such as defense R&D, development of economic policy and human resources, machinery, shipbuilding, chemicals, marine science, electronics, and electricity in order to satisfy various national and industrial needs under the supervision of related departments such as MOST, EPB, Ministry of Commerce (now MOTIE), and Ministry of Energy and Resources (now MOTIE) as shown in Table 4-1. The enactment of the Law for the Promotion of Specific Research Institutes in December 1973 was designed to drive the proliferation of GSRI.

[Table 4-1] Major GSRI Established in the 1970s

Year Established	Name	Field
1970	Agency for Defense Development (ADD)	Defense
1971	Korea Development Institute (KDI) Korea Advanced Institute of Science (KAIS)	Economic Policy Human Resources & R&D
1972	Korean Educational Development Institute (KEDI)	Human Resources
1973	Korea Ocean Research & Development Institute (KORDI) Korea Atomic Energy Research Institute (KAERI)	Marine Science Nuclear Energy
1975	Korea Standards Research Institute (KSRI)	Standards for Industry
1976	Korea Institute of Geoscience and Mineral Resources (KIGAM) Korea Institute of Communications Technology Korea Institute of Machinery & Materials (KIMM) Korea Research Institute of Ships (KRIS) Korea Research Institute of Chemical Technology (KRICT) Korea Electric Research & Testing Institute (KERTI)	Natural Resources Electronics Machinery Shipbuilding Chemical Technology Electricity

Year Established	Name	Field
1977	Korea Institute of Energy Conservation (KIEC) Korea Science and Engineering Foundation (KOSEF)	Energy Basic Science & Human Resources
1978	Sun Energy Research Institute (SERI) Korea Ginseng Research Institute Korea Tobacco Research Institute Regional Development Institute	Energy Ginseng Tobacco Regional Development

On the other hand, in order to monitor the government S&T policy and to coordinate its implementation across the relevant ministries, the National Council for Science and Technology (NCST) was established in 1973 and was placed at the Cabinet level and chaired by the Prime Minister. In reality, however, NCST did not work properly because ministers did not have any legal responsibility to comply with the Council's decisions (Kim, et al, 1997: 47). GSRI in fields related to natural science numbered 19 at the end of the 1970s, and they were answerable to 7 different departments (Ko, Y.J., 2005).

c) Funding

At the end of the 1970s the nation's total R&D investment was about 0.8% of GNP. Government expenditure accounted for nearly 70 % of the nation's total R&D investment at the end of the 1970s (Ko, Y.J., 2005). N/PRIs and GSRI constituted 86.4% of the total domestic R&D investment in 1969 and 56.4% in 1979, whereas the industry's share was 10.2% in 1969 and 34.1% in 1979. Of the 86.4% for which N/PRIs and GSRI were responsible, GSRI carried out 51.6%. Government financed GSRI through funding mechanisms. Grant-in-aid was used for facilities and operation of institutes. The other method was through contracts with departments. GSRI were also funded by contracts with the industry. All R&D work commissioned to GSRI was contractual, whether it came from public or private institutions (Ko, Y.J., 2005). The purposes were: first, to encourage industries to become actively involved in their projects; and second, to emphasize the responsibility of those in charge of a project. In 1973, the number of projects carried out in KIST was about 170 -- which was thrice as many as that at the early stage -- and the portion of R&D funds from the industry was 61%. KIST received rave reviews as a

successful story (MOST, 1997: 31; (Ko, Y.J., 2005).

d) Management

Since its creation in 1967, MOST has had the responsibility of monitoring and formulating science and technology policy across government departments and has published the annual survey of R&D activities (Ko, Y.J., 2005). This had been initiated by the Technology Development Bureau of EPB in 1963 in the name of “status survey of research institutes” and taken over by MOST in 1967. The coordination of S&T policy was the responsibility of NCST. The role and function of MOST and NCST, in practice, were not as originally planned. Economic and innovation policies were separately formulated by EPB and subordinate to the S&T policy. Other departments were higher in the hierarchy of position than MOST. NCST held meetings irregularly and had no power to make Ministers abide by its advice (Ko, Y.J., 2005).

N/PRI were part of the departments under the bureaucratic culture, and they lacked autonomy. GSRI including KIST were designed to guarantee not only the principle of autonomy but also financial support for stable R&D from the government. They were also self-financing in relation to the overhead costs of internal research divisions or labs as their basic management units. Individual labs had to earn enough from research contracts to cover their overhead expenses for human resources and facilities. Despite the principle of stability and autonomy, GSRI began to face two major problems at the start of the 1980s: less autonomy under the government departments, and less stability due to the decrease of research contracts from industries that had grown by themselves (Ko, Y.J., 2005).

1.1.2 Reorganization and Uniform Control in the 1980s

a) Objectives and Strategies

The Korean government converted its export drive policy with import and imitation of technology from advanced countries in the 1960s and 1970s into an indigenous technology drive policy in the 1980s with the following strategies: i) pan-national support for S&T development led by the president; ii) concentrated S&T administrative system and

construction of a technology infrastructure; iii) development of essentially strategic and high technology to strengthen industrial competitiveness through, for example, the import substitution of machinery parts; iv) building of a substantially private-led R&D system; v) nurturing of science and technology-related human resources; and vi) internationalization of S&T (MOST, 1997: 75-115). The main task of GSRI was to support the national goals and strategies by maximizing their efficiency and effectiveness (Ko, Y.J., 2005). They shifted their role from mainly direct contractual R&D with industry to carrying out national R&D projects for strategic technology development (Kim, et al, 1997; Ko, Y.J., 2005).

b) Governance Structure

Shortly before the 2nd military government in 1980, a standing committee for economic reform within the Board of National Security was launched to drive a wide range of socioeconomic change and to secure political power (Ko, Y.J., 2005); GSRI was comprehensively reviewed for the first time, with the following problems pointed out: i) low efficiency because they were too small to achieve economies of scale; ii) too many administrative staff; iii) duplication of R&D and excessive competition to obtain a contract; iv) little collaboration between GSRI stemming from their being under the authority of too many ministries; v) low level of coordination and effectiveness (Ko, Y.J., 2005). The report concluded that the restructuring of GSRI was needed to solve the problems and enhance efficiency and effectiveness (Ko, Y.J., 2005).

In November 1980, the government announced the reorganization of 16 out of 19 GSRI into 9 new ones and placed them under the supervision of MOST. The Korea Advanced Institute of Science and Technology (KAIST) was established through the merger of KIST and KAIS. The Korea Energy Research Institute (KERI) was created from the merger between KAERI and Korea Agency for the Development of Nuclear Fuels. The Korea Institute of Energy and Resources (KIER) was set up through the merger of KIGAM and KIEC. The Korea Standards Research Institute (KSRI) was transferred from the Office of Industrial Promotion (Ko, Y.J., 2005). The Korea Institute of Machinery and Materials (KIMM) was merged with KRIS. KERTI was transferred from the Ministry of Trade and Industry to MOST. The Korea Electricity & Communications Research Institute (KECRI) was

created through a merger between the Korea Institute of Communications Technology and KERTI. During the 1980s, however, particularly from the mid-1980s, there were some additional spin-offs, mergers, and new establishments to respond to the need for new areas of R&D in line with the growing national R&D programs across departments. The Bioengineering Center (now KRIBB) and the Electronics and Telecommunication Research Institute (ETRI) were created in 1985. The Policy Research and Evaluation Center for Science and Technology (now STEPI) and the Korea Food Research Institute (KFRI) were established under the Ministry of Agriculture, Forest, and Fisheries in 1987(Ko, Y.J., 2005). The Korea Institute of Construction Technology (KICT) was set up under the Ministry of Construction in 1988, and the Korea Institute of Industrial Technology (KITECH), under the Ministry of Trade and Industry in 1989(Ko, Y.J., 2005). KIST was again separated from KAIST, with the Korea Aerospace Research Institute (KARI) created in 1989. KERI was renamed KAERI in 1989. In 1990, science-related GSRI consisted of 13 organizations and 9 subsidiaries under an increased number of Departments. The main feature of governance structure related to GSRI in the 1980s was the strong central control by the rules and regulations made by MOST, which were similarly applied to all GSRI under other departments (Ko, Y.J., 2005).

In response to the growing problems and complaints about such strong and uniform control by the government, in August 1989, the EPB and a government “Committee on Administrative Reform” reviewed the new role and management of GSRI and, reflecting the changing environment, proposed as an alternative an “autonomous but more competitive R&D system” and the “transfer of R&D to industries, searching new areas” (MOST, 1997: 270). The recommendations were not put into practice, however. On the other hand, the Council of Technology Promotion (CTP) chaired by the President and a Standing Committee on Technology Promotion presided over by the Minister of S&T were set up in 1982 to drive the development of technology and coordinate policies. CTP was operated with around 250 members from the government, National Assembly, industry, scientific communities, and financial sector inter alia. The Council of Science and Technology Promotion (CSTP) replaced CTP in 1989(Ko, Y.J., 2005).

c) Funding

The gross domestic expenditure on R&D (GERD) increased rapidly in the 1980s from 0.58% of GNP in 1980 to 1.48% in 1985 and 1.91% in 1990. The rate of annual increase averaged 30.8%. Government expenditure on R&D also expanded throughout the 1980s with 17.6% average annual growth. Interestingly, majority of the funding went to development research during the period. The ratio of basic, applied, and development research was 17:25:58 in 1982, 17:29:54 in 1985, and 16:24:60 in 1990 (Ko, Y.J., 2005).

Meanwhile, despite the rapid increase of GERD, the proportion of R&D performed by the sector declined from 53.5% of GERD in 1981 to 19.4% in 1990 due to the growth of higher education and industrial sector. The industry accounted for 20 times more R&D activity in 1990 than in 1981, and the higher education sector also recorded 9 times more R&D activity in 1990 than in 1981, with GSRIIs performing 4 times more R&D in 1990 than in 1981. This meant that the R&D activities rapidly spread from the government sector to the industry and higher education sector throughout the 1980s. The relative proportion of R&D performed by GSRIIs was 26.8% in 1980 and 14.8% in 1990 (Ko, Y.J., 2005).

The funding of GSRIIs shifted from basic institutional funding and contractual R&D mainly from industry to a clearer dual system that consisted of grant-in-aid by referring to an annual budgetary procedure including the salary and operational cost of GSRIIs and R&D contracts mainly with departments. In particular, MOST's R&D program, which was launched in 1982 based on the Technology Development Promotion Act to develop essential industrial technology with the fifth Five-year Socioeconomic Development Plan, was a principal source of contractual R&D in GSRIIs (Ko, Y.J., 2005). The main reasons for introducing the national R&D program were: i) the need for a national R&D funding mechanism as a catalyst for industrial restructuring through domestic innovation; ii) the need for government intervention to enable using limited resources more efficiently; and iii) the need for expansion of public R&D in order to induce private firms to invest in their own R&D and to encourage collaborative R&D among industry, higher education sector, and GSRIIs (OECD, 1996). The scope of national R&D programs was expanded to other technological areas, e.g., the launch of the Industrial Generic Technology Development

Program in 1987 by the Ministry of Trade, Industry, and Energy (MOTIE) to complement MOST's program (Ko, Y.J., 2005). In 1988, the Ministry of Energy and Resource at the time started the Alternative Energy Development Program, aiming to reduce Korea's dependence on foreign energy resources while developing cleaner energy (Ko, Y.J., 2005). The participation of GSRI in the national R&D programs increased not only because that was one of the aims of the programs but also because the institutional funding, i.e., grant-in-aid, was insufficient.

d) Management

After the reorganization of GSRI into MOST in 1980, they were steered by administrative or highly bureaucratic management through uniform laws, guides, rules, and procedures in relation to both R&D and organizational operation (Ko, Y.J., 2005). The Board of Directors of each GSRI was governed by those in government, and they had no autonomy. GSRI were not able to change the rules on personnel, salary, and promotion system and anything related to internal management without permission from the government. Neither could the detailed items of R&D projects, which were fixed at the time of selection of proposals by the departments, be allocated to other items. All GSRI related to science and technology had similar characteristics of management such as insufficient institutional funding for salaries and operations, strong bureaucratic steering, input-based R&D management, and no autonomy. They were also the main performers of the government R&D program, increasingly dependent on government rather than industry (Ko, Y.J., 2005).

Despite the growth of departmental R&D programs and increasing government investment in R&D, the capability for coordinating or prioritizing the policy or R&D programs strategically and attempts to do so were very weak. MOST had the right to formulate and coordinate the science and technology policy, but its political role and power were very weak compared with the departments that decided the economic policy or the budget.

1.1.3 Series of Changes in GSRI in the 1990s

GSRI experienced a series of critical reviews and significant changes throughout the 1990s. Even though the reviews had little continuity and even different purposes, the resulting changes were significant, having consistent direction and common features. The main reviews and changes were:

- A review of GSRI to reset their function and enhance efficiency in 1991,
- A review of specialization and characterization of GSRI in 1993,
- Introduction of the Project-Based System (PBS) in 1996,
- A drive for management reform in 1998~1999 and the introduction of Research Councils in 1999.

1.1.3.1 Review of GSRI to Reset Functions and Enhance Efficiency in 1991

In March 1991, President Noh Tae-Woo ordered the comprehensive scrutiny of GSRI as a measure to strengthen industrial competitiveness (Ko, Y.J., 2005). Immediately, "a Joint Review Committee for the Evaluation of GSRI," chaired by the Vice Minister of Science and Technology, was set up with 27 members who were government senior officials and scientific experts mainly from the industry and academe. The across-the board review was carried out over 4 months and was the first comprehensive review since 1980. 22 GSRI -- 19 under MOST including subsidiaries and 1 each under the Ministry of Commerce and Industry, Ministry of Construction, and Ministry of Agriculture, Forestry, and Fisheries -- were evaluated with focus on the new role and function and efficiency of management. The results of the evaluation were reported in July 1991, and its proposals were as follows (Ko, Y.J., 2005):

i) Resetting the role and function of GSRI

- A role shift from industrial technology to generic technology or pre-competitive stage technology
- Promotion of institutes for research planning and management
- Partial consolidation to avoid duplication of R&D
- Merger of small GSRI such as the Supporting Center for Basic Research and the Institute of Astronomy and Space Science into other larger GSRI

- Strengthening of the function of training human resources in industrial technology

ii) Efficiency of Management

- A more flexible, competitive system of R&D
- Free technology transfer to industries from GSRI
- Strengthening of contractual employment
- Introduction of a performance-related wage system
- Introduction of an annual evaluation system for the management of GSRI by their presidents
- Improving R&D productivity
- Common utilization of GSRI across departments and the like
- Abolishment of the government control guide on personnel and payments
- Weakening of trade unions

Most of the recommendations of the report were approved by CCST in August and reported to the President. An Implementing Commission in MOST and a similar committee within GSRI were set up to monitor and drive the changes until June 1992 (MOST, 1991). During the process of review and implementation, which lasted for over a year, most GSRI suffered from the heavy burden of administrative work involved in reporting and changing. The most important thing was the shifting attitude to GSRI from considering them as important R&D performers to a situation wherein questions were asked about efficiency and effectiveness in relation to both their R&D and the operation (Ko, Y.J., 2005).

1.1.3.2 Specialization and Characterization of GSRI in 1993

After the trials and errors during the top-down implementing process above as well as strong resistance from scientific communities, the new and first non-military government in more than 20 years since 1961, led by President Kim Young Sam and inaugurated in February 1993, had to announce another plan “to strengthen autonomy, to improve the morale of R&D staff, and to establish a self-reform plan for the specialization and activation of GSRI in June 1993 (MOST, 1993). This plan was in line with the science and technology policies specified in the goals of technological development in the “New Economy 5-year Plan” for the period 1993 - 1997 announced in March 1993.

The New Economy Plan specifically includes strengthening the competitiveness of major

industries on a par with advanced countries by 1997, promoting first-class core technology in strategic specific areas, and expansion of basic scientific research that would improve the general level of technology for the benefit of the public (Kim, et al, 1997: 57). The Plan aimed to give the boards of directors and presidents of GSRI autonomy from the government - with the reduction in the number of directors originating from the government (MOST, 1993) - but more accountability and responsibility with regard to detailed plans and emphasis on bottom-up approaches. There were more positive changes to GSRI as reflected in some measures, i.e., boosting the morale of researchers and increasing funding and autonomy. Efficiency and responsibility were also emphasized. Microscopic changes in internal organizations to enhance research-centeredness and introduce matrix-style research units were enforced as well.

1.1.3.3 Introduction of the Project-Based System (PBS) in 1995~1996

After the president's address to Korea on the subject of globalization and the appointment of a new Minister of Science and Technology in late 1994, more radical restructuring plans emerged, including privatization, cutbacks, consolidation, and more competitive contractual system of R&D, particularly for those GSRI that had been working in areas closely connected with industry. These plans were implemented without any open or official reviews but mostly on the basis of the new minister's personal preferences (Ko, Y.J., 2005). In particular, the Korea Institute of Machinery and Materials (KIMM) and the Korea Research Institute of Chemical Technology (KRICT) were major targets for privatization through the removal of their names from the lists of GSRI in the Promotion Law of Special Research Institutes established in 1973 as the legal basis for government financial support to GSRI.

These attempts were met with strong resistance. In the end, in March 1995, MOST changed the plans for privatization and other organizational change that had caused strong resistance and problems. Instead, a new funding system, the so-called "Project-Based System" (PBS), was reported to the president in March and introduced first for the G7 program that was part of MOST's R&D program in August 1995 (Ko, Y.J., 2005). PBS was applied to all GSRI under MOST beginning January 1996 and was expanded to other

GSRIs under other departments (MOST, 1997). The new arrangement involved the abolishment of the “dual funding system,” which includes grant-in-aid for salary and operation of institutes and program funding by the national R&D program into a single funding system with contracts for projects based on full-cost pricing including the cost of staff’s wages and operational costs (Ko, Y.J., 2005). Such meant that, without sufficient projects, research units or teams or even an institute itself could not survive (Ko, Y.J., 2005).

PBS aimed to change not only the funding system but also management. It sought to give more autonomy and responsibility for R&D activities to the principal investigator of a project and emphasized the need for high transparency and performance in R&D activities (Ko, Y.J., 2005). The government believed that it could bring about better prioritization and allocation of government funds and higher productivity in GSRI (Lee, M.H., 2000). It was argued that PBS, in a sense, was a new approach to arranging GSRI in line with the national goals, replacing physical consolidation that had been criticized for failing (Ko, Y.J., 2005). In addition to the annual evaluation of institutions, which had continued since 1991, the evaluation of projects, particularly focusing on R&D performance rather than input factors, emerged as an important tool for managing PBS. At the same time, the annual institution evaluation was replaced by both annual evaluations by the GSRI themselves with other experts and a three-year comprehensive institution evaluation system. PBS was radical and uniform for all GSRI due to the extreme financial instability and lack of autonomy. Nonetheless, strong opposition as well as a wide range of resistance from GSRI and other related stakeholders emerged (Ko, Y.J., 2005).

The government decided to change the original PBS system, excluding a few GSRI that were not research-oriented and some national R&D programs from its application and introducing other measures that were intended to give a minimum level of stability (MOST, 1996).

1.1.3.4 Managerial Innovation and Introduction of Research Councils in 1998~1999

Since the foreign currency crisis in late 1997 and the inauguration of the new president,

Kim Dae Jung, stringent budget and public sector reform and restructuring have been crucial parts of the structural adjustment programs forced on Korea by IMF (Ko, Y.J., 2005). The Committee of Planning and Budget (CPB, now MOEF) newly spun off from the Ministry of Finance and Economy issued A Guideline for Management Reform in GSRI in early April 1998 to 59 GSRI involved in the areas of humanities and economic and social sciences in order to achieve change in their governance structure, funding, and management, etc. in line with the public sector reform (Ko, Y.J., 2005).

After the collection of plans from GSRI based on the guidelines, CPB outlined the output to the president in a report titled Measures for the Management Reform of GSRI after collaborative work with the Committee of Administrative Reform chaired by Lin Su Kim (Kim, L.S., 1998). The measures included four basic principles, i.e., adjustment of function and budget of GSRI instead of organizational consolidation, ensuring of specialization and objectiveness of GSRI, competition and collaboration with industry and universities, autonomy and accountability management, and plan to enact a new law to ensure that principles were implemented.

The detailed measures were as follows: partial merger of functions between GSRI that conducted similar R&D activities; 20% uniform cutback of budget allocation for the operation of GSRI with streamlining; division of existing grant-in-aid into two categories, i.e., fundamental funds and policy research funds, with the endowment of fundamental funds to GSRI according to the nature of their R&D, and competitive tendering for policy research funds between R&D performers including universities and industry (Ko, Y.J., 2005); abolishment of the existing board of directors of each GSRI and creation of five new Research Councils; regrouping of GSRI into new Research Councils free from the supervision of departments; introduction of business-style management including the contractual employment of researchers (Ko, Y.J., 2005); abolishment of audit by the relevant departments and introduction of an accounting-oriented audit system; introduction of a performance-related incentive system including a lump-sum annual payments system and a corresponding evaluation system; and activation of collaborative R&D (Ko, Y.J., 2005).

The measures were rapidly implemented by a Team for Implementation (Ko, Y.J., 2005). The Law for the Establishment, Management, and Promotion of GSRI was enacted in January 1999. According to the Law, 5 Research Councils were set up, and 43 GSRI were regrouped into the new Research Councils. The number of researchers and staff was to be reduced by nearly 20% across the board within a year in line with the 20% budget reduction for the operation of institutes. Other working conditions such as severance pay and support for tuition fees, etc. were also withdrawn (Ko, Y.J., 2005).

1.1.3.5 Changes from the reviews

The abovementioned four reviews conducted over a period of nearly 10 years resulted in changes in the objectives, strategies, governance structure, funding, and management of GSRI including the R&D culture.

a) Objectives and Strategies

According to the 7th Socioeconomic Five-Year Plan (1992-1996), the New Economy 5-year Plan (1993-1997), and the 5-Year Plan for Science and Technology Innovation (1998-2002) (MOST, et al, 1997), the main objectives and strategies of science, technology, and innovation policy in the 1990s were (Ko, Y.J., 2005): i) building of the industry-led technology innovation system and strengthening industrial competitiveness; ii) strategic drive of national R&D programs both in individual departments and across departments; iii) nurturing of all R&D performers and collaboration between them, iv) strengthening of the base for science and technology innovation; v) increasing government expenditure on R&D; vi) efficiency, effectiveness, and prioritization of government investment; vii) paying of greater attention to basic research; viii) promotion of scientific culture, globalization and localization. The main questions to GSRI were whether they had contributed to achieving the objectives and strategies above and how to make them deliver the national objectives (Ko, Y.J., 2005).

The abovementioned series of reviews for GSRI reflected the questions and recommended specific objectives and strategies. These were efficiency and effectiveness, competitive environment and emphasis on performance and R&D quality, accountability

and autonomous responsibility, utilization of R&D across departments, and collaboration with other R&D performers such as universities and industry (Ko, Y.J., 2005).

b) Governance Structure

Created in 1973 and presided over by the Prime Minister but rarely met, NCST was reactivated in April 1991 to coordinate policy and program for science and technology. The Presidential Advisory Council on Science and Technology (PACST), whose members were appointed by the president from scientific communities and industry, was set up in May 1991 to give advice to the president directly on matters of science and technology and related policies (Ko, Y.J., 2005). In 1996 CCST was replaced by the Ministers' Committee for Science and Technology (MCST) chaired by the Minister of Finance and Economy in order to achieve more effective coordination (Ko, Y.J., 2005).

In February 1998, MOST was upgraded to Cabinet level and mandated to, 1) provide central direction, leadership, and coordination of all science and technology activities in the country and 2) formulate S&T policies, programs, and projects including technology cooperation and atomic energy in support of national development priorities (Ko, Y.J., 2005). The Minister also chaired MCST instead of the Minister of Finance and Economy. Nonetheless, the intention did not work because of the traditionally weak position of MOST. The National Science and Technology Council (NSTC), chaired by the president, was launched in January 1999 to prioritize and coordinate national R&D programs and policies (Ko, Y.J., 2005). The Minister of Science and Technology was made the secretariat of NSTC, and KISTEP was created under MOST to evaluate national R&D programs as well as MOST's role of allocating and managing R&D programs.

Throughout the 1990s, GSRIIs were created or replaced under many departments in line with the increasing volume of R&D programs, increase in the number of responsible departments in specific and strategic areas, and growth of pan-department R&D programs (Ko, Y.J., 2005). Then in 1999, they were regrouped into new Research Councils in order to separate them from the bureaucratic culture that inevitably stemmed from their supervision by specific departments and to use them beyond departmental boundaries and missions (Ko, Y.J., 2005). The role of the Boards of Directors of GSRIIs was transferred

to the Boards of the relevant Research Councils, which have a wider range of responsibilities for matters related to GSRI. The main characteristics of the governance structure in the 1990s were to move toward a more coordinated approach with dynamic debates and more involvement of actors in policymaking and implementation process (Ko, Y.J., 2005).

c) Funding

The dual funding system for GSRI by the annual grant-in-aid and R&D program evaluation was continued until the Project-Based System (PBS) was introduced in January 1996. The PBS aimed to abolish the Grant-in-aid, which had been given in a sense regardless of R&D activities, and to introduce the new system of funding based on full-cost pricing for R&D including the salaries for staff participating in R&D projects and operation of GSRI with open competitive tendering for R&D projects.

After the establishment of the Research Councils, the funding system based on PBS was continued and spread to other departments. PBS, in a sense, brought about a highly unstable funding system and too much open competitive contracting. To remedy the situation of the least stable GSRI that lacked institutional funding, institutional core R&D projects allocated by EPB to GSRI in a project-based system but with no competition introduced (Ko, Y.J., 2005). One of the main issues has been the ratio of institutional core R&D projects to the total R&D contracts. As of 2003, the proportion of institutional core R&D projects averaged around 40%(Ko, Y.J., 2005). From a critical view, PBS seems to have failed and to have been distorted by debates and different practices between departments (Ko, Y.J., 2005).

d) Management

Much of the reviews of GSRI in the 1990s focused on management agenda. With regard to management, regardless of level, policy, programs, and institutions, laws were used in the 1990s as important instruments to drive the government policy goals (Ko, Y.J., 2005). For example, the “Special Law for S&T Innovation” was enacted in 1997 to specify several policy actions (OECD, 2000), notably: i) the creation of the National S&T Council to

improve the efficiency of government R&D activities through inter-ministerial coordination of R&D policy and investment; ii) the formulation and implementation of the five-year plan for S&T Innovation, which contains specific programs for R&D and proposals for human resource development and building of S&T infrastructure; and iii) the increase in government R&D investment to five percent of the total budget to support the programs proposed by the plan. In addition, the Five-year Plan for S&T Innovation was launched in 1997, containing specific plans for action to achieve those policy objectives (Ko, Y.J., 2005).

The Act on the Establishment, Management, and Promotion of Government-supported Research Institutes was passed to create Research Councils as well as to regroup GSRI and set up their role and missions (Ko, Y.J., 2005). The Fundamental Act for Science and Technology was enacted in 2001, replacing the Special Law that was in force for a limited period for five years (Ko, Y.J., 2005). The Act assigned a more influential role to NSTC and required government policy for science, technology, and innovation to have a long-term perspective and an overriding strategy to be established on a five-year basis. The Five-Year Plan for Science and Technology, which was formulated in 2002 by all government departments concerned, was related to the development of R&D based on the Act and was updated in 2003 by the new government inaugurated in February that year. The main traits of the management in GSRI were multiple levels of supervisory bodies, highly unstable and competitive R&D, and more coordinated and strategic methods of management and seemingly greater autonomy but still dominant bureaucratic culture *de facto*.

The annual evaluation of institutions, which was introduced in 1991 but changed to a dual system with annual self-evaluation and three-year term evaluation, has been used for performance-related budget allocation by Economic Planning Board (EPB), and the results of project evaluation have been utilized by NSTC to prioritize and coordinate departmental R&D programs under the supervision of the Research Councils; the autonomy of GSRI has increased particularly in terms of the discretionary power to change rules and statements on personnel, payments, accounting, and internal units for R&D. Still, the autonomy has been limited in extent particularly because of the financial pressure and control by the MPB and departments.

[Table 4-2] Changes in GSRI around the Establishments of Research Councils

	1960s – 1970s	1980s	1990s
Objectives	Support Industry National Security	Indigenous Technology	Innovation
Strategies	Tech Import & Imitation Government- supported Public R&D	Technology Push Demand Pull Government R&D Industry R&D	System & Networks S&T Push Industry-led R&D Coordination Collaboration
Governance Structure	NCST /MOST A Few Departments N/PRIs & GSRI Scientist-centered	MOST-GSRI Policymaker-centered	NSTC/MOST Departments – GSRI RCs/MOST – GSRI
Funding	Mainly Contractual R&D from Industry Basic Funds for Facilities & Salary	Dual Funding: Grant-in-aid & R&D Contracts	PBS Competitive Tendering
Management	Autonomy Independent within Research Unit-centered	Highly Bureaucratic Rules and Guidelines Institution-centered Internal Management	Performance-oriented Evaluation-centered Accountability Autonomy & Responsibility
Theoretical Rationales	Reverse Linear Model Market Failure	Reverse Linear Model Market Failure	Government Failure System & Network Failure

Source: (Ko, Y.J., 2005)

1.2 Evolution of Research Councils of Science & Technology Since their Establishments

1.2.1 Changes from 3 RCs to 1 RC for 20 years

Those three Research Councils in the fields of science and technology have been evolving over time through two RCs in 2008 to one RC in 2014. The GSRI under the Korea Research Council of Fundamental S&T had focused on developing fundamental or basic technology with relatively higher level of institutional funding. The GSRI belonging to the Korea Research Council of Public S&T had conducted more public good research or big science based on national demands such as space, energy, and natural resources. Other GSRI under the Korea Research Council for Industrial S&T had developed industrial technology with more involvement of industry or technology commercialization but lower level of institutional funding from the Government.

The GSRI of RCs had enjoyed higher level of managerial autonomy under the supervision of the Prime Minister's office where a few civil servants had worked on supporting RCs and GSRI (Ko, Y.J., 2005). They had no capability to manage or intervene in managerial issues and any policy change in GSRI and RCs.

There were some issues of high autonomy at that time. Their missions and the objectives of GSRI and distinction between RCs were unclear. There were some arguments as to what is public technology. Almost all GSRI had conducted all kinds of research such as basic, fundamental, and industrial technology. It was not that clear what were the criteria to distinguish between RCs. Additionally, the size of each RC was too small to support and manage GSRI. All of the GSRI had entered the highly competitive R&D market to get government projects under the new PBS regime.

Even though the Ministry of Science and Technology had gotten more power in the Kim Dae-jung government compared to the previous one, it still had not been able to coordinate the government R&D budget and to use GSRI as appropriate policy tools for solving many issues across government departments. Neither were there collaborations with the two RCs in the fields of humanity and socio-economy.

In order to solve those issues, the Roh Moo-hyun government put MOST in a new place in the position of Vice Prime Minister and transferred three S&T RCs and GSRI to MOST. The mission of MOST was expanded to microeconomic policy as well as S&T policy. MOST opened jobs of high position to some scientists from GSRI and some jobs to other department officials and reflected their opinions to S&T policy and budget coordination.

Those changes enabled better coordination of government R&D budget across departments and better use of GSRI as S&T policy tools. Despite such positive changes, however, there were also some issues that needed to be solved. The distinction issue between RCs was not solved. Each RC was still too small. The level of average institutional funding for GSRI was less than 40%. RCs had no power to increase their institutional funding. The autonomy of GSRI had decreased at that time because MOST has many officials who had the capability to intervene in managerial issues in GSRI. Collaborations among GSRI and with university and industry were not sufficient due mainly to competitive relations.

In 2008, the Lee Myung-bak government reorganized the government departments. MOST got merged with the Ministry of Education and became the Ministry of Education and Science and Technology (MEST). The Ministry of Industry and Trade was renamed Ministry of Knowledge Economy (MKE). The government also changed three RCs into two RCs, merging the Korea Research Council of Public S&T into other two RCs. The supervisory authority for two RCs was switched. The Korea Research Council of Fundamental S&T was transferred to MEST, and the Korea Research Council of Industrial S&T went to MKE.

MEST, where 13 GSRI belonged, tried to merge the Korea Research Institute of Bioscience and Biotechnology (KRIBB) with KAIST, emphasizing the integration of research and education and effective management of GSRI by university like the USA system. Had the MEST plan succeeded, a few GSRI would have been transferred to the relevant universities. Such kinds of plan failed due to the many arguments and conflicts among stakeholders. MKE, which also supervised 13 GSRI, tried to merge 12 GSRI into just one GSRI like AIST in Japan and the Research Council system of Germany and also decided to privatize the Korea Institute of Toxicology (KIT).

All of these plans did not succeed due to too many arguments and inappropriate short reviews by a global consulting company in a top-down way. In addition, in the case of KIT, the market condition was not sufficient, so no companies decided to buy it. After such kinds of conflicting experience and trial and error, public demand for the establishment of a new and higher level of governing body of GSRI had increased. MEST and MKE did not collaborate, and the National Council of S&T (NCST), which was established in January 1999, was too weak to coordinate them. It was also very difficult for GSRI under the two RCs each belonging to MEST and MKE to collaborate with each other. Thus, the Lee Myung-bak government changed the position and role of NCST from being a mere government committee into a standing administrative body with much more officials and experts in March 2011. GSRI wanted to move to the new NCST to collaborate more effectively, but such was not implemented.

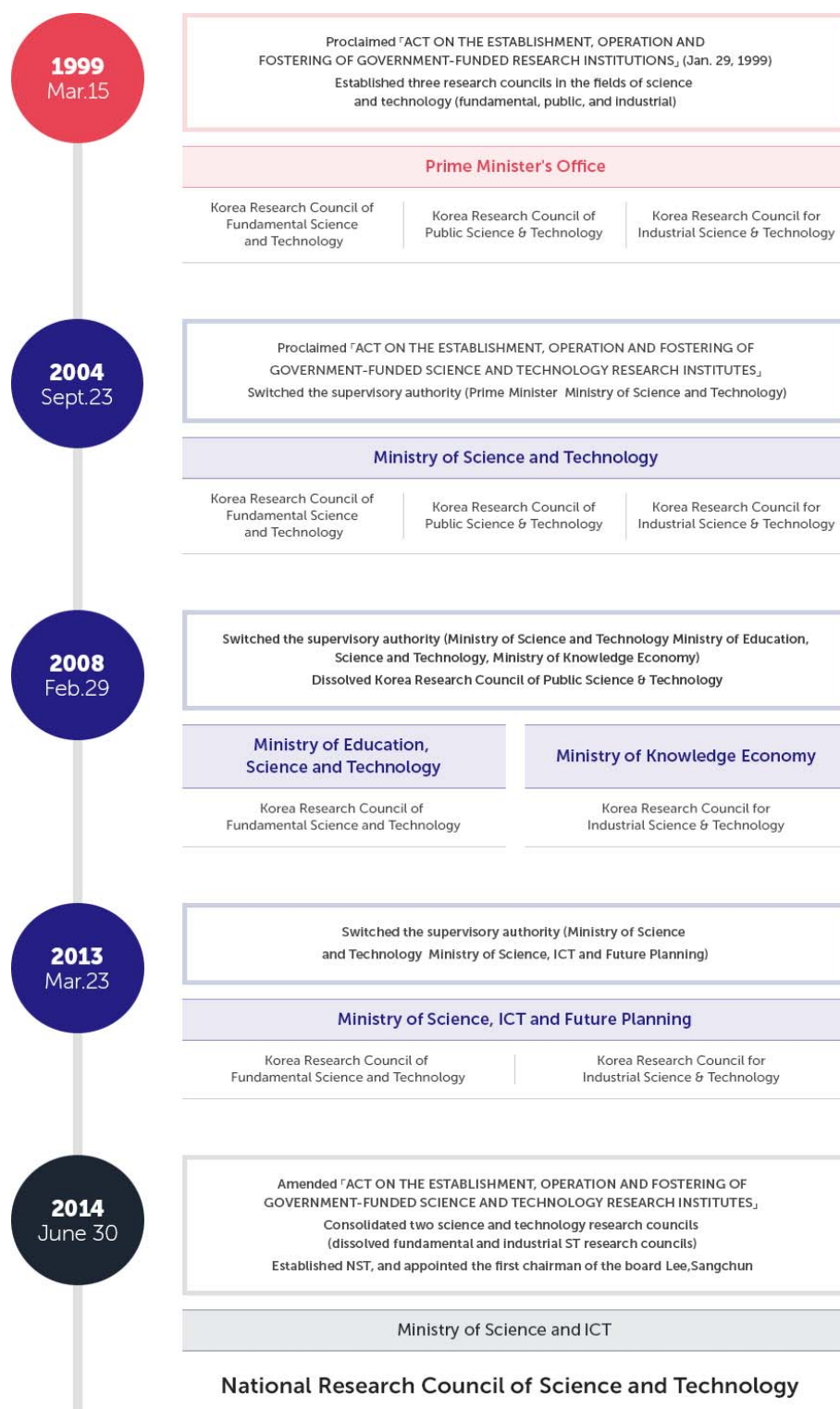
The Park Geun-hye government, which took office in February 2013, rearranged MEST into the Ministry of Education and the Ministry of Science, ICT, and Future Planning (MSIP). MSIP was set up as a top S&T policymaker, taking over the roles of NCST, which was abolished at the same time. A new experts' committee, the National S&T Committee (NSTC), was established to review and coordinate the R&D budget in keeping with the role of MSIP. The government also set up a new Division for S&T Strategy to produce S&T policy and R&D investment in a more strategic way.

Most of the GSRI and two RCs moved into MSIP for more collaboration and to develop convergent technology among GSRI under the harmonized supervision by a sole government department. It had not been easy for GSRI to make their relations closer and carry out more collaboration due to the long-time competition in the PBS market without good programs or projects for collaboration and two separate RCs. At that time, a strong policy trend was to create more convergence among different technological areas to lead technology innovation trend, instead of simply following those technologies developed in advanced countries. Two issues were then raised by many stakeholders: merger of two RCs, and merger of GSRI. There were a lot of discussions as to whether those issues could be feasible for nearly one year. In the end, the National Research Council of Science and Technology (NST) was established in June 2014, demolishing two RCs. 25 GSRI belonged to

NST. A new history of GSRI and RCs started in 2014 with a new convergent program model.

Since July 2017, the Moon Jae-in government has kept such governance except the new establishment of the Ministry of Science and ICT (MSIT) where NST belongs. In addition to two ministers, MSIT has another important vice minister for science, technology, and innovation as a position newly created by the Moon government. In addition, the Moon government merged NSTC into the Presidential Advisory Council on S&T (PACST) and newly established a Presidential Committee on the 4th Industrial Revolution (PCIR). MSIT has an important role of supporting those two bodies through the Vice Minister for science, technology, and innovation.

[Figure 4-1] History of NST



1.2.2 Current Status of NST

a) Mission and Objective Strategy

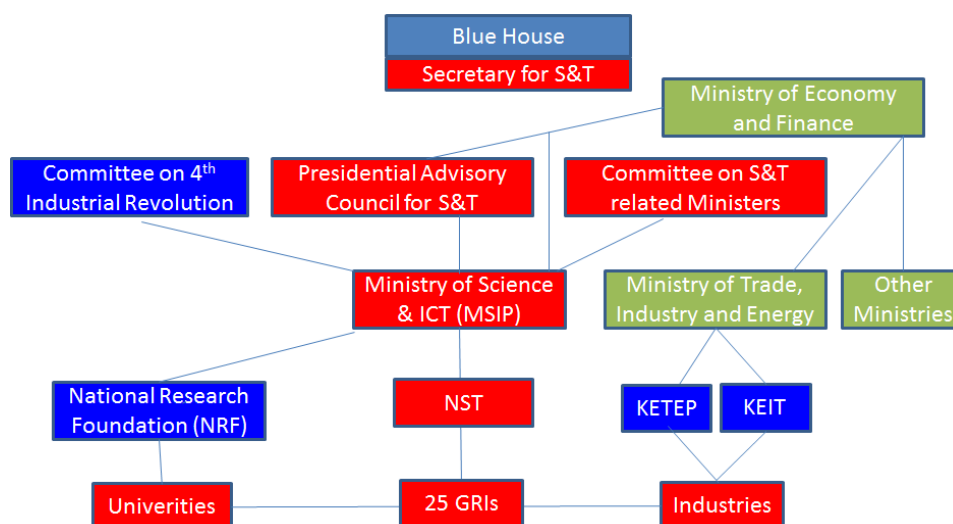
The vision of NST is to develop an open collaboration ecosystem for 25 GSRI, and its mission is to contribute to national R&D policymaking and S&T development including convergent technology by supporting and fostering GSRI. The main roles and strategy for realizing those vision and mission are to:

- Establish a supporting and fostering system for GSRI;
- Facilitate connections between GSRI and industry;
- Develop convergence technology between GSRI. The main functions are to plan policy and research for the development of GSRI, to adjust and align the functions of GSRI, to evaluate the research performance and management details of GSRI, to support cooperative research between GSRI, to improve research performance and to help spread the results, and to propose policy for innovation in national S&T and for strengthening competitiveness.

b) Governance

Since 2017, NST has been under MSIT, which set up a new division to manage and support NST and GSRI under the 1st Vice Minister, not the Vice Minister for S, T, and I.

[Figure 4-2] Expanded Governance of NST

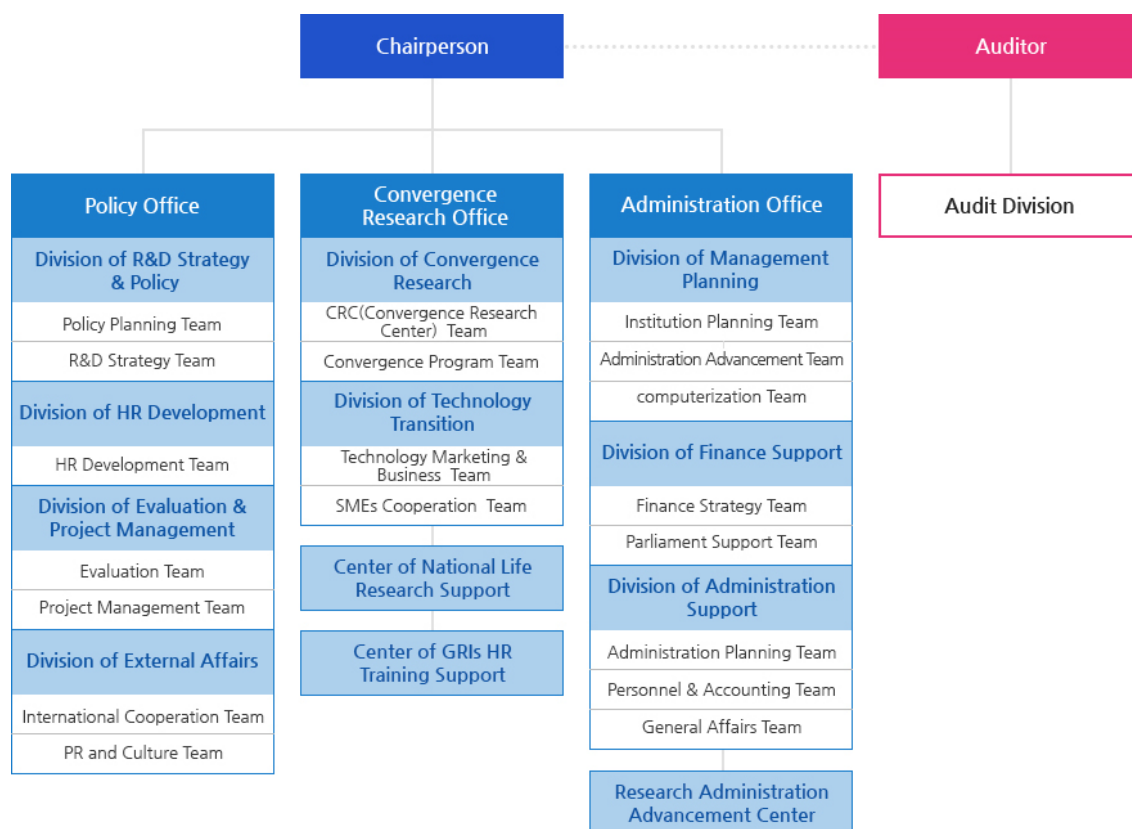


NST has three offices for internal works: policy office, convergence research office, and administrative office. The policy office covers the design of R&D strategy and policy for GSRI, human resources development, evaluation and grant-in-aid project management, and public relations and international affairs. The convergence research office handles convergence research, which was newly developed by NST as a convergent research model among GSRI, technology transfer and support for SMEs, development of living lab-style research to solve social and regional problems affecting the daily life of people, and young students training in preparation for the 4th industrial revolution. Lastly, the administrative office supports all kinds of administration matters of NST.

First of all, NST plays a crucial role to support GSRI through the Board of directors, which reviews and resolves the key pending issues of NST and member GSRI as the highest decision-making body regarding the operation of NST and GSRI. The president of NST is the chairman of the Board, with three fixed positions from Vice Minister I of Science and ICT, Vice Minister II of Strategy and Finance, and Vice Minister I of Trade, Industry, and Energy (MOTIE) and a circulated position every 1 year with two vice ministers among the Vice Minister of Education, Vice Minister of Agriculture, Food, and Rural Affairs, Vice Minister of Health and Welfare, and Vice Minister of Land, Infrastructure, and Transport. In addition, the Board has nine appointed positions from the academe, industry, and public sectors. All positions of the Board of Directors have a term of three years.

Additionally, NST operates two advisory bodies such as Planning and Evaluation Committee, which has 14 experts with three-year term, and a Committee for Management and Coordination consisting of the chairman of the Board of Directors, 3 office directors, and 25 presidents of GSRI. The latter committee plays an important advisory role on the major policy decisions of NST, major policy findings of NST, pending issues of GSRI, and discussion about the common agenda of GSRI and so on.

[Figure 4-3] Internal Organization of NST



※3 Offices, 10 Divisions, 18 Teams, 3 Centers

c) Funding

The annual budget of the 25 GSRI of NST is around 5 billion dollars. The institutional funding, as grant-in-aid, accounts for an average of 42% and comes from the Ministry of Economy and Finance (MOEF), with a limited number of researchers and staff decided annually. The PBS projects from government departments take up more than 50%. At least 5% of the budget comes from other sources including industry and self-earning finance. The institutional budget has to be submitted by GSRI to the NST Board of Directors and to MSIT. This goes to MOEF through the reviews of PACST at the end of June. MOEF submits the whole annual government budget including R&D investment to the National Assembly, which then decides the budget until the end of December. After finishing the Assembly process, MOEF evenly distributes the institutional funding to the GSRI every quarter.

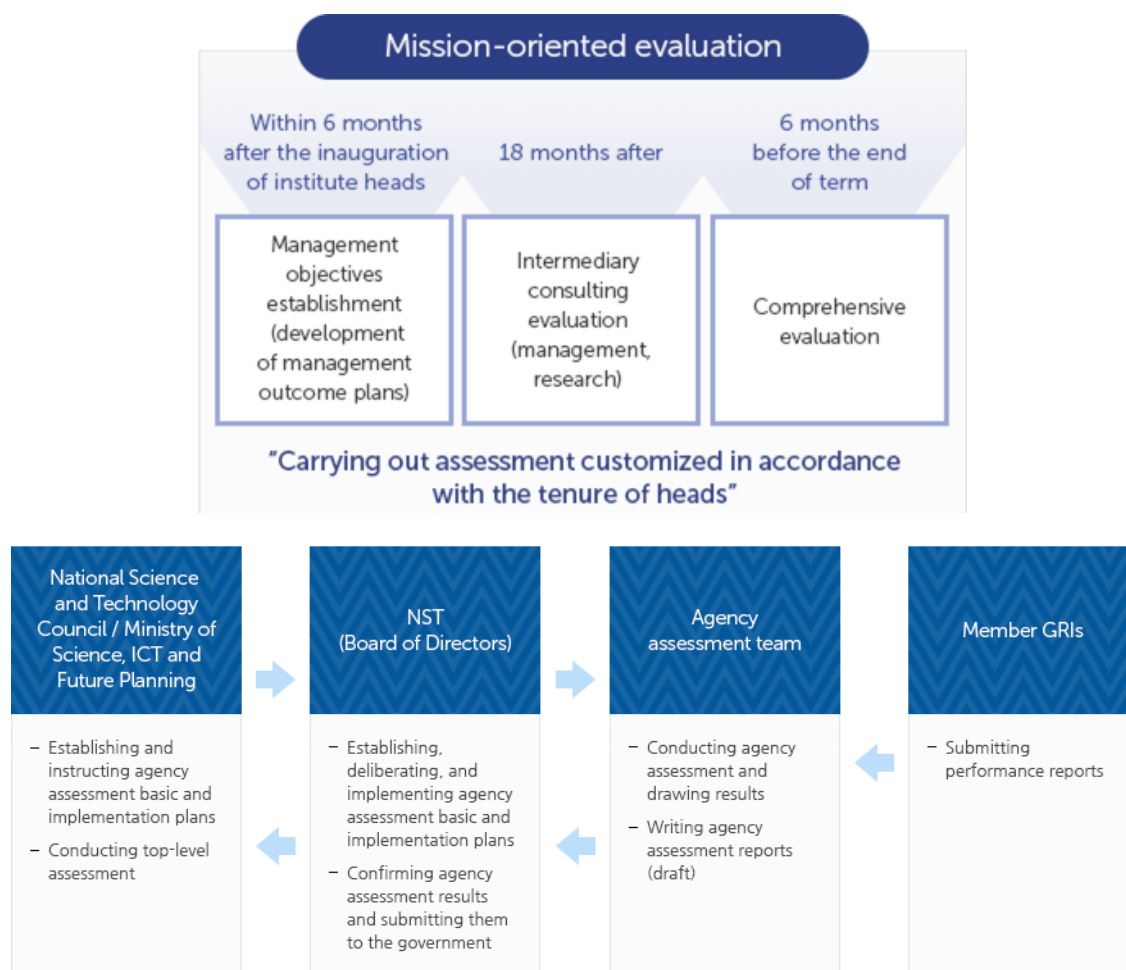
Other PBS projects from departments and industry depend on the programs anytime for a year.

NST has around 10 million dollars in self-funding budget for convergent research among GSRI as a new model for GSRI's research. All researchers have to conduct their research on the same site where the project manager's organization is located, with matching funds of a particular amount from all GSRI where all participants belong. Convergent research programs have several types as follows: future leading-type research for 3+3 years with 1 million dollars a year, commercialization-type research for 3 years, creative convergent research with several hundreds of thousands dollars, and those exclusive for network projects with 50,000 dollars. Except the last network project, all types of researches need matching fund from the participant GSRI.

d) Management

NST manages the comprehensive performance of GSRI every three years in line with the three-year term of the presidents of GSRI; mid-term consulting-style evaluation for management is also conducted. NST has to report to MSIT for a second round of meta-evaluation by a newly developed committee. The final results of evaluation are reflected to the annual salary of the GSRI president.

[Figure 4-4] Evaluation Scheme



NST also supports 61 regional branches and 12 international offices of 25 GSRI. There are a committee for regional innovation based on regional branches and a global task force for international collaboration using the international offices in Asia, Europe, and the USA. MSIT is going to establish a Korea-ASEAN Center soon in 2019 for more systemic and sustainable collaboration with ASEAN countries, probably related to NST.

NST has more than 100 staff - with experts dispatched from GSRI and public institutes - and operates issue-by ad-hoc committees. 25 GSRI have around 20,000 researchers and staff consisting of 80% of researchers and less than 20% of administrative staff, with around 500 post-doctoral scientists and 2,000 research students including master's and

PhD students of the University of Science and Technology established in 2003 by all of the GSRIIs.

[Figure 4-5] 25 GSRIIs

Fundamental S&T (11 GSRIIs)	Applied S&T (14 GSRIIs)
 Korea Institute of Science and Technology  Green Technology Center-Korea  Korea Basic Science Institute  National Fusion Research Institute  Korea Astronomy and Space Science Institute  Korea Research Institute of Bioscience and Biotechnology  Korea Institute of Science and Technology Information  Korea Institute of Oriental Medicine  Korea Research Institute of Standards and Science  Korea Aerospace Research Institute  Korea Atomic Energy Research Institute	 Korea Institute of Industrial Technology  Electronics and Telecommunications Research Institute  National Security Research Institute  Korea Institute of Civil Engineering and Building Technology  Korea Railroad Research Institute  Korea Food Research Institute  World Institute of Kimchi  Korea Institute of Geoscience and Mineral Resources  Korea Institute of Machinery and Materials  Korea Institute of Materials Science  Korea Institute of Energy Research  Korea Electrotechnology Research Institute  Korea Research Institute of Chemical Technology  Korea Institute of Toxicology

1.2.3 Major policy issues related to GSRIIs and NST

a) Knowledge demand

It has been estimated that knowledge demand has taken place in those knowledge-intensive areas such as energy, new growth engines, and quality of life. Korean companies have also increased their investment in those areas, desperately searching for highly talented HRST and knowledge for innovation as a sustainable way of adapting to societal challenges.

All of the ministries associated with plans have implemented them specifically. For example, 23 government departments have unveiled together the implementation plan based on the 4th five-year S&T Basic Plan. Despite the concerted efforts to produce those Plans by many government departments, the implementation process has not been

effective because of too many small projects scattered to internal divisions inside the Ministries and redundant review processes, which usually result from excessively competitive tendering system with little differentiation between industry, university, and public sector research institutes. GSRI have been affected by such policy environment of knowledge demand.

b) Knowledge circulation

Out of all the national R&D projects funded by the government, the proportion of collaborative R&D projects between industry, public research organizations, and universities has increased with the trends of technology convergence. The Korean government has tried to enhance collaborative R&D in the areas of new growth engines as well.

Knowledge circulation within the knowledge triangle has taken place thanks to private companies' funding for collaborative research with public research institutes and university. The level of investment has been very low, even decreasing since 2000 because Korea's big companies tended to invest in an internally closed form of R&D rather than open innovation.

The Korean government has also been forced to conduct collaborative R&D in GSRI under the so-called project base system (PBS) with low level of institutional funding of 42% on average in 2018. As of early 2019, more than 70% of R&D projects in GSRI were collaborative research with industry and university. The balance between the development of original technology and simple support for solving technological problems in SMEs has not yet been established.

Those policy instruments have resulted in superficially strong collaboration with industry and universities and focus of majority of R&D in GSRI on that of the developmental stage rather than on basic science and technology. The proportion of technology transfer produced by GSRI and universities out of the total R&D investment in them has remained lower than that of advanced countries. MOTIE produces a plan for promoting technology transfer and commercialization every three years instead of the previous five years, based

on the Law for the Promotion of Technology Transfer and Commercialization. The 1st Plan was introduced over the period 2000-2005 focusing on creating a technology market, and the 2nd Plan was implemented focusing on enlarging infrastructure for technology transfer and commercialization over the period 2006-2008.

MOTIE has been implementing the 3rd Plan, which consists of five core tasks: strengthening the finding and good management of national technology resources; establishment and expansion of technology finance system; establishment of the system for the total process in R&D and commercialization; support for entry into the global market by private companies; and expansion of infrastructure for technology transfer and commercialization. The Technology Licensing Office (TLO) system for universities and GSRI was introduced in 2006 with the objective of promoting and facilitating university and GSRI-based technology commercialization. The importance of utilizing research outcomes is deemed as important as technology development per se these days, so TLO's role in putting the research outcomes of universities and GSRI into commercialization is critical. The government's first-stage program to support leading TLOs in universities and GSRI has continued since 2006.

In order to enhance the technology transfer and commercialization from GSRI and universities to the market, the Korean government also introduced a Law for R&D Special Zone in 2005 wherein GSRI can spin off companies more easily and a technology holding company in the university is also promoted by the Law for the Promotion of Collaboration between industry and university. Since its commencement in 2006, the number of GSRI's companies has increased to more than 300 as of early 2019. The technology holding companies in universities have also been spreading thanks to the easing of regulations and government support such as financial support with the establishment of a new venture capital. Nonetheless, the criticism that basic research in GSRI and universities has been decreasing has been emerging because of too much emphasis on the entrepreneurial culture in the GSRI and universities through policy instrumental drivers.

The cluster development policy was implemented in part to promote science and technology innovation, particularly in state-of-the-art industries and SMEs through

knowledge circulation. This policy in part aims to promote balanced economic growth that will make a contribution to tackling regional disparities in the nation through the establishment of regional innovation systems. The cluster development policies are based on the “Special Law for Balanced National Development,” which was devised in 2003, the “3rd Comprehensive Plan for Promoting Regional S&T 2013-2018” laid out in 2013, and the “First Five-Year Balanced National Development Plan” devised in 2005.

The establishment of innovation clusters in regions involves: 1) creating clusters based on networks among major innovation players in local regions; 2) designating seven pilot clusters with the aim of linking existing industrial clusters with R&D centers, such as the Ulsan Automotive Components Cluster, Changwon Advanced Appliance Cluster, Gwangju Photonics Industry Cluster, Gumi Digital Electronics Industry Cluster, Wonju Advanced Medical Industry Cluster, Gunsan Automobile Appliance Components Cluster, and Banwol Sihwa Advanced Component Material Cluster; and 3) designating the "Daeduck R&D Special Zone" for expediting the commercialization of research outcomes of the Daeduck R&D Complex. The Daeduck R&D Special Zone expanded to Daegu, Gwangu, Busan, and Jonbuk at the end of 2017. The evaluation on how successful those clusters have been has yet to be conducted. That is why NST has tried to connect all of the regional branches of GSRI to contribute to regional innovation through connections with regional clusters and programs by local governments.

The other pending issues involving NST and GSRI are as follows:

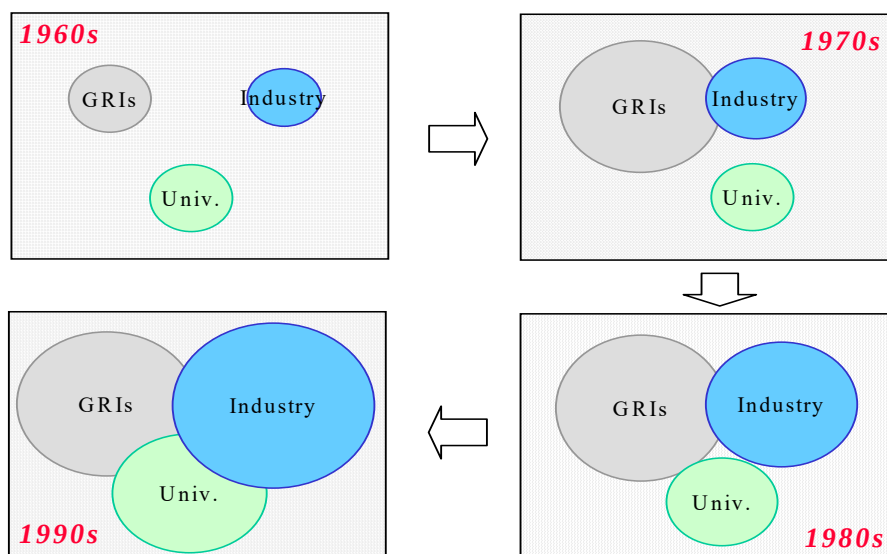
- Need for review on the 20-year evolutionary development of NST;
- How to pursue further collaboration and convergence between GSRI;
- What are the major Roles and Responsibilities in GSRI;
- How to make more connections and contributions to society and people;
- What is the appropriate volume of institutional funding for GSRI;
- How to contribute more to regional innovation and balanced growth;
- How to lead the so-called 4th Industrial Revolution paradigm;
- What would be the major Roles and Responsibility of NST.

2. Sharing Korea's Experience in STI Policy and Planning - Focusing on the 4th S&T Basic Plan 2018-2022

2.1 Introduction: Governance and Coordination in the Korean S&T Policy

The Korean innovation system has continued to evolve in terms of the major innovative role players since the 1960s. The Korean pathway can be summarized as follows: first, Government-funded Research Institutes ("GRIs")⁶ were established in the 1960s and 1970s; second, industrial activities for R&D were intensified in the 1980s, with GRIs playing a supporting role for the development of industrial technologies; third, measures for the development of research capability of universities were strengthened in the 1990s. Since the 2000s, the three players have developed in a balanced manner, and Korea is changing from focusing on benchmarking developed countries to becoming a pioneer that must create its own development models.

[Figure 4-6] Evolution of Major Players in the Korean Innovation System



6 In fact, the "GRI" is equal to "GSRI." There is no single official term so that the use of GRI or GSRI depends on the author.

The pathway was a strategic choice of the Korean government. As a result of the development, the government is required to institutionalize new mechanisms of policy coordination that can meet complex demands from the industrial and academic players and streamline the individual activities of many S&T-related ministries. The need for cross-government policy coordination has begun to grow from the early 1990s when most of the S&T-related ministries started to operate their own R&D programs. This is because the uncoordinated R&D programs of each ministry led to problems of duplicate investment and budget wastage.

In addition, the development of universities since the 1990s has required a new definition in the role sharing between emerging university research and existing public research. Public research -- which exists between university research and industrial research -- is mainly conducted by GRIs in Korea. The question of what should be the roles and responsibilities of these GRIs has always been an important issue in S&T governance since the 1990s.

2.2 Structure of S&T Governance

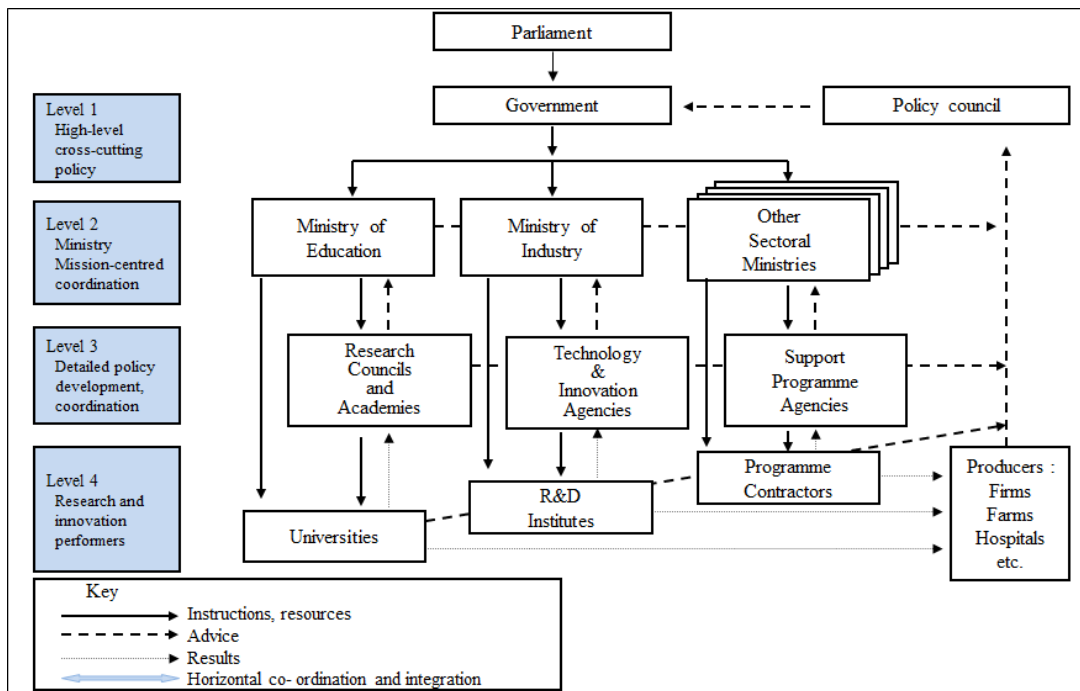
One of the major characteristics of S&T policy is that it is related not to a particular ministry but to many ministries carrying out various research programs and supportive measures. Many ministries and institutions are involved in the formulation, implementation, and evaluation of R&D programs, thereby giving rise to the problem of effective coordination among them.

Policy coordination to prevent program duplication between S&T-related ministries and to set priorities or guidelines for inter-ministerial resource allocation can be achieved in various ways and at different levels in the policy process of the government.

<Figure 4-7> shows the structure of S&T governance divided into four levels. Being a demonstration of an ideal type, not all administrative structures in different countries would have the same form. In this scheme, however, we can find a useful idea wherein S&T policy coordination takes place at four different levels.

- **Level 1** is the highest level. This involves setting inter-ministerial, overall directions, and priorities across the entire system of national innovation. It may be achieved through advice to ministries or through more binding means, such as budget allocation and program performance evaluation.
- **Level 2** is coordination among ministries whose sectoral responsibilities otherwise encourage them to pursue independent policies. In practice, this level of coordination may involve administrative aspects, policy issues, or both. Sometimes, an inter-ministerial group also functions as a Level 1 coordination mechanism.
- **Level 3** is more operational, in an attempt to make the actions of funding agencies into a coherent whole. This level, too, can involve administrative coordination as well as more substantive coordination of funding activities, such as co-programming.
- **Level 4** involves coordination among those that actually conduct research and innovation activities. Effective coordination at this level tends to be achieved through professional self-associations rather than external control.

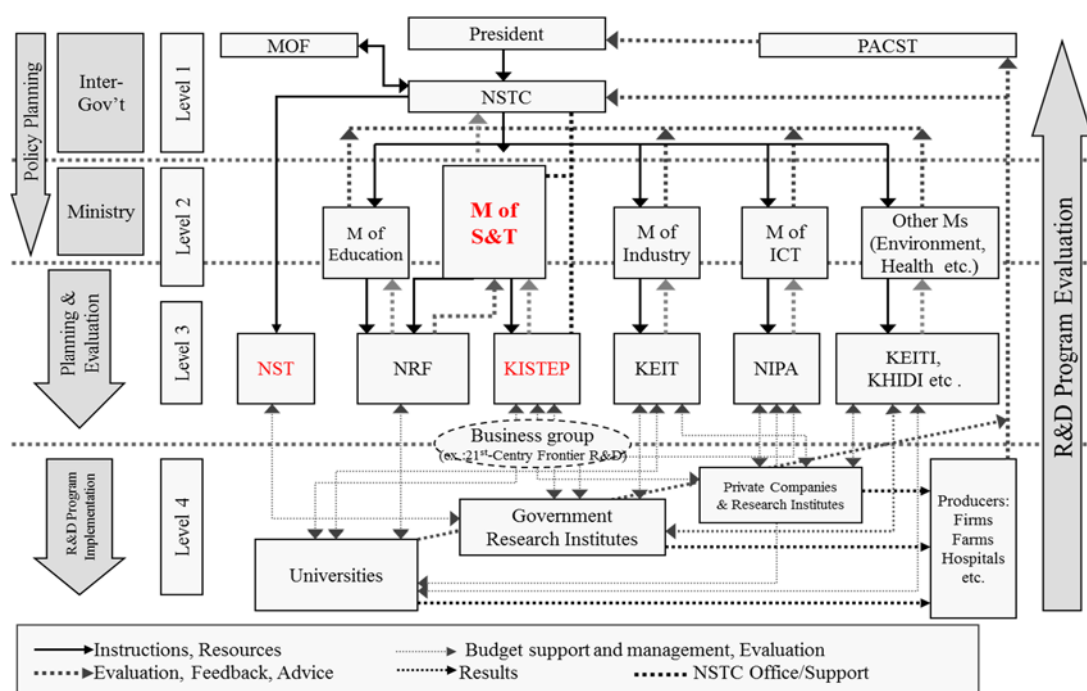
[Figure 4-7] Structure of S&T Governance



2.3 Structure of S&T Governance in Korea

Applying the previous diagram of S&T governance to the Korean science and technology system, we can draw the structure in <Figure 4-8>. At Level 1, the National Science and Technology Committee (“NSTC”) will focus on coordinating each ministry's policies, allocating budgets, and evaluating the ministry's policy performance.

[Figure 4-8] Overview of the Korean S&T Governance



At Level 2, individual ministries implement their own R&D programs. Among them, the Ministry of Science and Technology is responsible for overseeing national science and technology policy in terms of policy planning, resource allocation, and performance evaluations, with other ministries such as industry, education, and environment carrying out their own programs. Their activities are coordinated by NSTC at Level 1.

Level 3 is an R&D governing body supporting government ministries at Level 2. The staff at level 3 are not bureaucrats but doctoral academicians with expertise in their S&T fields. Their main duties are to support the ministries at Level 2 in policy planning, R&D program implementation and project evaluation, etc. As such, Level 3 plays the role of mediating

the scientific bureaucracy and academe.

Level 4 consists of research institutes and researchers who actually carry out R&D activities. It includes universities, public research institutes, and industrial research institutes.

With the discussion above in the background, <Figure 4-8> shows an overview of the major organizations from Level 1 to Level 3, including coordinating bodies at Level 1, S&T-related ministries at Level 2, and R&D management agencies at Level 3.

a) Cross-ministerial coordination institutions (Level 1)

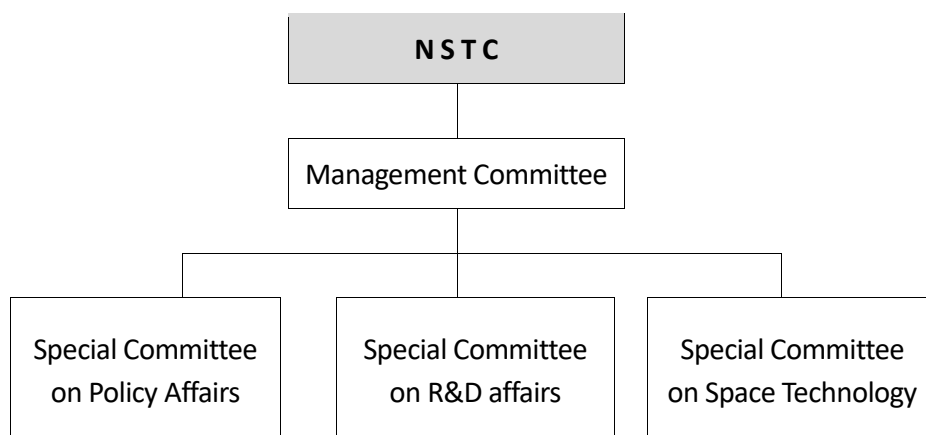
There are four government bodies that will play a role in resolving potential conflicts among the S&T-related ministries. First, there is NSTC, with the ministers of the S&T-related ministries as its members. Second is the Ministry of Finance and Economy ("MOFE"), which allocates R&D budgets to the ministries. Third, the Presidential Council on Science and Technology ("PACST") is a body of experts from outside government that advises the president on S&T affairs. Fourth is a President's Science and Technology Assistant at the office of the president (Blue House).

① National Science and Technology Council

The Korean government established NSTC in April 1999 in accordance with the Special Law for Scientific and Technological Innovation. Its mission is to review and coordinate national S&T policies, evaluate the implementation of R&D programs, and set principles of priority for the allocation of S&T budgets.

When NSTC was established in 1999, it was chaired by the president and composed of 15 ministers of S&T-related ministries and 3 representatives from the S&T community. The Ministry of Science and Technology ("MOST") has been a secretarial organization that supports the operation of NSTC. This means that NSTC is a critical governance framework for MOST to increase its influence.

[Figure 4-9] Structure of NSTC in Korea (early structure in 1999)



To back up the coordinating efforts efficiently, NSTC has one management committee, which in turn has three special sub-committees. The management committee is chaired by the minister of MOST and is composed of 15 vice ministers of related ministries and 8 civil experts. The 3 sub-committees each consist of about 30 directors and external experts from related ministries.

At the outset, NSTC is the coordinating body at the senior level that resolves conflicts among S&T ministries. In fact, it would be difficult for any ministerial-level committee to resolve differences among themselves without the help of a third party holistically overseeing the S&T governance. The chairmanship of NSTC by the president is expected to make achieving and effectively coordinating national S&T policies overall easier.

It should be mentioned that, in order to reach an agreement for substantially coordinated alternatives among conflicting ministries, an agenda should be discussed in advance and coordinated in the sub-committees before it is proposed to the NSTC meeting where the president sits as chairman. Since the NSTC meeting is held only three to four times a year, there is insufficient time to coordinate policies among ministries. Therefore, it is important that the sub-committees discuss inter-ministerial policy issues and prepare a consolidated agenda for the NSTC meeting. This means that sub-committee activities, rather than the NSTC meeting itself, are in effect more important in determining the performance of NSTC.

② Ministry of Finance and Economy

MOFE plays an important role in national S&T in two ways. First, the national S&T policy is designed and implemented in line with the national economic and social development policies as set by MOFE. Second, the budgetary control function of MOFE directly affects the scope and volume of ministerial investments in public R&D.

MOFE has acknowledged its lack of S&T expertise in allocating budget for S&T ministries. Therefore, it has shown a tendency to return to the ministries the responsibility for resolving conflicts and finding some kind of eclectic solutions. The position of MOFE is that it should set overall funding levels (total amount) for each ministry, and each ministry should be responsible for allocating budget in their individual programs within a given total amount in accordance with its respective set of priorities. Thus, individual ministries have a certain amount of discretion in their investment programs, which may give rise to the problem of redundant investment at the national level.

This is where MOST and MOEF can cooperate. Before the R&D budget for each ministry is finalized by MOFE, MOST makes a prior adjustment of the budget requested by S&T-related ministries. Then MOFE makes the final allocation to the ministries based on the result of preceding budget adjustment by MOST.

③ Presidential Advisory Council on Science & Technology

PACST enjoys a position that is similar to the president. Its chairman serves as science adviser to the president and occupies a full-time position in the government. In the current government, the council is composed of less than 30 members of PACST, including the President (Chairperson) or Vice Chairperson. All members are highly experienced and learned specialists from the public and private sectors, including the vice minister of science and technology as a permanent member. PACST receives secretarial support from MOST.

PACST has the function of advising the president on the following matters:

- To develop policy strategies for technology innovation, information, and human resources development

- To develop a system for the promotion of science and technology development
- To undertake special tasks requested by the president

PACST has a General Meeting, an Advisory Council, and a Deliberative Council. The General Meeting is composed of all council members of PACST who deliberate on general matters including the operation of PACST and matters that need the opinions of all council members in performing its duties. The Advisory Council performs the advisory function of PACST. It is composed of 13 civilian committee members including the President (Chairperson), Vice Chairperson, and secretaries as science and technology aides. It is assisted by 3 subcommittees: S&T infrastructure subcommittee, S&T innovation subcommittee, and S&T society subcommittee. The Deliberative Council performs the deliberative functions of PACST. It is composed of 11 civilian council members including the President (Chairperson), Vice Chairperson, and 5 government council members and aides for the secretariats for science and technology. It is supported by 4 subcommittees: operation committee, special committee, local S&T promotion council, and fundamental research promotion council.

In terms of the means of coordinating, PACST does not attempt to deal with internal government affairs such as resolving conflicts on the roles and missions of ministries at Level 2. Instead, it serves the president by preparing studies and briefings on background issues of importance and by helping the president interpret his S&T policies to the larger community of politics and society.

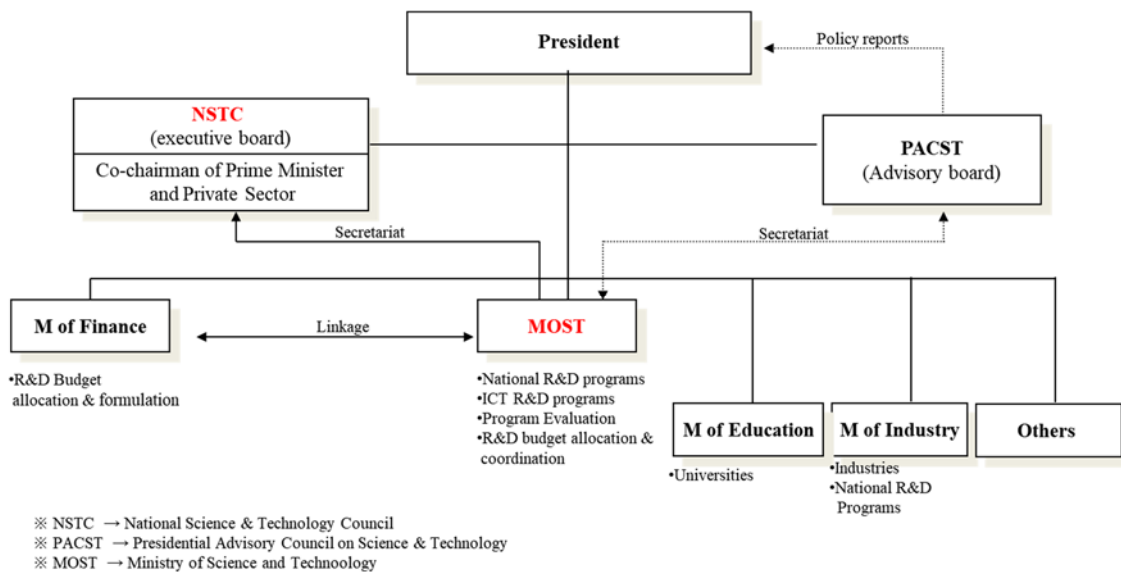
④ S&T advisory secretary to the president

The S&T advisory secretary to the president at the presidential office (Blue House) is one of the many government staff officers at Level 1. In the current government, the secretary plays the role of secretary member of PACST. This means that the S&T advisory secretary connects the President with the civilian and government council members of PACST. His or her main role is to bring various opinions from S&T ministries and expert communities to be reflected to the president's decision making.

b) S&T-related ministries (Level 2)

The figure below shows a simplified organization of government ministries related to science and technology in Korea. At Level 2, MOST, Ministry of Education, and Ministry of Industry are the main actors. In addition, there are many other ministries conducting their own R&D programs, each having R&D management agencies (Level 3) under their charge.

[Figure 4-10] S&T Government of Korea



① Ministry of Science and Technology

MOST has the responsibility of formulating national science and technology policies and plans. Since its establishment in 1967, MOST has acted as the main ministry for overseeing national S&T policy and administering affairs related to overall S&T policymaking. Its main functions are:

- To execute national programs for the development of core technologies, future-oriented technologies, big science, and multidisciplinary technologies, including aerospace, ocean, nuclear energy, etc.
- To support basic, applied R&D conducted by GRIs
- To predict the developmental trajectory in technology and to set up the basic policy for science and technology development and promotion
- To build a policy for R&D investment, human resources, information, and international cooperation in science and technology

- To promote public understanding of science and technology, etc.

Furthermore, its role has been strengthened to coordinate S&T-related ministries in terms of planning and implementing national R&D programs. MOST oversees and integrates the plans of other ministries for S&T development through two mechanisms. The first mechanism is NSTC, for which MOST serves a secretarial function. The second one relates to the budget allocation process. MOST evaluates major programs conducted by S&T-related ministries, with the results reflected to the budget allocation for the next fiscal year. Furthermore, the S&T-related ministries have to submit their request for R&D budget to MOST for review before the final adjustment made by the Ministry of Finance.

One of the controversies regarding the role of MOST is related to its duality of serving as both player and referee. In order to promote basic and applied technologies, MOST has had its own R&D programs that would need to have been evaluated in the same way as those of other ministries. On the other hand, MOST has been responsible for overall long-term planning for S&T and is authorized to coordinate the S&T programs of other ministries. The inherent conflict of interest in the function of MOST has drawn flak from other rival ministries. Such controversy has led to the establishment of NSTC in 1999.

② Ministry of Industry (MOI)

MOI is responsible for affairs on commerce, trade, industry, energy, and underground resources. Thus, one of MOI's main functions is to plan industrial policy and implement industrial R&D programs. In this light, MOI is also a major player in national S&T governance. The difference between MOST and MOI is that MOST's mission is associated with basic research and long-term science programs, whereas that of MOI focuses on applied research and industrial technology programs. MOI implements a technology policy with four aspects: development of strategic technologies; effective commercialization and deployment of new technologies; construction of the technology infrastructure; and promotion of international technological cooperation.

In the period of rapid economic growth in the 1980s and 1990s, MOI's industrial policy has made a significant contribution to the success of Korea's industrial development. The state

has targeted strategic industries and supported large companies in the selected sector. In response to the constraints on targeting specific industries imposed by GATT, however, the Korean industrial policy now emphasizes the building of technological infrastructure. MOI has embraced this new approach and developed industrial R&D programs including support for strategic technologies (not industries), infrastructures for R&D, and technology dissemination. This means that the focus of traditional industrial policy is shifting to the promotion of industrial technology, resulting in the convergence of industrial policy and technology policy. Its consequence was the policy competition between MOI and MOST in the early 1990s as to who should play the leading role in technology policy between them. Since then, MOST has shifted its focus to providing support for basic sciences and large-scale public sciences.

③ Ministry of Education

The Ministry of Education (MOE) is responsible for school education, lifelong education, and arts and sciences. Since it is in charge of the whole range of educational institutes, its influence on the development of human resources for S&T is substantial. In particular, MOE's policy on higher education institutions, colleges, and universities has a significant impact on the promotion of high-caliber S&T personnel. MOE also supports university research, which is managed by the National Research Foundation (NSF).

Brain Korea 21 (BK21) is one of the main programs designed to support university research. It provides financial support for graduate research and graduate students for a few selected centers of excellence (COE) based on the principle of excellence. The program overlapped the roles of MOE and MOST, thereby weakening MOST's influence on university research.

④ Other S&T-related ministries

In addition to MOE and MOI, about 20 ministries carry out their own R&D programs in Korea. Various ministries, such as environment, health & welfare, transportation, construction, and defense, carry out their own R&D programs and have R&D management agencies (Level 3) under their respective control. The activities of these ministries will be coordinated by NSTC at the upper level.

Among other ministries, the Ministry of Information and Communication (“MIC”) needs further explanation. The ministry had existed as an independent ministry until 2013 but was later abolished and merged with MOST. MIC has made a significant contribution to the development of information and communication technology in Korea. Through this, Korea could develop into an IT powerhouse in the world. As the IT industry expanded, however, the rationale for MIC to function as an independent ministry has weakened, and it was eventually integrated into MOST. The merger has given rise to an advantage, i.e., the combined MOST and MIC could secure the appropriate organizational size required to be a ministry in terms of the scope of works and number of staff because MOST and MIC each had a narrow range of work and a small number of employees.

c) R&D management agencies (Level 3)

One of the characteristics of Korea's science and technology administrative system is that major S&T-related ministries have their own R&D management agencies under their control. Therefore, there are specialized management bodies as many as the major S&T-related ministries. The most important agency is the Korea Institute of S&T Evaluation and Planning (KISTEP), which belongs to MOST. If we look into the activities of the R&D management agencies, focusing on KISTEP, it is known to be the only institute in Korea that encompasses all aspects of national S&T planning, coordination, and evaluation. Therefore, KISTEP's activities can explain the relationship between R&D management agencies and government bureaucracy in S&T. KISTEP basically performs five types of activities as follows:

① Foresight, S&T Policy Planning and Coordination

KISTEP performs activities such as future trend analysis, technology foresight studies, technology level evaluation, technology impact assessment, and development of the national standard S&T classification system in order to establish national S&T strategies.

- Foreseeing the scientific and technological outlook
- Supporting the national S&T strategy establishment
- Performing technology-level evaluation (every 2 years) and technology assessment (every year), establishing the national standard S&T classification system
- Policy planning for S&T
- Coordination of S&T policies
- Planning and running the operation system related to R&D programs

② Budget Allocation and Coordination of Government R&D Programs

KISTEP provides the outlays of the government's R&D budget allocation and coordination for strategic investment in government R&D programs, sets the directions and standards in overall government R&D investments and related budget deliberation, and evaluates the investment strategies in line with S&T policies and technology development status.

- Providing support for establishing the directions for effective government R&D investments and the standards of budget deliberation for government R&D programs
- Supporting budget allocation and its coordination of government R&D programs

③ Survey, Analysis, and Evaluation of Government R&D Programs

KISTEP conducts surveys and analyzes all R&D investments and performances in both public and private sectors, evaluates government R&D programs based on such surveys and analysis, further analyzes notable performances, and establishes effective measures to disseminate performance from government R&D programs.

- Surveying and analyzing government R&D programs and R&D activities in Korea
- Evaluating government R&D programs and GRIs
- Establishing the strategies for R&D performance dissemination and national intellectual property
- Publishing Composite S&T Innovation Index (COSTII)

④ Feasibility Study on Government R&D Programs

KISTEP performs feasibility studies on new large-scale government R&D program and conducts preliminary technological assessments to select suitable R&D programs that are qualified for feasibility study in order to verify the appropriateness of government investment prior to funding.

- Feasibility study on government R&D programs
- Preliminary technological assessment to select appropriate programs qualified for the feasibility study
- Research on improving the preliminary feasibility study system

⑤ Global Cooperation

KISTEP shares knowledge and experiences from national S&T-led innovation with other countries, makes various efforts to form the S&T network and S&T knowledge exchange hub with international communities, and collaborates with partner institutes.

- Establishing a global platform for cooperation and strengthening national S&T diplomatic capacities with developed and developing countries and international communities
- Sharing the experiences in national S&T-driven innovation in Korea with developing countries by providing S&T policy training programs and offering support to design their own S&T-guided master plans for building S&T innovation platforms
- Publishing S&T policy journals and magazines in English

The activities of KISTEP basically correspond to the activities of MOST. In this light, KISTEP can be regarded as a shadow ministry. MOST is a public organization composed of civil servants, whereas KISTEP is a private outfit made up of academicians most of whom are PhD holders. In the case of civil servants, their responsibilities are periodically changed by job rotations; since academic experts at KISTEP are free from such rotations, however, they can accumulate their expertise in the long run.

MOST is legally required to make the S&T plan every five years, and KISTEP's objective is to carry out the practical work of making the plan. In terms of S&T governance, the alignment of ministries at Levels 1 and 2 is important, but the planning process itself is mainly carried

out by agencies at Levels 2 and 3. For this reason, it is necessary to make an in-depth analysis on Level 3 agencies and their activities. With this fact in mind, the next chapter will describe the process of S&T planning in Korea's S&T policy, focusing on the 4th S&T basic plan.

2.4 Case study: 4th Science and Technology Basic Plan

2.4.1 Introduction

a) Nature of the plan

○ Significance

Mid- and long-term development strategy that presents the vision, goals, and direction of Korea's science and technology innovation policy for the next five years ('18 ~ '22).

Top-level plan that suggests the directions for establishing and promoting policies related to each ministry's science and technology.

○ Legal Basis

Article 7 (Science and Technology Master Plan) of the Framework Act on Science and Technology and Articles 3 ~ 5 of its Enforcement Decree;

- Establish and announce “mid- and long-term policy goals and directions” on science and technology development
- Establishment of the Basic Science and Technology Plan according to the mid- to long-term policy goals and directions

b) Direction of Planning

○ Presenting the future vision of science and technology in the long term

Analyzing the trends of change in global future society that we should pay attention to from the mid- to long-term perspective and the perspective of science and technology

- Analysis of the latest trends based on the Future Issues Analysis Report ('15) -- which analyzes the issues of the future society 10 years later -- and the 5th

Science and Technology Forecast Survey (17.3), which forecasts future changes until 2040.

Set a long-term vision up to 2040 and present specific goals for the future of science and technology.

- Revise and supplement the 2040 vision established in 2010 to reflect the environmental changes.

○ **Establishment of basic science and technology plan for the next five years according to the long-term vision**

Analyze our readiness to realize the long-term vision until 2040 and identify strategies and specific tasks to be pursued for the next five years.

Linking the long-term vision of science and technology with the five-year basic plan to enhance policy coherence and effectiveness and induce national capacities.

- Several long-term visions of science and technology have been established ('99, '06, '10), but they are not effective due to lack of linkage with the science and technology master plans.

○ **Induce the active participation of policy consumers, including citizens and researchers**

Establish an on-site plan by establishing an online channel to expand the participation of citizens and researchers in the basic science and technology planning process.

- Open Science and Technology Innovation Plus (www.scienceplus.kr), an online platform where users can freely suggest opinions on the science and technology policy.

2.4.2 S&T Policy Directions for the Future Vision

1) Focus on short-term performance and goals → R&D creating disruptive innovation
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a) Transition of R&D system from government-led to researcher-centered system

As technological innovation proceeds at a rapid pace, it is difficult to respond to environmental changes with the pursuit of excessively goal-oriented, short-term R&D

strategies.

- Since R&D has been led by the government, there is lack of investment led by researchers
- * Out of the 19.5 trillion-won government R&D in 2005, net R&D investment (excluding national defense, human resource development, and institutional operating expenses) was 10.2 trillion won. Among them, government-led R&D was 8.94 trillion won, and researcher-led R&D was 1.26 trillion won.

Need to transform government R&D systems into systems capable of disruptive innovation by enhancing researchers' autonomy, creativity, and zest for challenge.

- Switching from the technology supplier-focused & long time-fixed R&D system to the creative and frontier R&D system

b) Focusing on fostering creative and convergent human resources with problem-solving capabilities

As society advances with the development of intelligent information technology, problem-solving ability, critical thinking, and creativity that can create new areas and values are highlighted.

- Foster convergent human resources who will lead the 4th Industrial Revolution and actively seek out and support young researchers

2) Lack of convergence and cooperation → Creation of an active innovation ecosystem

a) Promote convergence and cooperation between actors and sectors

As the life cycle of the R&D cycle becomes shorter, independent technological innovations have shown limitations, but Korea lacks a culture of convergence and cooperation.

- A serious problem with "Not Invented Here syndrome," which refers to an exclusive organizational culture or attitude that does not recognize technology or research outcomes developed by third parties.
 - Boldly open the national R&D system, such as securing private R&D openness and establishing an organic industrial cooperation system, and transcend the growth limits

b) Creating an innovative ecosystem where challenges and growth are active

Quantitative indicators of entrepreneurship, such as the number of new corporations and venture firms, continue to rise, but lack of technologically innovative startups in terms of quality

- Opportunity-pursuing startups ('14): US 54.0%, Japan 46.0%, China 43.0%, Korea 21.0%
- Creating an entrepreneurship and venture ecosystem where outstanding talents with entrepreneurship actively challenge startups with an active “invest → recover” and “fail → rechallenge” cycle

Establish a system for public and local governments to participate actively in science and technology innovation

3) Lack of next growth engine → Accelerate new industry and job creation

a) Securing new growth engines to sustain the core industry

Korea has secured world-class industrial competitiveness based on the growth of science and technology, but the competitiveness of the flagship industry is weakening, and a next growth engine is lacking.

- * 1st in the world in semiconductors, 1st in displays, 3rd in information and communication devices ('15)
- ** Global market share of major exports: ('11) 5.7% → ('15) 5.3%

Adjusting investment portfolios and innovation ecosystems so that science and technology can play a leading role in driving innovation growth.

- Strengthen the strategic and innovation incentive system of R&D to foster promising industries through R&D.

b) Efforts to create jobs for science and technology and prepare for future job changes

To foster technologically innovative SMEs as a leader in economic development and job creation, and to create high-quality science and technology jobs.

Proactively prepare for changes in jobs in the future by predicting changes in the economic and social environments caused by technological developments.

4) Focus on economic growth → Expand contribution to improving the quality of life and solving human problems

a) Create practical results that people can feel in terms of quality of life

Nowadays, R&D projects aimed at improving the quality of life of the people are being promoted, but there is a limit to the actual performance creation.

- * Share of government R&D by economic and social purpose ('16): Economic development 42.7%, national defense 14%, health environment 12.7%
- * Ratio of product and service to social problem solving (Ministry of Science and ICT) and national benefit enhancement (Ministry of Industry) R&D is about 20%.
- Proactively respond to risk factors for improving the quality of life and future societies and expanding public participation in the R&D process and utilizing and spreading technology.

b) Expanding contribution to solving difficult social problems

Improving national status by playing a leading role in solving human problems such as disease, environment, climate change, and resource shortages.

2.4.3 Four Strategies

a) Expanding Science and Technology Capabilities for Future Challenges

- Reinventing support systems and management systems to enable researchers to conduct actively creative and challenging research that will lead to disruptive innovation
- Actively seek out and support outstanding talents so that they can be full of creative capabilities and ready to challenges that will lead the future society

b) Creation of innovation-friendly science and technology ecosystem

- Establish a system for creating excellent intellectual property that can lead the global market and quickly connecting research results to growth engines and commercialization
- Support R&D capacity building so that ventures, SMEs, and regions can play a leading role in innovation growth

c) Creation of new industries and jobs led by science and technology

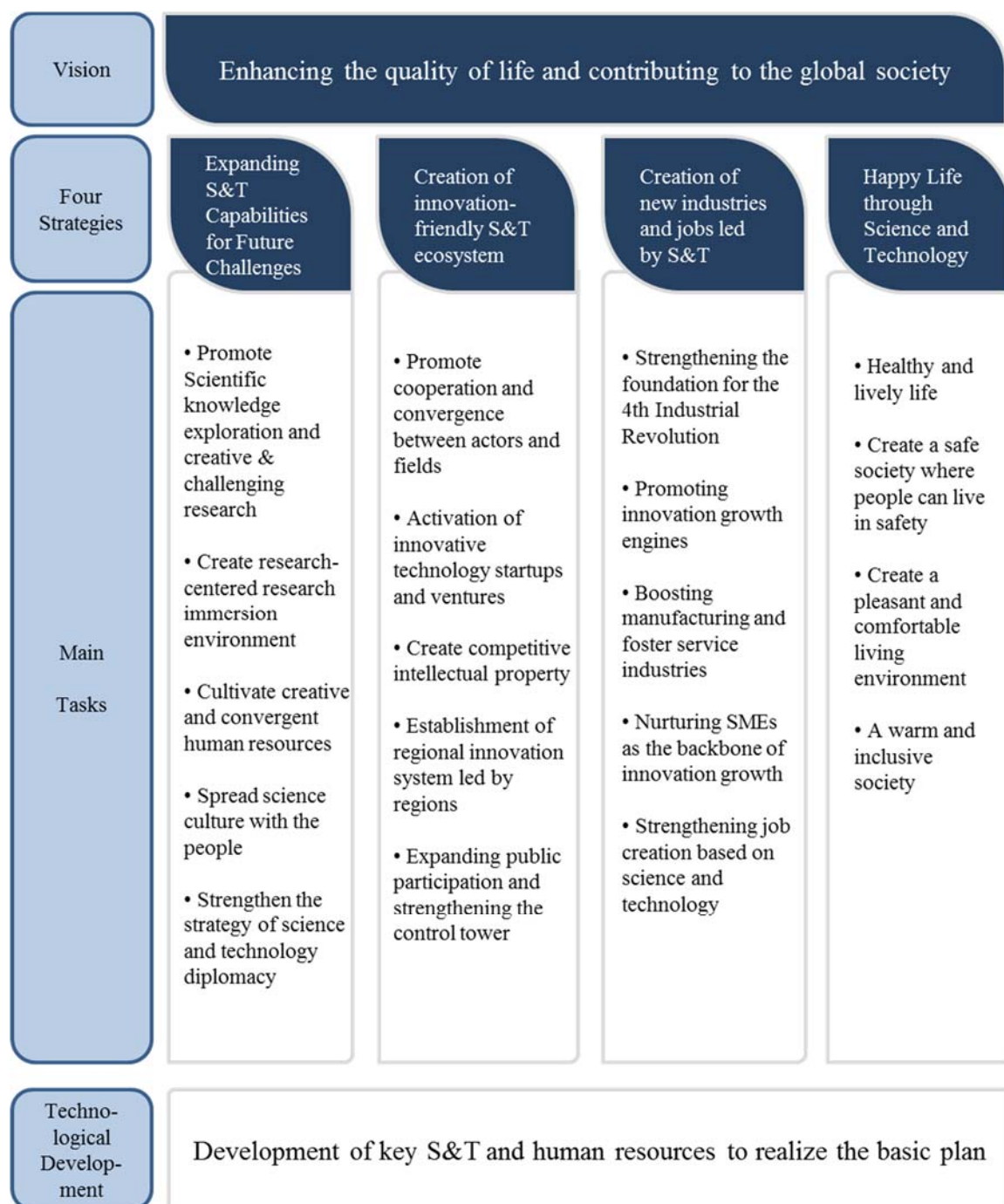
- Foster future growth engines by actively responding to the national challenges of preparing for the Fourth Industrial Revolution and expanding investment in future promising fields
- Create a virtuous ecosystem cycle where science and technology create quality jobs through growth engines and new industries

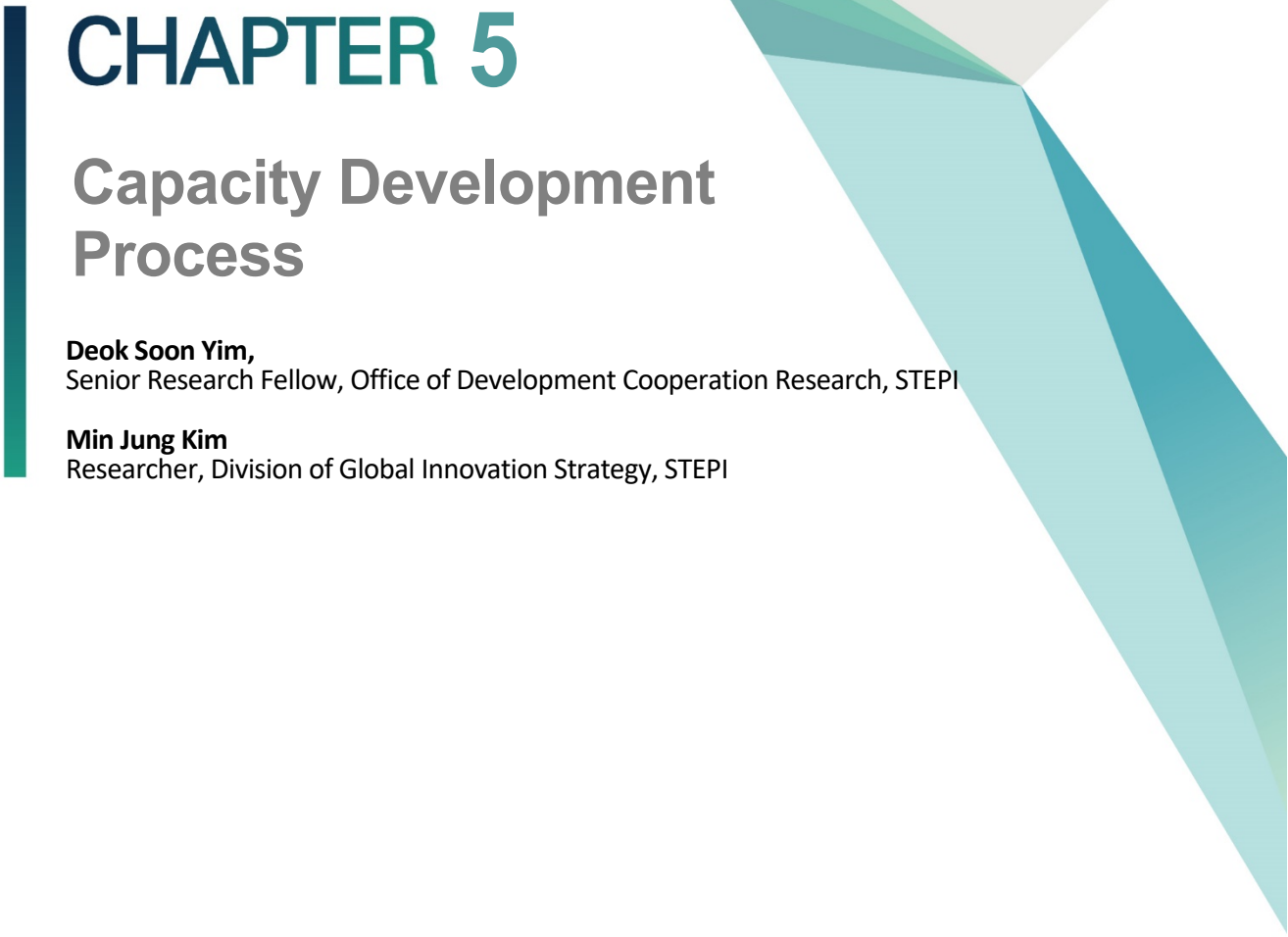
d) Happy Life through Science and Technology

- Spread ICT-based convergence technologies and services so that people can live healthy in a comfortable environment
- Strengthen the contributions of science and technology to solving problems such as disasters, safety, and environment

2.5 Construction of the 4th S&T Basic Plan: Strategies and Tasks

[Figure 4-11] Strategies and Tasks of the 4th S&T Basic Plan



A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and green, also pointing downwards.

CHAPTER 5

Capacity Development Process

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Chapter 5. Capacity Development Process

1. Kick-off Meeting and Feasibility Study

1.1 Introduction

Discussions were held between LIPI and STEPI about the topics to strengthen LIPI capacity for the workshop through business trips in December 2017 and pre-written consultations (Wangdong Kim et al., 2018).

1.2 Kick-off Meeting and Feasibility Study.

a) Research strategy

- Chi-Woong Song, research fellow of STEPI, attended the 6th National Conference on Science and Technology in Indonesia (Indonesia) to present on the experience and implications of Korean science and technology policy; policy advisory projects were discussed with the LIPI researchers during the conference (Wangdong Kim et al., 2018)
- From November 20 to December 3, 2016, two researchers from LIPI (two persons) stayed for two weeks in STEPI-IICC visiting researcher programs and discussing consultation on policy advisory projects (Wangdong Kim et al., 2018)
- Consultation on the STP (Science and Technology Park) policy for regional innovation
- Conducting interviews and surveys with Indonesian government officials and researchers

The STEPI team conducted demand analyses and feasibility studies for Indonesia from December 23, 2017 to February 8, 2018. The study sought to analyze the feasibility of the R&D Strategic Plan Evaluation consulting project through detailed demand analysis and literature research (Wangdong Kim et al., 2018).

b) The research team consisted of three STEPI internal experts and one external expert.

[Table 5-1] Research Team for Kick-off Meeting and Feasible Study

	Name	Organization & Position
Internal Experts	Wang Dong Kim	Research Fellow of STEPI
	Seong Won Park	Research Fellow of STEPI
	Boyan Lee	Researcher of STEPI
External Expert	Bum Cheol Shin	Professor of Kyunggi University

1.3 Wrap-up Meeting

After the kick-off discussion and feasibility study, the purpose, scope, demand realization, and specific expected schedule of the project were confirmed (Wangdong Kim et al., 2018).

The existing plan and schedule in 2018 before discussion were as follows (Wangdong Kim et al., 2018):

<Program Scope in 2018 Project Concept Paper>

- LIPI's R&D Strategic Planning: Comments for the R&D program progress evaluation
- 1. Diagnosis and evaluation of LIPI's R&D Strategic Plan (2015-2019)
(literature review)
- 2. Supporting the development of new LIPI R&D Strategic Plan (2020-2025)
- 3. Supporting the development of science and technology complex (STPs) under LIPI management

The modified plan and schedule after discussion were as follows:

<Reform of Program Scope in 2018>

* First half of 2018 (March ~ May)

- Field Trip of Korean experts (2 persons, 1 week in Jakarta)
 - 1) LIPI R&D Strategic Planning: Review and draft new plan guide
 - 2) Together with experts in technology park, visit to local TP

* Second half of 2018 (July ~ September)

- Draft of a new R&D strategy plan by LIPI
- Review and comment by Korean experts
- Final Report (2 people, 3 days in Jakarta)

2. Local Capacity Development Workshop

2.1 Introduction

The 1st field study and workshop for the K-innovation Indonesia ODA Program was conducted from May 1 to May 6 — including the departure and arrival dates (Wangdong Kim et al., 2018).

The objective of the field study and workshop was to carry out a workshop on the consultation and evaluation of LIPI's R&D Strategic Planning 2015-2019 and establishment of STP (Science and Technology Park) (Wangdong Kim et al., 2018).

The visiting research team consisted of five researchers (Wangdong Kim et al., 2018).

[Table 5-2] Research Team for Field Study and Workshop

Name	Organization & Position	Roles
Seong Won Park	Research Fellow of STEPI	Project Manager
Jaehoon Rhee	CEO of Gyengbuk Technopark	Management strategies of the science and technology complex
Young Sub Kwon	Senior Research Fellow of the Korea Research Institute for Human Settlements	Evaluation of the outputs of the science and technology complex
Kibeom Park	Research Fellow of STEPI	Sharing Korea's experience in the management of human resources
Seona Lee	Researcher of STEPI	Project Coordinator

2.2 Major Themes (activities)

The research team consisted of 5 people, who carried out the 1st dispatch and held the local capacity-building workshop on May 1-6, 2018 (including the departure / arrival dates) (Wangdong Kim et al., 2018).

There were three main purposes of the workshop: 1) to focus on selected topics that were organized during business trips in December 2017; 2) to derive the future strategic plan and human resource management methods of the 2020-2024 LIPI; and 3) to consult on the establishment and operation of LIPI's Science and Technology Park (STP) (Wangdong Kim et al., 2018).

STP (Science and Technology Park) diagnosis was an important topic of the workshop at the same time. Additionally, it included simulation and strategy forecast methods to identify potential problems that may arise during the advisory stage and establish solutions for those potential problems (Wangdong Kim et al., 2018).

The theme of the workshop was largely divided into three types of content: (1) presentation and discussion on the strategy of LIPI's Human Resource Management; (2) LIPI's R&D Strategic Plan; and (3) STP (Science and Technology Park in Indonesia) (Wangdong Kim et al., 2018).

The workshop was held for two days with LIPI and STEPI experts discussing together, followed by an internal STEPI meeting (Wangdong Kim et al., 2018).

a) On the first day of the Workshop (May 2, 2018)

A briefing was provided by a researcher from the LIPI, covering the status and limits of LIPI's Institutional Strategy and Science Technology Park (Cibinong STP, which is under the jurisdiction of LIPI). There was a discussion on the need for linkage with a research institute already established around the science and technology park (Wangdong Kim et al., 2018).

This was followed by the main lecture, which introduced the experience of Korea. At that

time, LIPI did not have a strategic place to undertake researcher human resource management. In that respect, the lecture was provided to share Korea's successful case of manpower management and discuss the Cibinong STP and HRM of LIPI. In the Korean case, industrial development was rooted in a "selection and concentration strategy" and a 5-year economic development plan as the main sources of the industry. As a result of this industrial development and the policy's effects, manpower was also cultivated. For LIPI, we can conclude that continuous assessment and monitoring of LIPI research personnel are required. Likewise, LIPI might be able to collect a survey with the sample of all researchers in Indonesia for constructing basic data to support the national development strategy of Indonesia (Wangdong Kim et al., 2018).

b) On the second day of the Workshop (May 3, 2018)

The strategic planning team had a Korean case presentation and a discussion on the National R&D innovation via public participation. A Research Fellow of STEPI provided a presentation about the X-Project, which was initiated in 2015 as a STEPI-led R&D innovation project. Discussing the citizen participation methodology was meaningful considering the history of Indonesia with its various ethnic backgrounds. Stakeholders were engaged to improve LIPI's Strategic Plan 2020-2024, and four future scenarios of LIPI were derived using a forecasting methodology. Additionally, the need to revise the vision and strategy of past LIPI was discussed (Wangdong Kim et al., 2018).

On the other hand, the Science and Technology Complex Team gave a presentation and led discussion on the selection and evaluation of the location of STP. The policy suggestions provided by the experts from the STEPI research team establish STP relevant to regional strategic industries. Moreover, the results of this research indicated that the main success factors of the STP are the location of STP, clear vision, innovation system, adequate human resources, trust in enterprise and government, and existence of proper networks (Wangdong Kim et al., 2018).

c) On the third day of the Workshop (May 4, 2018)

A final meeting was held. The capacity-building workshop was meaningful because it

presented a new perspective and a basic framework for strengthening LIPI's organizational capabilities and reflecting Korea's experience in human resource management and strategy for STP (Wangdong Kim et al., 2018).

The schedule of the field studies and workshops was as follows:

Date	Time	Contents	
May 1 (Tue.)	15:30	Researchers depart from Incheon	
	20:30	Arriving in Jakarta	
May 2 (Wed.)	First day of the Workshop Venue: Jakarta		
	09:00-12:00	Conference with researchers of LIPI Science and Technology Development Research Center - LIPI Strategic Planning Team and LIPI Computer Engineering Team	Conference with researchers of LIPI Science and Technology Development Research Center (PAPPIPTEK), Innovation Center (PUSINOV) - Including the presentation on the current status of LIPI’s science and technology complex
	12:00-13:00	Lunch Hour	
	13:00-16:00	Presentation and Discussion on the strategy of LIPI’s Human Resource Management - Presentation from LIPI Science and Technology Development Research Center (PAPPIPTEK) LIPI - Presentation by Kibeom Park, research fellow of STEPI - Presentation by Dr. Trina Fizzanty of LIPI, PAPPIPTEK - Presentation by Dr. Heru Santoso, LIPI, Organization Personnel Department	
May 3 (Thu.)	Second day of the Workshop Venue: Jakarta		
		LIPI Strategic Planning Team	LIPI Science and Technology Complex Team
	09:00 – 12:00	STEPI – LIPI Presentation and Discussion - LIPI Strategic Planning Team	STEPI – LIPI Presentation and Discussion - Theories on the locational conditions of science and technology complex

Date	Time	Contents	
		- LIPI Selection Research Priority Team - Kibeom Park, research fellow of STEPI	and Korea’s case - Young Sub Kwon, Senior Research Fellow of the Korea Research Institute for Human Settlements
	12:00-13:00	Lunch Hour	
	13:00-16:00	STEPI – LIPI Presentation and Discussion - LIPI Research and Development Bureau - LIPI Strategic Planning Team - LIPI Selection Research Priority Team - Seong Won Park, Research Fellow of STEPI	STEPI – LIPI Presentation and Discussion - Management of science and technology complex - Jaehoon Rhee, CEO of Gyeongbuk Technopark
	17:00-19:00	Visiting the Indonesian Embassy and introducing business activities	
May 4 (Fri.)	Third day of the Workshop Venue: Jakarta		
	09:00-12:00	Discussion on the 2020-2025 plan and agreement on the next discussion topic between LIPI and STEPI - PAPPIPTEK LIPI – STEPI - Proposal of further schedule	
	12:00-14:00	Lunch Hour	
	14:00-16:00	Interview with the director of LIPI	
	17:00-19:00	In-depth discussion with LIPI	
May 5 (Sat.)	09:30-11:00	Internal researchers’ team meeting	
	22:05	Researchers depart from Jakarta	
May 6 (Sun.)	07:15	Arrival in Incheon	

[Table 5-3] Photos of the Local ODA Workshop in Indonesia

Research Presentation of LIPI	Lecture on HRM (Kibeom Park, Research Fellow of STEPI)
	
Strategic Planning Team Workshop	Science and Technology Complex Team Workshop
	
Interview with the Director of LIPI	Farewell Speeches and Closing Session
	

2.3 Evaluation Survey

A satisfaction survey was conducted among 12 persons as a complete enumeration on those who participated in the capacity-building workshop held in Indonesia from May 2 to 3.

LIPI researchers who participated in the workshop were 41.75 years old on average (standard deviation: 8.51). Among the 12 respondents, males accounted for 45%, and females accounted for 55%. The average tenure of office in a position was 9.08 years.

The following table summarizes the results of the survey:

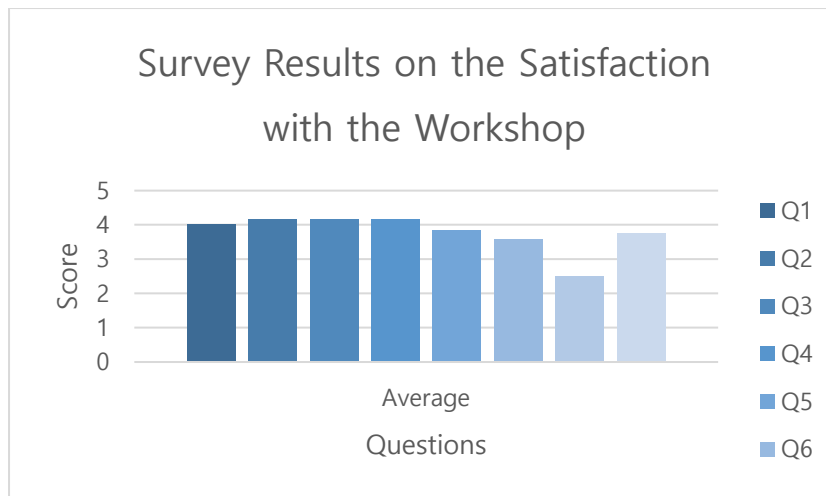
[Table 5-4] Results of the Satisfaction Survey

Questions	Average	Standard Deviation
1. The program was helpful in enhancing my capacity to perform the duties.	4	0.43
2. The program met the objective of institutional capacity building.	4.17	0.58
3. The materials have been conducive for understanding the contents of the program.	4.17	0.58
4. The topics covered were relevant to my institution's needs for STI development.	4.17	0.72
5. Enough participation opportunities were provided throughout the program.	3.83	0.72
6. The program has influenced me in changing my mindset.	3.58	0.51
7. The duration of the program (2 days) was sufficient.	2.50	0.67
8. I would recommend this program to others.	3.75	0.87

All 8 questions were surveyed on a 5-point Likert scale. 5 = Strongly agree, 4 = Agree, 3 = Neutral, 2 = Disagree 1 = Strongly disagree.

The average score of 8 survey questions with the same weight is 3.77, which can be regarded as satisfactory overall. Regarding questions 1 to 4, the scores were higher than 4, and the results reveal that the workshop was conducive for improving job performance and institutional capacity building as well as LIPI's STI development.

With regard to the results of question 7, however, the average was 2.50, which was the lowest. It shows that the participants found the 2-day workshop too short. It can be inferred that the demand for such capacity-building workshops was high.



The average scores of questions 5, 6, and 8 were favorable at higher than 3 points. If we increase the opportunities for participants and improve the program next time, the satisfaction of the workshop's participants is expected to be enhanced. In particular, it is necessary to extend the duration of the workshop to strengthen the operation further.

In addition, descriptive questions were asked to the participants regarding the further topic of the workshop and suggestions for program improvement.

The answers to these descriptive questions are shown in the table below.

[Table 5-5] Results of the Descriptive Satisfaction Questions

	Answers
Q 9. What other topics/contents would you suggest we deal with in the future?	- Human Resources devoted to Science and Technology
	- R&D Investment
	- Innovation contribution
	- Leadership in R&D institutions toward Industry 4.0 or higher versions
	- Efficiency of public spending and national policies
	- Digital economy, Industry 4.0
	- Horizon scanning, issue mapping
	- Strategic and Operational Management of Science and Technology Parks
	- Business Development of a Science Park
	- Policies on Creating a Strong Innovation Ecosystem in a country or in a region
Q 10. Are there any suggestions for improvement or requests on the overall procedure? If so, please explain in detail (program process, contents, results, sustainability, etc.).	- Technology Commercialization
	- Identification of recent status of organizations compared with the expected future conditions, including how to adapt to this
	- Policy coordination for STI
	- Policy advocacy for STI
	- Future industry research
	- Entrepreneurship and innovation studies
	- Science communication
	- Duration of more than two days
	- Add technical matters, such as gathering and using data for S&T policy making
	- Longer duration is better
	- Joint research is probably more efficient for knowledge transfer
	- It would be good if the participants are required to prepare information in advance in certain areas that they would like to improve. During the training program, the participants can apply their knowledge and skill to do the case of his or her interest
	- The material used for training such as ppt presentations and other materials can also be given to the participants -- for example, two weeks before the training. This could help make the discussion more productive
	- It is better if we can summarize the discussion in a paper

We will reflect the suggestions proposed in these survey results to the upcoming invitation training in December 2018. Specifically, the improvement points are as follows:

- To extend the duration of invitation training to five days from two days as in the past capacity-building workshop
- To introduce more room for participants to participate. On the first day of the invitation training, the participants will each be required to give a presentation on their research relevant to the STP policy or R&D strategic plan of Indonesia. On the last day, a wrap-up group discussion will be conducted.
- The lecture materials will be shared prior to the start of the workshop via e-mail so that the participants can be prepared and the workshop quality will improve.

3. Capacity-building Workshop

3.1 Introduction

The Korea Institute of Science and Technology Policy (STEPI) held a capacity-building training session for seven days from December 2 to 8, 2018. The training program, which was conducted with seven researchers from the Indonesian Institute of Sciences (LIPI) and two professors from Kathmandu University (KU), is part of K-Innovation 2018, which is aimed at strengthening the capacity of LIPI and KU.

As part of the “STEPI-LIPI Project 2018 Policy Consultation on 1. R&D Strategy Plan 2. Science and Technology Park” and the “Project for the Integrated Rural Development of Nepal by Strengthening the Research and Development Capacity of Kathmandu University,” the team from LIPI and NTIC was invited to Korea to participate in the capacity-building workshop. Through the workshop, the participants were able to deepen their understanding — particularly on the role of the technopark (Korean-version Science Technology Park) in Korea and R&D strategy in general – including how their functions and programs can be applied to LIPI (Indonesian Institute of Sciences) and to KU (Kathmandu University) as well as the necessary policy measures (Wangdong Kim et al., 2018).

The STEPI research team and the invited participants from LIPI and Kathmandu University are described below (Wangdong Kim et al., 2018).

3.2 Major Themes (activities)

There were two major themes in the Capacity-building Workshop in Korea with LIPI and KU. First of all, the workshop focused on providing knowledge on R&D Strategy Planning and experiencing practical fields regarding Science and Technology Parks. The second theme is on the project for the Integrated Rural Development of Nepal by Strengthening the Research and Development Capacity of Kathmandu University with the operation of a systematic education program (Wangdong Kim et al., 2018).

The capacity-building workshop above consisted of four in-depth lectures on science and technology innovation policies, regional innovation systems, and technology commercialization as well as five site visits, including a vivid training program through field visits. The lectures featured a theoretical and practical intensive course designed for Nepalese government officials and qualified practitioners looking to broaden their knowledge in the areas of Science Technology Parks by participating in practical training (Wangdong Kim et al., 2018).

On the first day of the workshop, Dr. Deok Soon Yim provided the opening remarks, and Dr. Ki Kook Kim gave a welcome speech. After that, Dr. Sunyang Chung, professor at Konkuk University, gave a lecture titled “An Introduction to Korean S&T and Innovation Policy.” Dr. Jang Jae Lee, director of the Research Institute of S&T Innovation and Strategy, KISTEP, then delivered a lecture titled “Regional (Sub-national) Innovation System in Korea – Search for Best Practice.” Dr. Jong-In Choi, professor of Hanbat University, then delivered a lecture titled “Technology Commercialization and Entrepreneurial University.” Dr. Gwang Hwi Park, CEO of Rehoboth Business Incubator, gave a lecture on “Tech Commercialization Concepts and Startup Case Studies.” Dr. Jaehoon Lee, CEO of Gyeongbuk Technopark, gave a lecture on the “Roles of Science and Technology Parks in Regional Innovations.” Dr. Jae Yong, Choung, professor of KAIST, delivered a lecture titled ‘Korea’s Technology Catch-up and Transition’ (Wangdong Kim et al., 2018).

Our invited trainees visited five STI-related institutes including Gyeongbuk TP, Daegu TP, Daedeok Innopolis, and Hanbat Incubator Center and experienced Korea’s R&D strategic planning and management at the Science and Technology Park (Wangdong Kim et al., 2018).

The participants in the workshop were required to give individual, formal presentations during the workshop. This additional improvement to expand the chances to participate in the workshop reflected the suggestions of participants in a former capacity-building workshop held in May 2018. At that time, few LIPI researchers who participated in the workshop felt short-changed as there was less room for participation during the workshop. Therefore, we decided to open the research presentation session on the first day of the

workshop, and have the wrap-up group discussion and presentation time on the last day of the workshop (Wangdong Kim et al., 2018).

The list of participants in the workshop is as follows:

[Table 5-6] Participants in the Capacity-building Workshop in Korea with LIPI and KU

Organization	Name	Position
Science and Technology Policy Institute (STEPI)	Deok Soon Yim	Project Manager
	Ki Kook Kim	Research Fellow
	Wang Dong Kim	Senior Research Fellow
	Seona Lee	Researcher
	Min Jung Kim	Researcher
Indonesian Institute of Sciences (LIPI)	Manaek Simamora	Program Coordinator of Cibinong Science and Technology Park
	Ardanawati Ayu Pitaloka	Junior Researcher of the Research Center for Science & Technology Development
	Hadi Kardoyo	Researcher of the Research Center for the Development of Sciences and Technology
	Setiowiji Handoyo	Researcher of the Research Center for Science & Technology Development
	Karlina Sari	Junior Researcher of the Research Center for Science & Technology Development
	Anggini Dinaseviani	Junior Researcher of the Research Center for Science & Technology Development
	Andi Budiansyah	Junior Researcher of the Center for Innovation
Kathmandu University	Bim Shrestha	Professor, School of Engineering
	Rajeev Shrestha	Professor, School of Medical Science

The specific workshop schedule was as follows:

[Table 5-7] Program and Schedule of the Capacity-building Workshop in Korea with LIPI and KU

Date	Time		Place	Key Activity
12/1 (Sat.)	20:20	05:30 +1	Nepal Tribhuvan International Airport	Departure from Kathmandu (Korean Air, KE696)
	21:50	6:50 +1	Soekarno–Hatta International Airport (CGK)	Departure from Jakarta (Korean Air, KE0628)
12/2 (Sun.)	05:30		Incheon Airport	Arrival at Incheon Airport (Korean Air, KE696)
	07:00	12:00	Courtyard Marriott Namdaemoon	Move to Seoul & Check-in (Rest)
	12:00	13:00	Courtyard Marriott Namdaemoon in Seoul	Lunch
	13:00	18:00	Courtyard Marriott Namdaemoon	Rest & Preparation for the Presentation
12/3 (Mon.)	9:30	10:00	Courtyard Marriott Namdaemoon	Orientation - Opening Remarks - Introduction of Workshop
	10:00	12:00	Courtyard Marriott Namdaemoon	[Lecture 1] - Introduction to Science Technology Innovation Policy (Prof. Sunyang Chung) (lecture and Q&A session)
	12:00	13:00	Courtyard Marriott Namdaemoon	Welcome Lunch
	14:00	16:00	Courtyard Marriott Namdaemoon	Research Presentation - Subtheme 1. R&D Strategic Plan - Subtheme 2. Science and Technology Park - Subtheme 3. NTIC Management Strategy (sustainability, KU brand project, startup support, and research management) - Subtheme 4. STI Park Policy in Nepal (20~30 minutes' PPT and 10 minutes'

Date	Time		Place	Key Activity
				discussion per participant)
	16:00	18:00	Courtyard Marriott Namdaemoon	[Lecture 2] - Regional Innovation System (Dr. Jang Jae Lee) (lecture and Q&A session)
	18:00	19:00	Courtyard Marriott Namdaemoon	Dinner
12/4 (Tues.)	07:30	11:00		Move to Daegu
	11:00	13:00	Gyeongbuk Technopark	[Field Visit 1] Gyeongbuk Technopark
	13:00	14:00		Lunch
	14:00	16:00		[Field Visit 2] Asia Science Park Association (visit to related Smart Health Technology Firm)
	16:00	18:00		Move to Daejeon & Check in
	18:00	19:00		Dinner
12/5 (Wed.)	10:00	12:00		[Lecture 3] Hanbat University (Prof. Jong-In Choi) Startup promotion in University
	12:00	13:30		Lunch
	13:30	15:30	KAIST	[Field Visit 3] KAIST (Prof. Jae Yong, Choung) Visit to Global Commercialization Center (GCC)
	17:00	18:00	STEPI	[Field Visit 4] Science and Technology Policy Institute (STEPI)
	18:00	19:00		Dinner

Date	Time		Place	Key Activity
12/6 (Thurs.)	9:30	11:00	Daedeok Innopolis	[Field Visit 5] Daedeok Innopolis
	11:00	13:00		Move to Seoul
	13:00	14:00		Lunch
	15:30	17:30	Seoul	[Lecture 4] Tech Commercialization Concepts and Startup Case Studies (Rehoboth Business Incubator Chairman, Gwang Hwi Park)
	17:00	18:00		Dinner
12/7 (Fri.)	10:00	12:00	Courtyard Marriott Namdaemoon	[Wrap-up and Follow-up] Group Presentation
	12:00	13:00	Seoul	Lunch
	13:00	18:00		[Cultural Tour]
	18:00	19:00		Dinner
12/8 (Sat.)	10:00	11:00		Move to Airport
	13:25	17:50	Incheon Airport (ICN)	Departure from Incheon Airport (Korean Air, KE695)
	15:05	20:15	Incheon Airport (ICN)	Departure from Incheon Airport (Korean Air, KE0627)

[Table 5-8] Photos of the Capacity-Building Workshop in Korea with LIPI and KU

Opening Ceremony and Lecture by Dr. Sunyang Chung, Professor at Konkuk University	Lecture by Dr. Jaehoon Lee, CEO of Gyeongbuk Technopark
	
Field Visit to Gyeongbuk Technopark	Field Visit to Daeduk Innopolis
	
Meeting with Hwang-Hee Cho, President of STEPI	Incubating Center of Hanbat National University
	

3.3 Evaluation Survey

A satisfaction survey was conducted among twelve persons as a complete enumeration of those who participated in the capacity-building workshop held in Indonesia from May 2 to 3 (Wangdong Kim et al., 2018).

LIPI researchers who participated in the workshop were 41.75 years old on average (standard deviation: 8.51). Among the 12 respondents, males accounted for 45%, and females constituted 55%. The average tenure of office in a position was 9.08 years (Wangdong Kim et al., 2018).

The following table summarizes the survey results:

[Table 5-9] Results of the Satisfaction Survey

Questions	Average of Capacity-building Workshop in December 2018	Average of Former Local Capacity-building Workshop (May 2018)
1. The program was helpful in enhancing my capacity to perform my duties.	4.86	4
2. The program met the objective of institutional capacity building.	4.71	4.17
3. The materials were conducive for understanding the contents of the program.	4.71	4.17
4. The lecturer's expertise and lectures on the topic were good.	4.86	-
5. The passion of the instructor enabled me to concentrate on the lesson.	4.71	-
6. The topics covered were relevant to my institution's needs for STI development.	4.71	4.17
7. The topics covered were relevant to my institution's needs for Science Technology Park development.	4.71	-
8. Enough participation opportunities were provided throughout the program.	4.57	3.83
9. The program has influenced me in changing my mindset.	4.71	3.58
10. The duration of the program was sufficient.	3.86	2.50
11. I would recommend this program to others.	4.57	3.75
Total Average	4.64	3.77

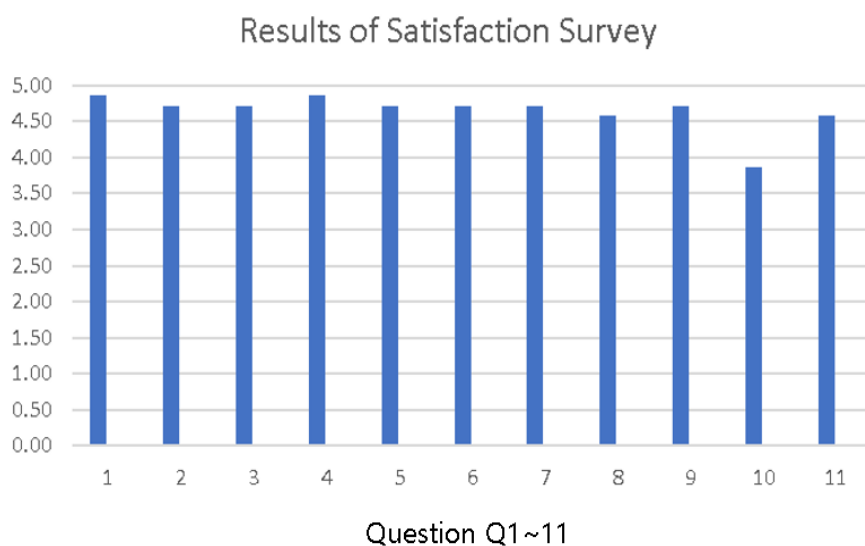
All 8 questions were surveyed on a 5-point Likert scale: 5 = Strongly agree, 4 = Agree, 3 = Neutral, 2 = Disagree 1 = Strongly disagree.

The average score of the eight survey questions with the same weight is 4.64, which shows that the overall satisfaction with the capacity-building workshop increased compared to the former local capacity-building workshop (average = 3.77). It demonstrated the improvement in quality of the capacity-building workshops. Notably, the participants in capacity-building workshops with LIPI and KU agreed that the lecturer's expertise and lectures on the topic were good (average score=4.86). This time, the capacity-building workshop ran for five days instead of two days, and it led to an increase in the satisfaction of the participants (from 2.50 → 3.86). Most of all, the workshop can be said to have been evaluated positively (average = 4.71) because the participants agreed that the topics covered were relevant to their institution's needs for STI development and STP (Science and Technology Park). Many participants acknowledged that the program had contributed to changing their mindsets (average = 4.71). Still, the workshop has limits that need to be addressed, as the score for the question on sufficient participation opportunities was slightly low compared to the responses to other questions (average = 4.57) (Wangdong Kim et al., 2018).

[Figure 5-1] Survey Results on the Satisfaction with the Local Capacity-building Workshop (May 2018)



[Figure 5-2] Survey Results on the Satisfaction with the Capacity-building Workshop (December 2018)



In addition, descriptive questions were asked to the participants regarding the topic of further workshops and suggestions for program improvement.

The answers to these descriptive questions are shown in the table below. As a reflection of these perspectives, future workshops will be modified to allow more time for each field visit, and we will fix the time schedule to introduce 1-2 hours of group discussion for sharing ideas and discoveries as to what each participant has learned and discovered, instead of a full one-way lecture.

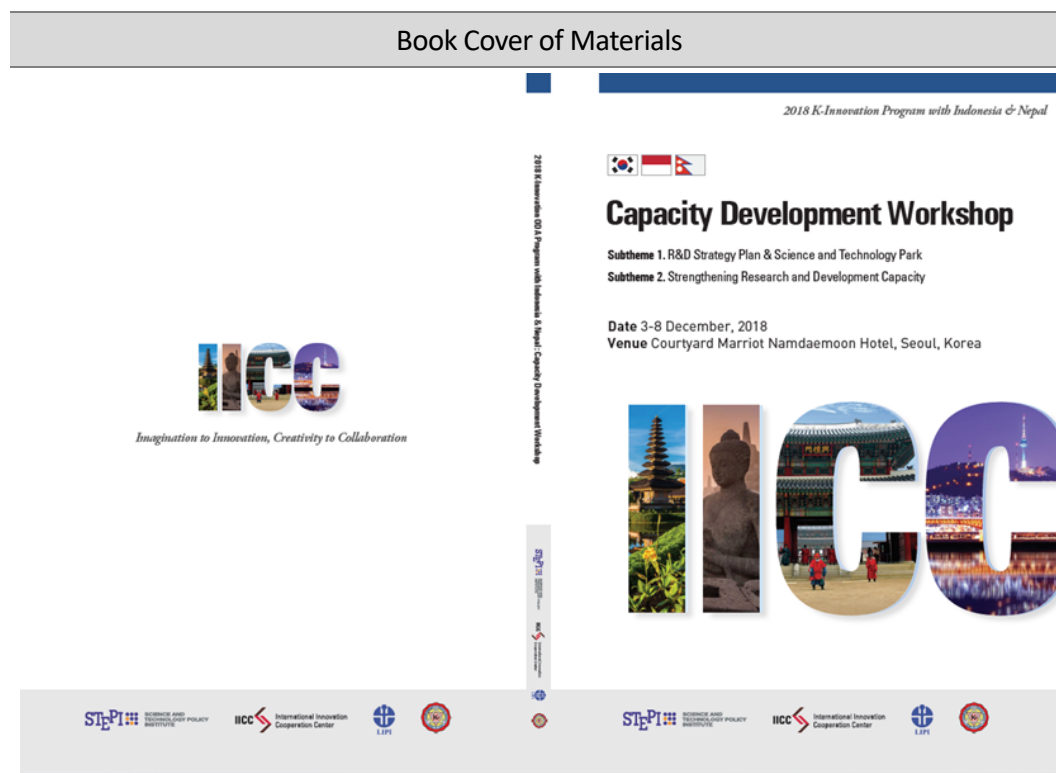
We will adopt several suggested topics in the next capacity-building workshop.

[Table 5-10] Results of the Descriptive Satisfaction Questions

	Answers
Q 12. How do you think this program will help your job at LIPI or Kathmandu University in the future?	- It is useful since it helps empower us in our programming process; setting policy resource; programming strategy; managing STP
	- The capacity-building workshop is deepening our knowledge of the STI policy, and we have implementation for research policy in Indonesia
	- This program helped me in understanding issues, strengthening my knowledge of policy approaches for technology development
	- The program is strongly helpful in making me become open-minded about the STI process businesses in Korea, and I gained many insights from this opportunity
	- I think this program would give another perspective on how to conduct science on the marketing/commercialization of technology that successfully worked in Korea, especially from the STI policy
	- This program has enabled me to broaden my knowledge and shape my mindset in viewing the STP policy
	- This program opened my eyes, and I was able to learn about innovation management in Korea, which I wish to adopt in our team in Indonesia

	Answers
Q 13. What other topics/contents or field visit institutions would you suggest we deal with in the future?	- Executive Program for STP Management
	- STI Policy Paper & Policy Brief
	- Promoting SMEs and Entrepreneurship for Future Marketing
	- How to apply the STI policy to the local & central governments
	- How to encourage society to have a high scientific culture
	- Management of financial stability of STP
	- How they can link the startup with funding agency
Q 14. Are there any suggestions for improvement or requests on the overall procedure? If so, please explain in detail. (Program process, contents, results, sustainability, etc.)	- 4th industrial revolution and digital economy
	- About IP management in Korea, how Korea is making IP, from imitation to innovation
	- More time allocation for each field visit please
	- There should be more time for discussion
	- Collaboration on resources in the future hopefully
	- Duration may increase, and it would be good to sustain every session or year
	- Everything was great. Thanks for inviting us, and see you again.

[Figure 5-3] Materials of the Capacity-building Workshop in Korea with LIPI and KU



3.4 Publicity and Dissemination

■ STEPI Press Release (reported on December 7, 2018)

(<http://www.stepi.re.kr/app/movements/view.jsp?cmsCd=CM0027&ntNo=27515&src=&srcTemp=&currPg=1>)

STEPI 동정 HOME > 연구원 소식 > STEPI 동정 Print + - 원대대로

제목	인도네시아 국립과학원 네팔 카트만두대학교 역량강화 초청연수 개최(12.3~12.8)		
작성일	2018-12-07	조회수	21






과학기술정책연구원은 이달 12월 3~8일 6일 간 역량강화 연수를 개최하였다. 인도네시아 국립과학원(LIPI) 연구진 7명, 카트만두대학교 교수 2명을 대상으로 진행된 해당 연수 프로그램은 인도네시아 국립과학원과 카트만두대학교의 자체 역량강화를 위한 '국제기술혁신협력 사업(K-Innovation) 2018'의 일환으로 진행되었다.

R&D 전략기획 수립, 과학기술단지 관한 한국 전문가 초청 강연과 정책TP, 대구TP, 대덕이노폴리스, 한밭대 창업보육센터 등 현장에 직접 방문하여 한국의 STI 시스템과 발전, 과학기술단지의 역할과 성공사례 등의 노하우 전수를 제공하였다.

초청연수는 과학기술혁신 정책, 지역 혁신 시스템, 기술 사업화 등에 관한 네 차례의 심도 깊은 강연과 경북 테크노파크, 대구 테크노파크 웰스케어실용센터, 대덕이노폴리스 연구개발특구진흥재단 등의 5개 현장 기관 방문을 통한 생생한 연수 프로그램으로 구성되었다.

■ STEPI-IICC FACEBOOK (reported on December 7, 2018)

(<https://www.facebook.com/iicc.stepi>)

 **Stepi iicc**님이 Deok Soon Yim님 외 2명과 있습니다. 22분 · 

[2018 K-Innovation Capacity Development Workshop with Indonesia & Nepal]

The Korea Institute of Science and Technology Policy (STEPi) held a capacity building training session for five days from December 10 to 14 this month. The training program, which was conducted with 7 researchers from Indonesian Institute of Sciences (LIPI) and 2 professors from Kathmandu University(KU), is part of K-Innovation 2018 to strengthen the capacity of LIPI and KU. ... 더 보기




INNOPOLIS Foundation

4.

Joint Capacity Development Workshop for Uganda and Indonesia

4.1 Background

As part of the K-Innovation Program with Uganda and Indonesia, experts from the Ministry of Science, Technology, and Innovation of Uganda, Uganda National Council for Science and Technology (UNCST), and Indonesian Institute of Sciences (LIPI) are invited to participate in the joint capacity development workshop. Through the workshop, participants are expected to deepen their understanding of the themes covered by the K-Innovation Program 2019 for Indonesia and Uganda. Major themes for the workshop include:

- Structure and history of Korean S&T policy in general
- STI policy and necessary policy measures
- Industry-University cooperation, incubation, and commercialization of R&D outputs

The four-day workshop is composed of two elements: 1) lectures and group discussions, and 2) field visits. Participants are expected to participate actively in the course and present the action plan for developing an innovative science and technology policy that could be applied to Indonesia and Uganda based on the Korean experiences.

Theoretical and practical intensive courses are designed for the Indonesian and Ugandan government officials and practitioners looking to broaden their knowledge through group discussions. Field visits to public research institutes and other related agencies are arranged as well.

4.2 Overview of the capacity-building workshop

- Dates: 8 ~ 11 July 2019
 - Arrival on 7 July and Departure on 11 July 2019 for the Ugandan team
 - Arrival on 7 July and Departure on 12 July 2019 for the Indonesian team
- Venue: Shilla Stay Yeoksam, Seoul and Daejeon, Republic of Korea
- Participants: STEPI research team and 10-12 invited participants from LIPI and UNCST and Ministry of Science, Technology, and Innovation of Uganda

Organization	Name	Position/Department
Science and Technology Policy Institute (STEPI)	Dr. Deok Soon Yim	Project Manager
	Dr. Wang Dong Kim	Senior Research Fellow
	Ms. Elly Hyanghee Lee	Researcher
	Ms. Min Jung Kim	Researcher
Indonesian Institute of Sciences (LIPI)	Dr. Mego Pinandito	Deputy Chairman for Scientific Services
	Dr. Trina Fizzanty	Senior Researcher
	Mr. Qinan Maulana Binu Soesanto	Junior Researcher
	Ms. Indah Purwaningsih	Junior Researcher
Agency for the Assessment and Application of Technology (BPPT)	Mr. Boni Benyamin	Engineer (Perekayasa Madya)
Ministry of Research, Technology, and Higher Education (Ristekdikti)	Dr. Adawiah Muhammad Hasan	Deputy Director of Harmonization of Innovation Policy and Programs
Ministry of National Development Planning (BAPPENAS)	Dr. Ade Faisal	Acting Deputy Director for Research and Development in Science and Technology

Organization	Name	Position/Department
Ministry of Science, Technology, and Innovation, Uganda	Dr. Maxwell Otim Onapa	Director
	Ms. Monica Rose Nyakaisiki	Assistant Commissioner
	Mr. Morrish Ochen	Assistant Commissioner
Uganda National Council for Science and Technology	Mr. Bashir Rajab Kagere	Head, Policy Analysis, Planning, and M&E
	Mr. Geoffrey Sempiri	Head, STI Infrastructure and Development

4.3 Program

Date	Time		Place	Key Activity
6 July (Sat.)	22:05	7:15 +1	Soekarno–Hatta International Airport (CGK)	Departure from Jakarta (Korean Air, KE0628)
	16:20		Entebbe Airport	Departure from Entebbe (EK730) Via Dubai
7 July (Sun.)	05:30		Incheon	Arrival at Incheon Airport (Korean Air, KE696)
	07:00	12:00	Shilla Stay Yeoksam	Move to Seoul & Check-in (rest)
	12:00	13:00		Lunch
	13:00	18:00		Rest & Preparation for the Workshop
	16:55			Arrival at Incheon Airport (EK322)
	17:00	18:30		Move to Seoul & Check-in (rest)
	19:00			Dinner
	20:00			Rest

Date	Time		Place	Key Activity
8 July (Mon.)	9:30	10:00	Shilla Stay Yeoksam	Orientation - Opening Remarks (Dr. Chi Ung Song) - Introduction of Workshop (Dr. Deok Soon Yim)
	10:00	11:00		Lecture 1: Understanding the National Research Council of S&T by Dr. Youngjoo Ko
	11:00	11:15		Presentation 1: Status and Challenges of Science Technology and Innovation in Indonesia by Dr. Adawiah Muhammad Hasan
	11:15	11:30		Presentation 2: Status and Challenges of S&T Governance in Uganda by Mr. Geoffrey Sempri
	11:30	12:00		Q & A and Group Discussion
	12:00	13:30		Welcome Lunch
	14:00	15:00		Lecture 2: S&T Policies and Governance in Korea by Prof. Seong-soo Kim
	15:00	15:15		Presentation 3: Status and Challenges of STI Policy in Indonesia by Dr. Mego Pinandito
	15:15	15:30		Presentation 4: Science and Technology Innovation Policy in Uganda by Mr. Bashir Kagere
	15:30	16:00		Q & A and Group Discussion
	16:00	17:00		Lecture 3: Technology Commercialization: "How it Works to Catch Up" by Dr. Jongheung Park, Executive Director, ETRI
	17:00	17:30		Q&A
	18:00			Move to Han River
	18:30	20:00		Dinner (Near Han River)
	09:00	10:00	Smartwork center for	Lecture 4: Facts and Truth about Technology Transfer by Prof. Johng-Ihl

Date	Time		Place	Key Activity
9 July (Tues.)			National Research Institutes in Yangjae	Lee, State University of New York, Korea
	10:00	10:15		Presentation 5: Status and Challenges of Technology Commercialization in Uganda by Mr. Morrish Ochen
	10:15	10:30		Presentation 6: Model of Technology Commercialization from GRI to Private Sector in Indonesia by Mr. Boni Benyamin
	10:30	10:45		Presentation 7: Emerging Technologies with Focus on ICT: A Case for Uganda by Ms. Monica Rose Nyakaisiki
	10:45	11:15		Q & A and Group Discussion
	11:30	12:30		Lunch
	12:30			Move to KIST
	13:30	14:30	KIST	Lecture 5: How KIST develops its Technologies into Business by Dr. Yoohyung Won
	14:30	15:00	KIST	Field Visit 1: Korea Institute of Science and Technology (KIST), KIST History Hall
	15:30	18:00	Seoul	Sightseeing Around the Palace and Insadong, Seoul
	18:00		Seoul	Dinner
	20:30			Move to the Hotel
10 July (Wed.)	08:00	10:00	Seoul to Daejeon	Move to Daejeon
	10:00	11:00	Hanbat National University	Lecture 6: Industry-University Collaboration by Prof. Jongin Choi, Hanbat National University (60')
	11:00	12:00		Lecture 7: Biotechnology Industry in Korea by Prof. Byunghwan Hyun, Daejeon University (60')
	12:00	12:15		Q & A (15'):

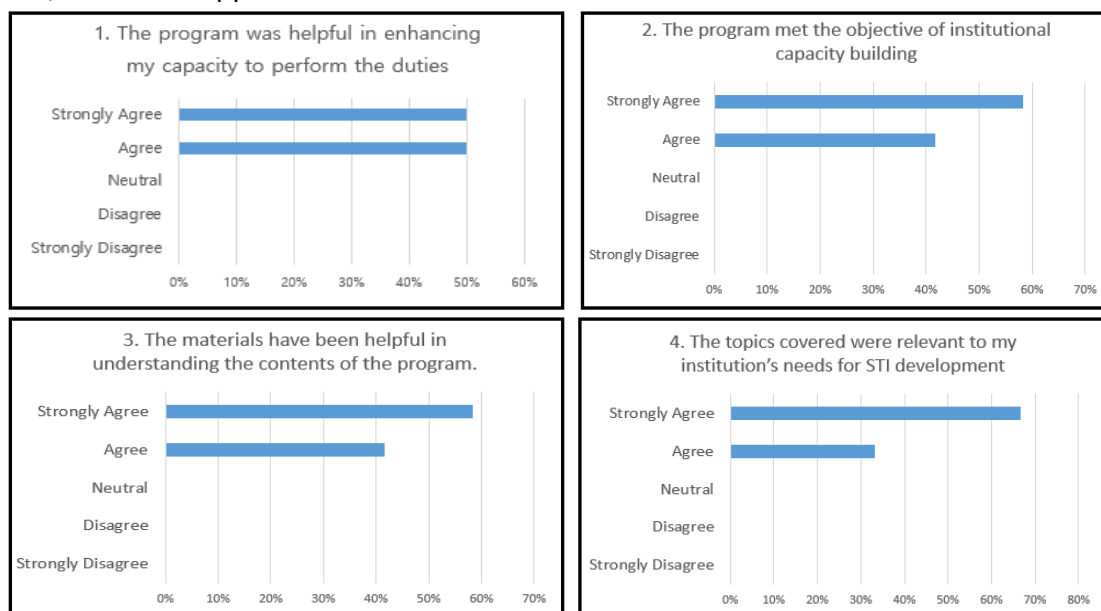
Date	Time		Place	Key Activity
	12:30	13:30	Near the Univ.	Lunch
	13:30	14:00		Move to KRIBB
	14:00	15:00	Daejeon	Field Visit 2: Korean Research Institute of Bioscience and Biotechnology (KRIBB)
	15:30	16:30		Field Visit 3: Daejeon Technopark (Dr. Sangho Choi)
	17:00	18:00		Field Visit 4: Advanced Nano Products Co., Ltd. (Prof. Changwoo Park, CEO)
	18:30	19:30		Dinner
	19:30		Move to Seoul	Move to Hotel
11 July (Thurs.)	09:00	10:00	Shilla Stay Yeoksam	Lecture 8: Key Factors of Korea's STI Development by Dr. Youngrak Choi, Senior Research Fellow Emeritus, STEPI
	10:00	10:15		Presentation 7. Industry-University Linkage: Issues and Current Status by Maxwell Otim Onapa
	10:15	10:30		Presentation 8. Industry-University Policy Linkage in Indonesia by Dr. Ade Faisal
	10:30	11:30		Q & A and Group Discussion -Overview of Indonesia's Science and Technology Innovation in Indonesia by Dr. Trina Fizzanty -Overview of STI Policy in Indonesia by Mr. Qinan Maulana Binu Soesanto -Overview of S&T Governance in Indonesia by Ms. Indah Purwaningsih
	12:00	13:30		Lunch
	13:30	14:30		[Wrap-up and Follow-up] Group Presentations
	14:30	15:00		Closing Ceremony
	15:00	17:00		Rest

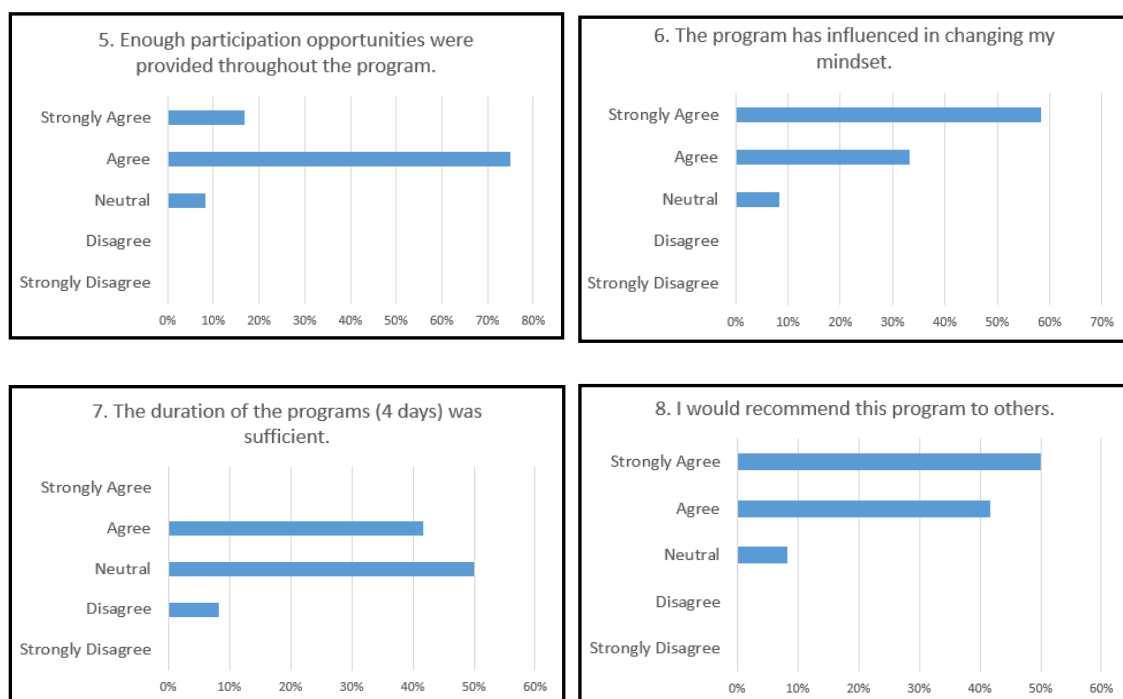
Date	Time		Place	Key Activity
	17:00	18:00		Dinner
	18:30			Move to Airport (Uganda)
	23:55		Incheon Airport (ICN)	Departure from Incheon Airport (EX323)
12 July (Fri.)	11:00			Move to Airport (Indonesia)
	15:30	20:30	Incheon Airport (ICN)	Departure from Incheon Airport (Korean Air, KE0627)

4.4 Evaluation Survey

The survey indicates the participants' opinions including (1) Relevance, (2) Efficiency, (3) Effectiveness, (3) Impact, (4) Coordination, and (5) Sustainability.

The Indonesian and Ugandan participants were highly satisfied with the course and class quality. For the next invitational workshop, they would like to have more site visits and more time for discussion. They expressed special interest in the following areas: demand-side policy; public-private-people collaboration/partnership in STI; designing technology foresight for STI development, particularly for realizing industry 4.0; and R&D support mechanism.





4.5 Lessons learned from the workshop and future actions

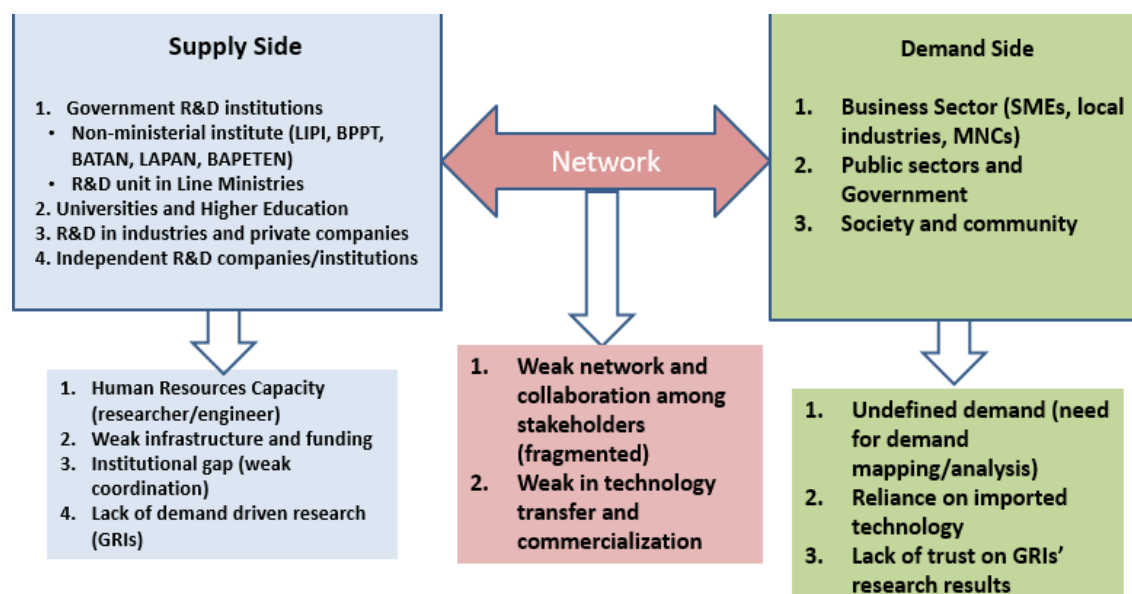
4.5.1 Lessons learned

- Leader's and people's vision (shared vision)/paradigm/sociotechnical imaginaries about the urgency and importance of STI – market mechanism, economics paradigm still apply
- Demand commitment and political will from the government for STI development in Indonesia
- Institutional transformation on Indonesia's innovation system (strengthening coordination and collaboration)
Ex.: Strengthening the role of MoRTHE (Ristekdikti), NRA (DRN); developing a new agency for think tank and coordinating R&D activities in Indonesia
- Clear division of labor (government, GRIs, university, industries) – shared clear roles and contributions according to their main task

4.5.2 Future actions

- Promoting university/GRI – industry collaboration through incentives, grants scheme - clear Target, ex. Product or winning the market
- Encouraging the industry to invest more in STI development by providing incentives, grants
- Providing a good incentive scheme for researchers/engineers (ex. share of patent royalty/opportunity to grow startup) to motivate their productivity
- Promoting technology transfer by conducting best practices in the implementation of GRI/university research results in several industries (SMEs, MNCs)/regions (ex. Local government)
- Identifying the comparative advantage (ex. biodiversity, socioeconomics, geopolitics) to nurture competitiveness in facing Industry 4.0
- Paying more attention to environmental issues (climate change, waste management, etc.), global development and looking for existing opportunities

4.5.3 Annex. Current Status of Indonesia's Innovation System



Main issues

- Lack of political will on S&T Development
- Policy implementation and capability issues
- Weak Coordination & Convergence among GRIs and other Ministries/Institutions

- Consistencies in policy
- Rigid on regulation to attract researchers and engineers to become a startup owner from their own result of R&D activities



5.

Collaboration Process for Capacity Development

5.1 Drafting of Science and Technology Park Policy by LIPI

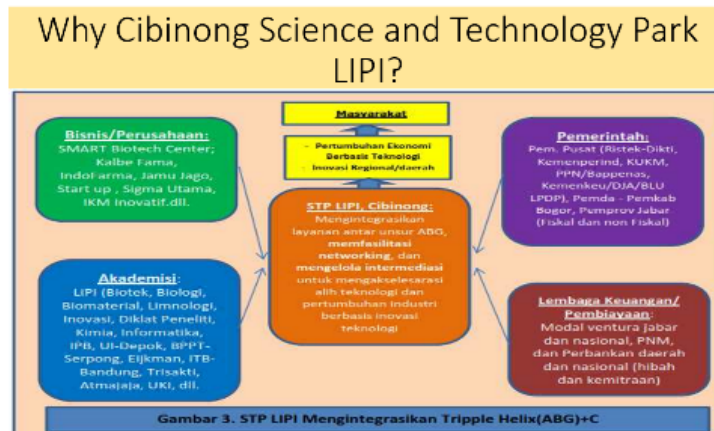
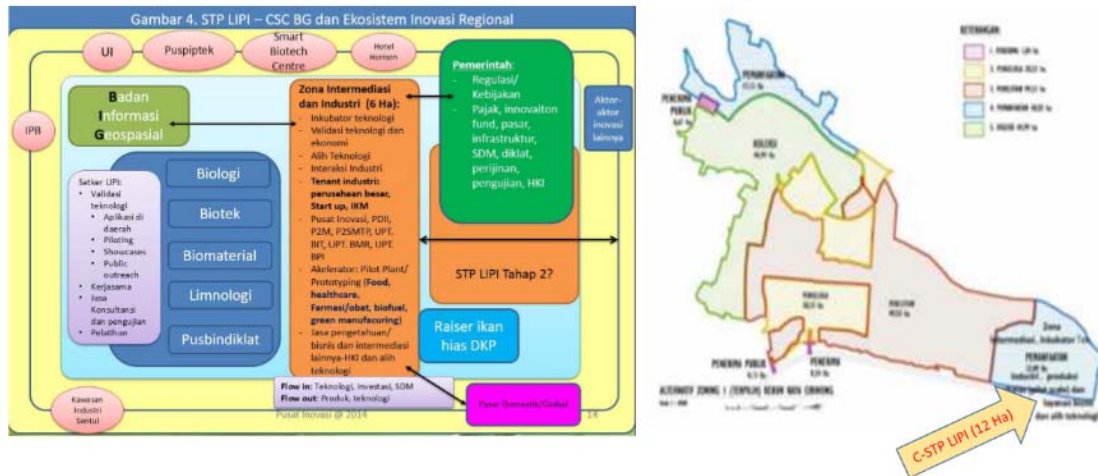
Dr. Manaek Simamora presented on 19 December 2017 our capacity development process under the title “Cibinong Science and Technology Park (C-STP) LIPI: Current Condition and Proposed Joint Research on STP Policy Implementation.” The major contents of the presentation are as follows:

Policy direction of national science and technology park.

Development of National Science Techno Park, with N–STP directed to function as: center for S&T Development; center for hi tech-based entrepreneurship; and center for hi-tech service providers for business and industry.

STP is defined as a professionally managed vehicle (wahana) to develop and encourage a sustainable economy through development, application of science and technology, and development of growth of new technology-based firms/startups. The objective of STP is to develop and utilize S&T to promote economic growth.

[Figure 5-4] Why Cibinong Science and Technology Park LIPI?



Source: Simamora (2017), presentation ppt (p.5-6)

a) Vision of the Cibinong Science and Technology Park

The vision of the Cibinong Science and Technology Park is to deliver excellent innovation services for the acceleration of technology commercialization and growth of competitive knowledge-based enterprise in enhancing economic development.

In addition, the Cibinong Science and Technology Park has four missions:

- To facilitate the acceleration of development, utilization, and commercialization of technology into startup and industrial applications in the areas of health and medicine, medical devices, food, cosmetics, green energy, agriculture, and ICT.
- To provide technological services to enhance SMEs and industrial competitiveness.
- To nurture talents to accelerate Science and Technology commercialization and technology-based enterprise development.
- To promote global innovation networks in strengthening collaboration in the technology-based enterprise development.

C-STP LIPI facilitates and synergizes the elements of the regional innovation system in nurturing and developing knowledge-based industries with the main objectives of enhancing industry competitiveness and improving productivity in society.

b) Proposed Areas of Joint Research/Study of STP Policy Implementation (LIPI and STEPI)

5.2 Feedback from Partner Organizations

Title: Reflection and Feedback on Workshop on Strategic Plan and STP Management and Policy Organized by STEPI and PAPPIPTK LIPI (May 2018)

Written by Mr. Manaek Simamora, Cibinong Science and Technology Park Coordinator, Indonesian Institute of Sciences (Wangdong Kim et al., 2018).

5.2.1 Workshop on Policy and Strategic Planning

a) Insights

The workshop on the trajectory of the Korea Innovation System from the 1970s up to the present time shows a gradual change of policy on Science, Technology, and Innovation. Initially, in the 1970s, the Korean government put emphasis on R&D carried out by Government Research Institutes to support industrial development. Gradually, R&D

activity was directed to strengthen basic science development and hi-tech orientation (Wangdong Kim et al., 2018).

Korea's experience shows a clear mission, i.e., in order to strengthen basic science capacity and hi tech-based industry, it is essential for the nation to show the contribution of R&D to the economy. Parallel to the growth of the economy where R&D plays an important role in it, investment in basic science and hi-tech areas become more feasible. It is difficult to comprehend a nation where the contribution of R&D to the economy has been insignificant or very little, but the nation focuses its GRI research on producing scientific publication and gives very little attention to R&D contribution to industrial development (Wangdong Kim et al., 2018).

b) Points to Ponder

This workshop has been able to share the experience of Korea in its innovation trajectory. Focus should be placed on how to make sure that the participants of this workshop can take action from the lessons learned. Knowing that, in the 1970s, Korea's R&D in GRI was directed at supporting industrial development (not to focus on scientific publication and ignore its impact on industrial development). In another context, Indonesia in general and LIPI in particular still have an unclear position as to the role of R&D in industrial development — even up to present time. I hope this kind of workshop can trigger a change in participants to take action to try to adopt the good lessons obtained as well (Wangdong Kim et al., 2018).

Likewise, the LIPI Strategic Plan also needs to have a clear position on how its research program will contribute to industrial development in Indonesia. This short workshop could have encouraged LIPI to determine its R&D position in industrial development, not only in contributions to scientific publication. The Workshop methodology may need to be reviewed in order to be able to persuade the participants to make a change based on their learnings from the Workshop. This includes the types and position of the participants, deliverable targets of the Workshop, and follow-up to the Workshop (Wangdong Kim et al., 2018).

c) Proposed Topics on Policy Issues in the future

I hope that STEPI and LIPI can cooperate to undertake joint research on policy implementation both at the national and institutional levels. At the national level are policy issues such as R&D and its role in national industrial development, and incentive policy on innovation adoption includes the implementation of Presidential Regulation No. 106/2017 especially on incentives for Science and Technology Parks. The lists of incentives already stipulated in the Presidential Regulation, such as how to make it operational, including tax incentive, are still unclear (Wangdong Kim et al., 2018).

The second area of cooperation that could be considered is joint research on the promotion of technology and innovation partnerships between Indonesia and Taiwan. The objective of this collaboration is to identify common interests in the area of Technology and Innovation, which can be beneficial for both countries in the mid-term future. This collaboration should also be able to create strategy to materialize the common interests that will be developed (Wangdong Kim et al., 2018).

5.2.2 Workshop on Science and Technology Park

a) Insights

It is interesting to note that the Gyeongbuk Technopark development in Korea also has a task of building research centers in the science park. It will be useful to know how these research centers contribute to the attainment of its mission -- does these research centers function as cost centers, or can they create their own funding? How can these research centers be integrated into science park management and operations (Wangdong Kim et al., 2018)?

SMEs are important target customers of this Gyeongbuk Technopark. In the context of Indonesia, where the supply of startups into a science park is still quite minimal, providing services to SMEs becomes very crucial (Wangdong Kim et al., 2018).

Design of the management of Gyeongbuk Technopark seems to be quite lean. I also found that the structure of Gyeongbuk Technopark is very interesting as it also consists of 12

teams assigned to achieve specific tasks, outputs, and outcomes (Wangdong Kim et al., 2018).

b) Points to Ponder

Cibinong STP was developed in 2015 and is still under development up to now. Just next to this CSTP are five R&D Centers, including the Research Center for Biology, Research Center for Biotechnology, Research Center for Biomaterials, and Research Center for Limnology. There is also the National Center for Researchers Development, and they all belong to LIPI. If Gyeongbuk Technopark establishes its own Research Center within its park, how can those centers next to CSTP in one compound be integrated into the CSTP? What should be integrated and how? It is noteworthy that the research centers were already established for quite a long time before the CSTP came to being (Wangdong Kim et al., 2018).

How can the Gyeongbuk Technopark experience be adopted into the CSTP LIPI? How can finance services be provided to SMEs, and what will be the business model? What are the key objectives and key performance indicators of Gyeongbuk Technopark's services to SMEs? (Wangdong Kim et al., 2018)

A lean organizational structure supported by a group of functional teams in Gyeongbuk Technopark seems to provide an agile mode of operation, not a bureaucratic mode. I am of the opinion that the existence of a functional team working in CSTP will help make its operation much more proactive and responsive in response to customers' needs. It will be very interesting if we could introduce this management structure in CSTP so as to prevent it from behaving like a genuine bureaucrat with lots of red tape (Wangdong Kim et al., 2018).

c) Proposed Topics on Technoparks in the Future

One proposed topic is joint research on STP's strategic industrial direction for CSTP LIPI. The objectives of this research are: to analyze and determine what type of strategic industry will become the focus of CSTP in the next 5 ~ 15 years, which will have significant impact on Indonesia's economy, and; to explore the common interests of Indonesia and

Korea in the identified strategic industrial areas where STEPI and LIPI can collaborate to undertake this research. Therefore, this research seeks to identify not only strategic industrial areas to focus on but also some areas of strategic industry that can benefit Indonesia and Korea (Wangdong Kim et al., 2018).

The second more practical cooperation that can be considered, say over the next five years, is to explore a kind of sister-technoparks between CSTP and a particular Technopark in Korea such as Gyeongbuk Technopark. STEPI and LIPI can make recommendations on how to manage such sister-technopark. I would suggest that this cooperation be undertaken soon. For example, STEPI and LIPI can conduct a joint study on how to develop such sister-technopark and provide recommendation on how to manage sister-technoparks effectively (Wangdong Kim et al., 2018).

Title: Feedback on the 2018 K-Innovation Program with Indonesia: Capacity-building Workshop on National STI Policy Development (May 2-3)

The workshop held on May 2-3 about developing Science & Technology Parks (S&T) has given us valuable knowledge on how Korea established and improved S&T over more than 30 years. Prof. Young Sub Kwon delivered a presentation on the concept and practices of Korean Technoparks, which gave us lessons on the determining factors and constraints in building technoparks. Prof. Jaehoon Rhee, CEO of GyeongBuk Technopark, provided us with a history of GyeongBuk Technopark including how they developed the technopark and how they will improve the technopark in the future. As an academician and a practitioner, both were excellent lecturers on the STP topic (Wangdong Kim et al., 2018).

We find that this program has been helpful in enhancing our researchers' capacity in studying and developing STP. The PPT materials have been helpful in delivering the contents of STP development. Since one of our research teams is currently working on the research project on Cibinong STP, this topic is very relevant to our institution's needs. There were enough participation opportunities provided throughout the program, since our researchers were each given a time slot to present their projects and the problems faced in developing STP; active discussion was also facilitated after Prof. Young and Prof.

Rhee presented their materials (Wangdong Kim et al., 2018).

In order to improve subsequent programs, we hope that the next workshop can be conducted over a longer period, i.e., 5 days, so that knowledge transfer can take place more effectively. Intensive supervision in developing our STP would also be very much welcome (Wangdong Kim et al., 2018). In addition, we hope for a joint research program (probably 9-12 months' long) so that our young researchers can gain experience under the STEPI researchers' supervision. The next topic we would suggest is Industry 4.0 and the digital economy, since our world is now entering a new era, and hopefully we can learn from Korea about how to adapt to the rapid changes caused by digitalization (Wangdong Kim et al., 2018).

Finally, we would like to address our deepest appreciation to STEPI for sending us your best academics and practitioners and transferring your knowledge to us (Wangdong Kim et al., 2018).

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing left, and several overlapping triangles in shades of teal and light blue pointing right.

CHAPTER 6

Conclusion

Deok Soon Yim,

Senior Research Fellow, Office of Development Cooperation Research, STEPI

Seong-soo Kim,

Professor of Hankuk University of Foreign Studies (HUFS)

Youngjoo Ko

Principal Researcher of the Korea Research Institute of Chemical Technology (KRICT)

Min Jung Kim

Researcher, Division of Global Innovation Strategy, STEPI

Chapter 6. Conclusion

1. Policy Implications

1.1 Comprehensive implications for Indonesia from Korea's GSRI and the NST cases

Korea's recent policy changes have mainly aimed at enhancing strengths and opportunities in Korea's research and innovation system and reducing or eliminating system weaknesses and threats faced by Korea. Korea has had strengths in many aspects such as strong commitment of successive governments to technology-based economic development, national consensus on the importance of science, technology, and innovation as drivers of future socioeconomic growth, high levels of GERD and BERD, highly educated labor force, good and improving framework conditions for innovation, high degree of interest in science and technology, large firms that are internationally competitive, capability to produce talent, and strong ICT infrastructure and the like.

In terms of the levels of GERD and ratio of BERD, Korea has reached one of the highest levels in the world. Several leading companies such as Samsung, LG, and Hyundai Motors and the like have become renowned and competitive transnational corporations with operations across the globe and leading-edge technologies. The number of papers and patents produced in research and innovation systems has rapidly increased to world-class levels. The concerted efforts to come up with new plans and to implement specific schemes based on the plans between many ministries, particularly for the development of technology convergence, have been strengthened. More and more national R&D projects have evolved into a more collaborative style among industry, university, and GSRI. The

number of TLOs, Technoparks, regional clusters, and university- and GSRI-companies for technology transfer and commercialization has been significantly increasing. National R&D investments by both government and business enterprise sector have moved toward more public, welfare-oriented R&D to respond to social challenges compared to five or ten years earlier, when it mainly focused on the development of industrial R&D. The various tax incentives and procurement system have been incentives for the SMEs.

Nonetheless, there are barriers, weaknesses, and problems. Despite the high level of research mobilization for national R&D and the concerted efforts of many ministries for policy and program planning, the efficiency and effectiveness of R&D investment and policy implementation are still not high due to excessive competition between industry, university, and public sector research institutes in the selection process of projects with little differentiation in their role for research and innovation, weak role of a central body for coordination, and low level of institutional funding for GRIs. Additionally, the evaluation of R&D outcome has focused on the quantitative aspects such as the number of papers and patents and superficial number of participants for collaboration between industry, university, and public sector research organizations.

In spite of the significantly increasing number of papers and patents, their quality is still low because the evaluation system is focused on quantitative outputs, with weak capacity to conduct fundamental research in both university and GSRI, superficial collaboration for obtaining a government project without substantial knowledge circulation between industry, university, and GSRI and so on.

Despite the increasing number of research institutes in SMEs and high level of private investment, the innovation gap between the chaebol, or Korean big conglomerate, and SME has widened due to too much concentrated and closed R&D system in chaebol, weak HRST in SMEs, and low technology transfer from universities and GSRI to SMEs.

In spite of the R&D globalization, the overall level of Korea's investment for the internationalization of S&T and global open innovation is very low. The main investment has usually gone to small projects base including persistently minimal participation in

overseas R&D programs and organizations.

Indonesia first needs to focus on using and expanding its strength in an innovative way such as relative abundance of natural resources and relatively large population compared to other ASEAN countries. Many government R&D organizations should collaborate to make these strengths much stronger by providing innovative technology and ideas to the related industry, government, and society. The central government could take into account how government R&D organizations work in a more effective way with industry, government, society, and among themselves. New rules and regulations for reform and new governance are needed to move in a more proactive way and communicate well with experts in government R&D organizations and outside. Especially there is a need for new governance such as NST Korea and new programs and evaluation system to catalyze collaborations with industry and other innovators, including new rules and cultural approaches for researchers and workers to contribute more to society and industry communication with an open innovation approach. Technology innovation for industry and society is no longer a linear model. A new system approach based on co-platform among government R&D organizations, industry, society, and academe is needed. The STI policy target should go there.

1.2 Specific implications for Indonesia from Korea's experiences

Another issue is R&D productivity and more effective commercialization. In order to improve them, industry and society should be involved more with government R&D organizations from the planning stage and R&D conducting stage as well as technology transfer stage. The government needs to expand R&D investment in new programs and different evaluation criteria to expand such involvement of industry in the whole R&D stage. These kinds of activities should be increased overtime with priority setting in R&D and industry areas.

One of the most important things is that environment and societal problems should be considered as well from the R&D planning stage. Indonesia has a characteristic geography with lots of islands and complex society with diverse people. If the government can solve

those issues with research and innovation, there could be societal innovation, and the next new development of industry the Indonesian way with better quality of life can be expected.

If the Indonesian Government takes into consideration the new governance for government R&D organizations like NST in Korea, there are some points to requiring further thought. The new governing body, first of all, has to get funding power with autonomy to coordinate and collaborate with more R&D organizations in a flexible, sustainable way. Without such kinds of autonomous funding power, the reduction of bureaucratic intervention from government officials and innovative change in bureaucratic culture inside R&D organizations could not succeed. Of course, flexible collaborations between R&D organizations to solve any problems are very important so that the role of the new governing body with funding power could be strengthened. In that case, more democratic managerial governance of the new research council should also be taken into account with mutual communication among stakeholders. Another point is that the integration of R&D policy and industry policy and many societal policies is needed beyond the linear model of innovation with a more open innovation ecosystem. In that process, the policy for expanding the limited number of talented human resources should be the priority in case of a need to increase human collaborations with other countries such as Korea.

1.3 More consideration to public sector research

Public sector research organizations have been developed in many countries to achieve national or regional goals, develop the strategies of science, technology, and innovation policy, or to solve social and political problems, through research. They produce knowledge and services as crucial elements of national research and innovation systems and of the public services governments deliver to people and society. They have, however, been understood and configured in some different theoretical rationales which result in different strategies, governance structures, funding schemes and management styles etc.

Public sector research organizations are, as a whole, an area where complicated

interactions take place among economy, society and polity, necessarily engendering various approaches to them. This thesis showed that three different conceptual approaches to public sector research organizations based on different focuses (namely institutions, activities, and a networked system) and reviewed the various theoretical rationales for public research organizations (namely linear model approach, market failure approach, government failure approach, and system or network failure approach). Those different perceptions and rationales provide different objectives, strategies, governance structure, funding schemes and management tools in public sector research organizations, all of which are interactively connected to one another, shaping a public sector research organizations' regime.

Like any other organizations, public sector research organizations themselves always try to change their instrumental settings such as institutional objectives, strategies to achieve the objectives, internal governance, funding priorities for their R&D and management tools in order both to conduct their R&D and operate institutions better. Such kinds of changes usually take the form of incremental change which alters their settings partially by reference to the evaluation concerning the existing settings through an existing normal process within the same or similar regime (Ko, Y.J., 2005). Such incremental change in public sector research organizations can also take place by incremental change of government policy for them through normal a policy cycle which takes place by reference to the evaluation of the stages in an existing policy process or policy cycle which usually consists of agenda setting, policy formulation and legitimatization and implementation, within the same or similar theoretical rationales.

However, under any circumstances in which the existing settings become abnormal, public sector research organizations become unstable and a government policy can change the settings abruptly and lead the whole aspects of public sector research organizations into non-incremental transformation (Ko, Y.J., 2005). The non-incremental change is usually initiated not by reference to evaluations of the stages in an existing policy process or policy cycle but by new policy options stemming from external perturbations (Ko, Y.J., 2005). The non-incremental change in the end completes, not by a single action within a short time but by a series of actions and events over a long time-span and makes public sector

research organizations shift away from an existing theoretical rationale for policy in relation to them to a new theoretical one, accordingly replacing an old regime with a new regime of public sector research organizations (Ko, Y.J., 2005). Shifting from one theoretical rationale for policy or public sector research organizations' regime to another is not a minor policy change inside a normal evaluation process, but a large one which takes place through dynamic interactions within many factors and variables over a long time-span (Ko, Y.J., 2005).

There has also been a wide diversity of perceptions on science, technology, and innovation ranging from techno-scientific optimism to a view that sees them as sources of risk and problems to a new social contract (Ko, Y.J., 2005). Science, technology, and innovation are highly uncertain and complex concepts and activities, in that they are becoming increasingly active both among themselves and within society in a number of different ways, even though the processes involved are still simple linear and hierarchical ones in the minds of many people. Science, technology, and innovation policy are areas of government intervention which aim both to reduce the questions related to the negative externalities of science, technology, and innovation by society and, more actively, to enhance the positive benefits from science, technology, and innovation in a broader sense. Science, technology, and innovation policy is different from other policy domains due to the inherent features of the scientific and technological knowledge, mainly such as uncertainties of research outputs and a long time perspective of science (Ko, Y.J., 2005). Particularly, in advanced countries, policy is usually nowadays forged through the involvement of various levels of social actors, institutions and ideas, based on different views within complex socioeconomic and political environments at sub-national, national, supra-national and even global levels. That is where the theoretical rationales or justification of government policies' intervention in science, technology, and innovation come to the fore. Science, technology, and innovation policy is formulated and implemented on the basis of four theoretical rationales, namely: the linear model, the market failure, government failure, and the system failure approach. Each of these gives specific guides for instrumental settings of public sector research organizations (Ko, Y.J., 2005).

1.4 Conclusion

The science, technology, and innovation policy can be characterized by referring to the dynamic aspects, and it can change over time and vary from country to country. Policymaking and implementation reflect various perceptions depending on the dynamics within specific situations in particular countries (Ko, Y.J., 2005). This is because making and implementing a science, technology, and innovation policy, even though the normal policy process or cycle, is by no means easy (Ko, Y.J., 2005). What is more, non-incremental policy changes that may be similar or different from country to country can take place outside the normal policy process or cycle with complex interactions among institutional arrangements, competing interests, emerging ideas, socioeconomic situations, and political preferences at regional, sectoral, national, and global levels. Therefore, it is important to address issues in the wider cultural, political, and economic milieu when formulating a science, technology, and innovation policy (Ko, Y.J., 2005).

Public sector research organizations themselves are one of the crucial vehicles implementing the science, technology, and innovation policy. Major changes in science, technology, and innovation policy usually influence the public-sector research organizations' regime, which is comprised of objectives, strategies, governance structure, funding schemes, and management styles (Ko, Y.J., 2005). Many governments usually design new science, technology, and innovation policy in order to reformulate the public sector research organizations, aiming at achieving national goals or strengthening national innovative capacities and industrial competitiveness and so on (Ko, Y.J., 2005). During the processes, many actors and ideas embedded in institutions and socioeconomic and political environments are necessarily involved (Ko, Y.J., 2005).

If the Indonesian government wants to introduce the research council system, it might require a non-incremental policy change in government R&D organizations as well as S&T policy. It needs to be a long, strategic approach for successful policy rather than a short-term, top-down approach (Ko, Y.J., 2005).

STI policy issues and R&D governance problems need a "fitting way" (with the Indonesian

STI realities) of thinking to provide policy alternative and governance solution. From the viewpoint of institution and public policy, this paper constructed an operational model STI policy and an R&D governance system, which have three characteristics. (i) First is long-term strategic perspective, top STI policy direction, and synergistic R&D governance. The model deals with STI policy coordination through the function of the STI policy council led by president and which is capable of coordinating STI resources in an integrated manner. (ii) Second is the instrument of STI policy through the function R&D governance body, which executes the coordinated, synergized national R&D program and activities involving cross-sectoral ministries/institutions. (iii) Third is putting the couple elements -- STI policy council and R&D governance body – to actual use by measurable key performance index (KPI).

The expected outcomes of the model implementation contribute to the following: (i) societal transformation by strengthening STI capacity, which has the power of providing solutions to eradicate poverty, improve social and economic conditions, increase resilience to natural hazards, and preserve natural resources to meet the Sustainable Development Goals (SDGs); (ii) the state of STI-based economy in the long run, i.e., economic prosperity supported by technological excellence, industrial development, and growth of the manufacturing sector, which would all be achieved through science advancement, technology development, and continuous innovation; and (iii) the application of industrial policy for the promotion of R&D in the specific technology-intensive sector (new materials and life science–based industry) is expected to bring Indonesia out of the trap of middle income country in 2030.

The leverage factor of effective and productive R&D governance in Indonesia is the availability of highly qualified researchers. Therefore, universities and R&D institutions should: (i) invest in sophisticated research laboratory infrastructures and recruit highly competent researchers; (ii) upgrade the quality and quantity of the country's higher education to produce graduates and postgraduates; (iii) utilize competitive brain gain by applying internationally comparative rewards to attract the return of highly qualified researchers from abroad; (iv) implement conducive legal and institutional support to maintain the positions of highly competent researchers in industrial R&D; and (v) manage

new technology mastery by giving strong support to the prioritized field of sciences, where new materials and life science-based R&D for industrial technology development will be important for Indonesia's future competitiveness.

Abbreviations

- (1) BAPPENAS : Badan Perencanaan Pembangunan Nasional (National Development Planning Agency)
- (2) BATAN : Badan Tenaga Atom Nasional (National Nuclear Energy Agency)
- (3) BPPT : Badan Pengkajian dan Penerapan Teknologi (Agency for the Assessment and Implementation of Technology)
- (4) BSN : Badan Standarisasi Nasional (National Standardization Agency)
- (5) LAPAN : Lembaga Antariksa dan Penerbangan Nasional (National Institute of Aeronautics and Space)
- (6) LIPI : Lembaga Ilmu Pengetahuan Indonesia (Indonesian Institute of Sciences)
- (7) MoI : Kementerian Perindustrian (Ministry of Industry)
- (8) MoRTHE : Kementerian Riset, Teknologi dan Pendidikan Tinggi (Ministry of Research, Technology and Higher Education)
- (9) RIRN : Rencana Induk Riset Nasional (National Research Master Plan)
- (10) RPJMN : Rencana Pembangunan Jangka Menengah Nasional (National Medium-Term Development Plan)
- (11) RPJPN : Rencana Pembangunan Jangka Panjang Nasional (Long-Term National Development Plan)
- (12) R&D : Research and Development
- (13) S&T : Science and Technology

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